

The Effects of Tax Incentives to Donate Wealth to Charity: Evidence from Qualified Charitable Distributions*

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Abstract

Can tax incentives encourage people to donate wealth to charity? Using regression discontinuity, I estimate the causal effects of tax rules that incentivize transfers of funds from retirement savings accounts to charities once individuals turn $70\frac{1}{2}$. These transfers, called qualified charitable distributions (QCDs), are excluded from taxable income and have additional tax benefits. I find that QCD eligibility significantly increased donations after the Tax Cuts and Jobs Act, which reduced incentives to itemize deductions and thus strengthened incentives to use QCDs to facilitate tax-advantaged giving. The increase occurred on the intensive margin and reflects increased likelihoods of donating large sums.

Keywords: Tax Policy, Charitable Giving, Qualified Charitable Distributions, Retirement Savings, Individual Retirement Accounts

JEL codes: H24, H31, H41, D14, D64, J14, J26

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1 Introduction

The tax treatment of wealth might influence how people manage and consume their assets (Kopczuk, 2013; Saez and Stantcheva, 2018; Scheuer and Slemrod, 2021). In the U.S., tax-advantaged retirement savings accounts are a dominant asset. For instance, the Investment Company Institute (2025) estimates that about \$12.2 trillion are held in defined-contribution plans like 401(k)s and an additional \$16.8 trillion are held in Individual Retirement Accounts (IRAs). Typically, contributions to these accounts are tax-deductible, earnings within the accounts are not taxed, and then withdrawals from the accounts are taxed as regular income. However, special tax rules allow people who have reached a certain age to make tax-free withdrawals from IRAs if the distributed funds are used for a particular type of consumption: donating to charity. Do these tax incentives for donating retirement wealth to charity increase charitable giving? In this paper, I provide novel evidence on the answer to this question.

The tax rules that I study stipulate that IRA holders can make tax-free donations—known as IRA charitable rollovers or qualified charitable distributions (QCDs)—once they reach age $70\frac{1}{2}$. These rules are closely linked to another set of tax rules called required minimum distributions (RMDs), which during my main analysis period mandated withdrawals from IRAs at the same time by instituting penalties for failing to withdraw sufficient funds.¹ The combination of these rules meant that once account holders turned $70\frac{1}{2}$, they had to begin withdrawing funds from IRAs, but they could avoid paying taxes on those withdrawals if they were donated to charity. Importantly, while donations to charity are in general tax-deductible for people who itemize their deductions, QCDs allow people who take the standard deduction to make tax-free donations as well. Moreover, QCDs have additional tax benefits. First, they do not count towards the maximum charitable deduction limit, which thus increases the amount of money that itemizers could potentially donate tax-free. Second, they do count towards RMDs and can therefore be used to avoid under-withdrawal penalties, and by reducing IRA balances they reduce future RMDs, which could be valuable for people looking to avoid tax implications of greater withdrawals later.

Using a regression discontinuity design and detailed, biennial survey data from the Health and Retirement Study (HRS), I estimate the causal effects of aging into this bundle of charitable-giving incentives on donation decisions. Survey data are crucial for my study because administrative tax data only measure donations recorded on a tax form, like deductible donations for people who itemize, but a key advantage of QCDs is that they also incentivize

¹The RMD age recently increased to 72 and then again to 73, but the QCD age has remained at $70\frac{1}{2}$.

donations for people who take the standard deduction. I need to be able to measure donations for these people as well, which the HRS data allow me to do. My research design exploits the discontinuous nature of the policy for identification. The basic idea is to test for whether charitable giving changes discontinuously once people reach $70\frac{1}{2}$ and become eligible to make the tax-advantaged donations. Specifically, for a given tax year, I compare the charitable giving of IRA holders who turn $70\frac{1}{2}$ before the end of the year, who are thus eligible to make QCDs, to otherwise-similar households who do not turn $70\frac{1}{2}$ until the start of the next year, who are thus ineligible to make QCDs.

I use this approach to document effects over two distinct time horizons, before and after the Tax Cuts and Jobs Act (TCJA) of 2017. This major tax reform had important implications for charitable giving and for QCD policy. Crucially, it increased the standard deduction, which increases the attractiveness of using QCDs to donate to charity. The key insight is that IRA holders who itemize deductions can replicate an advantage of QCDs by withdrawing money from their IRA, donating the money to charity, and then claiming the charitable deduction. By substantially increasing the standard deduction and reducing incentives to itemize, the TCJA eliminated the typical tax benefits from donating to charity for many. This change should therefore be expected to increase the use of QCDs.

I find clear evidence of a discontinuous increase in charitable donations at $70\frac{1}{2}$ after the TCJA. The increase occurred on the intensive margin. I find no evidence of changes in the extensive-margin decision about whether to give to charity at all. For people who did give, the baseline estimate indicates that QCD eligibility increased total donations by \$2,229 in 2019, on average. This estimate corresponds to a large, 53.9% increase when compared to average donations for people who gave to charity but were just-ineligible for QCDs, although it is modest when compared to the amount of wealth people hold in IRAs. In contrast, evidence from earlier years suggests little to no effects on charitable giving before the TCJA. The point estimates for earlier years are generally smaller and not statistically different from zero.

I then show that the estimates are robust to many specification checks and are indeed driven by incentives from QCD rules. A primary interpretation issue relates to RMD rules inducing IRA withdrawals at the same time. If mandated withdrawals from IRAs have their own, independent effect on charitable giving, then my RD estimates would capture the combination of this effect and that of QCDs. To investigate this possibility, I use pre-period data, when RMDs were in place but QCDs were not. I first use my RD design to estimate the effects of reaching $70\frac{1}{2}$ on charitable giving in the pre-period and find no statistical evidence of

an impact. Even so, I then pool pre-period and post-period data to estimate a difference-in-discontinuities design, which takes the difference between post-period discontinuities (when QCDs and RMDs were in place) and pre-period discontinuities (when only RMDs were in place). The difference-in-discontinuities estimates are very similar to the baseline, year-by-year RD estimates. The evidence thus indicates that the main results are indeed driven by QCD eligibility, not by mandated withdrawals themselves. However, I emphasize that the effects of QCD eligibility could still be different when also exposed to mandated drawdown through RMDs compared to when not.²

Finally, I conclude my analysis by looking at additional outcomes and subsamples that help to unpack the overall estimates and shed light on mechanisms. First, to better understand behaviors underlying the \$2,229 increase in average donations, I estimate the effects on indicators for making donations of various sizes. I find considerable increases in the likelihood of donating large sums. For example, the estimates imply a near-doubling of the number of people who donated more than \$10,000. (For comparison, mean donations for people with IRAs conditional on giving in 2019 were just under \$5,000.) Second, to shed more light on the role of the standard deduction in influencing the effects, I conduct a subsample analysis based on tax filing status. I find larger and statistically significant effects for people who claimed the standard deduction, whereas the estimates for people who itemized deductions are smaller and not statistically different from zero. Third, I also estimate separate effects for subsamples based on IRA balances. I find larger effects for people with larger IRAs, which might suggest that people with stronger incentives to reduce future RMDs by using QCDs to lower IRA balances were more likely to increase donations.

My paper relates primarily to the longstanding literature that studies charitable giving responses to tax incentives (e.g., Feldstein and Clotfelter, 1976; Feldstein and Taylor, 1976; Boskin and Feldstein, 1977; Clotfelter, 1980; Auten, Cilke and Randolph, 1992; Randolph, 1995; Barrett, McGuirk and Steinberg, 1997; Auten, Sieg and Clotfelter, 2002; Fack and Landais, 2010; Feldman, 2010; Bakija and Heim, 2011; Meer, 2014; Duquette, 2016; Fack and Landais, 2016; Hickey, Minaker and Payne, 2019; Almunia et al., 2020; Meer and Priday, 2020; Hungerman and Ottoni-Wilhelm, 2021; Hickey et al., 2023; Han, Hungerman and Ottoni-Wilhelm, 2024), which informs optimal tax policy (Saez, 2004; Diamond, 2006).³

²Indeed, a supplementary analysis using data from 2021, when the RMD age had increased to 72, suggests that donations increased more after age 72 than $70\frac{1}{2}$, which emphasizes the relevance of QCDs allowing people to offset associated taxes with required drawdown. However, because of these two nearby age cutoffs, other changes in tax policy on charitable giving, and the Covid pandemic in general, I interpret the evidence from this additional year as much more suggestive.

³For reviews of the literature, see Andreoni (2006), List (2011), Andreoni and Payne (2013), Clotfelter

One of the most common incentives studied is the tax-deductibility of charitable gifts for people who itemize. Many papers focus on estimating an after-tax price elasticity of giving and often exploit variation in the price of giving due to variation in marginal tax rates, either from changes in income or tax reforms. In general, the degree to which giving responds to tax incentives differs across studies and may be quite context-dependent.

A few papers are particularly relevant for my study. Two provide important information on the broader policy-and-charitable-giving environment in which I conduct my analysis. Meer and Priday (2020) and Han, Hungerman and Ottoni-Wilhelm (2024) project and estimate, respectively, the overall effects of the TCJA on charitable giving and find that this major tax reform substantially reduced donations. To help contextualize the magnitudes of my estimates, note that Han, Hungerman and Ottoni-Wilhelm (2024) find that the TCJA decreased average donations by about \$900 dollars for people induced to switch from itemizing to taking the standard deduction, who therefore lost access to the typical route to tax-advantaged giving. My baseline estimate indicates a \$2,229 increase in average donations for people who aged into QCDs in 2019, when many took the standard deduction and thus gained access to tax-advantaged giving at $70\frac{1}{2}$. My estimate thus suggests a relatively-high degree of responsiveness. Moreover, since QCDs apply to retirement wealth, my study connects to a few recent papers on how donations respond to other types of wealth-related tax incentives. Ring and Thoresen (2025) show that indirect incentives to donate from wealth taxation in Norway (specifically the tax treatment of secondary homes) impacts charitable giving, Cagé and Guillot (2025) leverage a wealth tax reform in France to study the substitutability of charitable and political donations, and Essig Aberg (2025) examines how high-wealth donors in the U.S. using family foundations to donate make charitable giving decisions.

I contribute by providing new evidence on the causal effects of qualified charitable distributions on giving. To the best of my knowledge, no other paper has estimated the effects of these important tax rules and incentives, despite the fact that the regulations have been around for almost two decades and apply to older people with assets, a group who may be particularly likely to give to charity (see List, 2011; Meer and Priday, 2021). My estimates on the effects of these tax rules provide unique evidence on how individuals respond to direct incentives to use accumulated assets to make monetary donations to charity.

My research design and setting have three additional advantages worth emphasizing.

(2016), and Andreoni and Smith (2021). For examples of relevant, yet distinct, papers using experiments to study matching grants, see Eckel and Grossman (2003), Karlan and List (2007), Huck and Rasul (2011), Karlan, List and Shafir (2011), Huck, Rasul and Shephard (2015), and Karlan and List (2020).

One is that the identifying variation in incentives to donate is particularly clean, as it comes from an individual’s age at the end of the year, and allows me to use a powerful regression discontinuity design. Two is that the nature of QCD rules allows me to study responses of people regardless of their itemization status, whereas previous work has tended to focus on people who itemize. Three is that I study both extensive and intensive margin responses, which is less common in the literature (Almunia et al., 2020).

My paper also relates to a broader literature on the decumulation of wealth in retirement (Poterba, Venti and Wise, 2011; Banks and Crawford, 2022) and specifically an emerging strand of this literature that estimates the effects of tax rules governing the drawdown of retirement savings accounts. Understanding how these rules impact people is increasingly important against the backdrop of population aging and in light of recent and substantial growth in distributions from retirement accounts; total withdrawals from IRAs increased from about \$295 billion in 2015 to just-under \$500 billion in 2022 (IRS Statistics of Income, 2024). Existing evidence indicates that additional taxes on early withdrawals (Goda, Jones and Ramnath, 2022; Stuart and Bryant, 2024) and required minimum distributions (Poterba, Venti and Wise, 2013; Brown, Poterba and Richardson, 2017; Mortenson, Schramm and Whitten, 2019; Horneff, Maurer and Mitchell, 2023; Stuart and Bryant, 2024) strongly influence drawdown, but there is little causal evidence on how drawdown policies impact broader financial behaviors. One exception is Leganza (2024), which shows that aging into required minimum distributions increases inter vivos transfers to children. By showing that the ability to make tax-free distributions from IRAs if the funds are transferred to charity can increase charitable giving, my findings support the idea that tax rules that encourage people to use retirement accounts in specific ways can indeed influence household behaviors.⁴

2 Policy Environment

To understand qualified charitable distributions, one needs to understand the tax treatment of charitable gifts more generally, as well as the tax treatment of retirement savings accounts and required minimum distributions (a closely-related, yet distinct, set of tax rules). In this section, I discuss these relevant institutional details, provide more information on QCD rules, and overview the TCJA, which is also important for my study.

⁴For examples of other behaviors the government might influence with tax rules that apply to the drawdown of retirement savings, consider the tax treatment of early withdrawals. Distributions from retirement accounts before age $59\frac{1}{2}$ are generally subject to income tax and an additional 10% tax, but this additional tax is waived—and thus funds are less costly to access—if, for example, the distribution is for qualified education expenses, a first-time home purchase, or expenses related to childbirth or adoption.

2.1 Typical Tax Treatment of Charitable Gifts

The U.S. government incentivizes gifts to charitable organizations through the tax code. Since 1917, taxpayers who itemize their deductions have been able to deduct contributions to qualifying charities from taxable income. This regulation encourages giving by reducing the effective price of donating to charity. For an individual who itemizes, each dollar donated to charity reduces their tax liability by their marginal tax rate. So, for example, an individual who faces a marginal tax rate of 32% can donate \$1 to charity at a price of 68 cents. Importantly, this incentive does not apply to people who do not itemize and who instead take the standard deduction.

2.2 Typical Tax Treatment of Retirement Savings Accounts

The U.S. government incentivizes saving for retirement with tax-advantaged retirement savings accounts. Some retirement accounts like 401(k)s are sponsored by employers, whereas others, called Individual Retirement Accounts (IRAs), are not. Traditionally, contributions to these accounts are tax-deductible, earnings within the accounts are tax-free, and withdrawals from the accounts are taxed as regular income. In contrast, there are Roth accounts that are treated differently, with contributions that not tax-deductible, earnings that are tax-free, and withdrawals that are not taxed.

I focus my analysis on IRAs because qualified charitable distribution policy applies to IRAs, not employer retirement accounts. Notably, many people accumulate funds in employer retirement accounts while working but then roll over the assets to IRAs later in life, which likely explains why IRA assets are greater than those in employer retirement plans in cross sectional data like those provided by Investment Company Institute (2025).

2.3 Required Minimum Distributions

A key set of rules related to QCDs are RMDs, which apply to retirement savings accounts and form the backdrop to QCD policy. RMD rules require households to begin making withdrawals from IRAs once account holders reach the calendar year that they turn a specified age. Traditionally, the RMD age was $70\frac{1}{2}$, and this age is the relevant one for most of my analysis. However, the RMD age was recently increased by the Setting Every Community Up for Retirement Enhancement (SECURE) Act of 2019 and then again by the SECURE 2.0 Act of 2022. RMDs aim to limit revenue losses from IRAs and apply to traditional IRAs, but not Roth IRAs, since contributions to Roth IRAs are made on an after-tax basis.

The amount that an account holder is required to withdraw depends on the balances of their accounts. For a given IRA, the RMD is calculated by dividing the balance of the account on December 31 of the previous year by a distribution period taken from IRS life expectancy tables. If a person owns multiple IRAs, their total RMD is the sum of their RMDs for each account. Required distributions typically start at roughly 4% of the IRA balance and increase to about 5.5% after 10 years.⁵

While RMDs officially begin in the year the account holder turns $70\frac{1}{2}$, the first required withdrawal is subject to a grace period and is not due until April 1 of the next year. All other RMDs are due by December 31 of the calendar year to which the RMD applies, and the penalty for not taking an RMD is a 50% tax on the required-but-undistributed amount.

It is important to note that these rules coincide almost perfectly with QCD rules, described in more detail in Section 2.4. I therefore take these rules into careful consideration when implementing my regression discontinuity design and interpreting the resulting estimates. If RMD policy has its own impact on charitable giving, then my estimates would reflect a combination of RMD and QCD regulations. Crucially, QCDs came into place long after RMDs were implemented, which allows me to use data from years when RMDs were in place but QCDs were not to test and account for any potential impacts of RMDs themselves on charitable giving. I describe the test in more detail later, when assessing threats to the validity and interpretation of my research design, but I essentially (i) show that there is no evidence that RMDs impacted charitable giving in the pre-period and (ii) show that difference-in-discontinuity estimates that difference out any pure effects of RMDs are very similar to the main cross-sectional regression discontinuity estimates. Together, these exercises support the interpretation of my main estimates as being driven by QCD incentives.

I also note two instances in which RMDs were waived that are relevant for my analysis. First, in response to the Great Recession, the Worker, Retiree, and Employer Recovery Act of 2008 suspended all RMDs for 2009. This suspension means that people in my study who were surveyed about their donations in 2009 were not facing mandated drawdown in that year. Second, the Coronavirus Aid, Relief, and Economic Security (CARES) Act, which was enacted in March of 2020 and suspended RMDs in that year, also contained a provision

⁵The specific distribution period divisor a person faces is determined by their age (as of their birthday in that year) and marital status. For most people, the divisor comes from the IRS Uniform Lifetime Table, although a person with a spouse who is the sole beneficiary of their IRA and who is more than 10 years younger than them use distribution periods from the Joint Life and Last Survivor Expectancy Table. For a simple example, consider an unmarried individual who turned 71 in 2019 and who had one IRA with a balance that was \$100,000 on December 31 of 2018. The IRS Uniform Lifetime Table indicates this person faced a distribution period of 26.5. Their required distributions for 2019 were thus $\$100,000/26.5 = \$3,773.58$.

relevant for initial RMDs in 2019, which is the final tax year in my analysis. Specifically, the CARES Act waived initial RMDs for 2019 that had not yet been taken. Recall that the grace period for initial RMDs means that people facing their first required withdrawals for 2019 were actually allowed to wait until April of 2020 to take the distributions. This waiver means that it is possible that some people in my sample who were surveyed during 2020 and asked about their 2019 donations were planning on taking their 2019 RMDs in the early months of 2020 but then ultimately had the requirements waived by the CARES Act.

2.4 Qualified Charitable Distributions

Part of the Pension Protection Act of 2006 established qualified charitable distributions. The rules allow people who have reached age $70\frac{1}{2}$ to exclude IRA withdrawals given directly to a qualified charity from taxable income. The maximum QCD for an individual is \$100,000, but each member of a couple filing jointly can make a QCD. In addition, QCDs do not apply to the overall charitable deduction limit, which is typically 60% of adjusted gross income, and they count towards any RMDs the individual needs to take.

For a donation to qualify as a QCD, several conditions must be met. The person making the donation must be older than $70\frac{1}{2}$ when the donation is made, the funds must come from a traditional IRA, and the withdrawn funds must have been otherwise included in taxable income.⁶ Moreover, the distribution must be a trustee-to-trustee transfer of funds directly from the IRA to the receiving charity, and the charity must be eligible to receive tax-deductible charitable distributions.

The original legislation created QCDs on a temporary basis: people could make QCDs in tax years 2006 and 2007. However, several acts extended the policy in two-year increments so that it was in place through 2014. Finally, the Protecting Americans from Tax Hikes Act of 2015 made QCDs permanent.

QCDs provide several tax benefits that encourage people to donate. Perhaps most importantly, they allow people who take the standard deduction to make tax-advantaged gifts. In the absence of QCDs, IRA holders at advanced ages would need to itemize in order to deduct their charitable gifts. People who itemize can benefit from QCDs as well though. Without QCDs, these individuals could still make a tax-advantaged gift by withdrawing

⁶Typically distributions from traditional IRAs are considered taxable income. However, sometimes people make non-deductible contributions to traditional accounts. For instance, contributions to traditional IRAs are not deductible for individuals who are covered by a workplace retirement plan and have income above a certain threshold. The earnings on any non-deductible contributions are taxed as income upon withdrawal, but the non-deductible contributions themselves are not, since the account holder already paid income taxes on them.

funds from their IRA, donating the funds to a charity, and then taking a deduction for the gift. However, QCDs allow people who itemize to make greater tax-free donations because QCDs do not count towards the charitable deduction limit. For both groups of people, the fact that QCDs count towards RMDs provides some additional incentives to give: donating required distributions reduces tax liability today and reduces IRA balances and thus also reduces future RMDs.

Moreover, I note that QCDs may lead to reductions in tax liability for people who would have itemized in the absence of the rules, but who instead choose to use QCDs and take the standard deduction because of the rules. For example, suppose a single individual faces a standard deduction of \$12,000, would like to donate \$20,000 to charity, and this gift would be their only tax deduction. Without being able to make a QCD, this person would itemize and thus reduce their taxable income by \$20,000. With QCDs though, this person could make a \$20,000 QCD and then take the standard deduction, which would together reduce their taxable income by \$32,000. This type of incentive means that QCDs might impact itemization status.

2.5 The Tax Cuts and Jobs Act

The TCJA was passed in December of 2017 and was implemented in 2018. This major tax reform had important implications for charitable giving in general (Meer and Priday, 2020; Han, Hungerman and Ottoni-Wilhelm, 2024). The most prominent change brought about by the TCJA as it relates to charitable giving is the increase in the standard deduction. In the years before the TCJA, the standard deductions for single taxpayers and those who are married filing jointly were roughly \$6,000 and \$12,000, respectively. After the TCJA, those standard deductions were increased to \$12,000 and \$24,000, respectively (in 2018) and \$12,200 and \$24,400 (in 2019, the main post-TCJA year in my analysis). Moreover, the TCJA limited deductions for state and local taxes and for mortgage interest payments.

These key changes resulted in a large shift in the share of people who took the standard deduction. The fraction of tax returns with itemized deductions fell abruptly after the TCJA from over 30% to under 10% (Tax Policy Center, 2024). Meer and Priday (2020) and Han, Hungerman and Ottoni-Wilhelm (2024) study the effects of the TCJA on charitable giving and provide in depth discussions of how the TCJA and the increase in the standard deduction could theoretically impact donations. On the one hand, the increase in the standard deduction removes the typical incentive for itemizers to donate because their donations will no longer be tax-deductible. On the other hand, the increase in the standard deduction can

also represent a positive income effect, which could induce some people to donate more.

My focus is not on estimating the effects of the TCJA, but rather on the effects of QCDs both before and after the TCJA was implemented. The key insight for my analysis is that QCDs should become more attractive after the TCJA. Before the TCJA, when the standard deduction was lower, many older people with IRAs could make tax-free donations by itemizing and deducting their charitable gifts. After the TCJA, the increase in the standard deduction limits that route to tax-free giving, but older people with IRAs who no longer find it advantageous to itemize can instead use QCDs to make tax-free donations. Moreover, to the extent that income effects from an increase in the standard deduction induce people to donate more, the fact that QCDs allow people to make greater amounts of tax-free gifts should also make giving through QCDs more attractive.

2.6 Summary of Incentives and Time Horizons

Becoming eligible for QCDs should be expected to increase charitable giving. Both itemizers and non-itemizers who own IRAs experience additional tax incentives to give once they reach 70.5. My analysis thus focuses on estimating changes in giving once people age into QCD eligibility.

For reference, Figure 1 presents a policy timeline that highlights key dates relevant for my analysis and previews the timing of my data. As I describe in more detail in the next section, the data allow me to study odd-numbered tax years between 1999 and 2021. I use data from 1999 through 2005 for pre-period analyses; RMDs were in place then, but QCDs were not introduced until 2006. I use data from 2007 through 2019 for my main analysis. Importantly, this period contains several pre-TCJA tax years (2005-2017) and one post-TCJA tax year (2019). Because the TCJA created major changes, both to the charitable giving environment in general and to the relative benefits of using QCDs, I estimate the year-by-year impacts of aging into QCD eligibility and categorize responses into two distinct periods: post-TCJA and pre-TCJA. After presenting the main results from these periods, I provide additional, suggestive evidence from 2021, although as the timeline indicates and as I discuss in more detail in Section 5.5, the implementations of the CARES Act and the SECURE Act, as well as Covid in general, complicate the 2021 analysis.

3 Data

I use individual-level, panel data from the HRS, a biennial survey of older households in the U.S.⁷ The data contain detailed information on household finances and consist of reasonably large samples of people around the QCD-eligibility age. The data are thus well-suited for my study.

To conduct my analysis, I use the HRS datasets produced by the RAND Center for the Study of Aging (Bugliari et al., 2023). These datasets provide researchers with processed versions of the HRS data. Specifically, I use the RAND HRS Longitudinal File 2022 V1 and the RAND Fat Files. The RAND HRS Longitudinal File contains cleaned demographic and economic variables for every individual to appear in the HRS data and thus forms the base of my analysis data. The RAND Fat Files are versions of the raw HRS data and contain variables absent from the Longitudinal File that are necessary for my analysis: the charitable giving outcomes and variables that allow me to determine which individuals within coupled households own an IRA.

I use data from survey waves 5 through 16, which correspond to surveys taking place during the even-numbered years between 2000 and 2022. Because the charitable giving survey questions ask about donations in the previous calendar year, discussed in more detail in Section 3.1, this time horizon allows me to look at donations in several, odd-numbered years over the three time periods discussed earlier: the pre-QCD period (1999 through 2005), the post-QCD-but-pre-TCJA period (2007 through 2017), and the post-TCJA period (2019, with additional, suggestive evidence from 2021).

3.1 Key Variables

My outcome variables capture charitable giving behaviors. The HRS survey asks respondents if either they or their spouse donated more than \$500 of money, property, or possessions to charity in the last calendar year. The survey then also asks for the approximate total value of these donations. To prevent my analysis from being affected by outliers, I winsorize the donation amounts variable at the 99.5th percentile. Using these variables, I define three outcomes: (i) an indicator for donating to charity, (ii) donation amounts (including zeros for people who do not donate), and (iii) donation amounts conditional on giving. I express donation amounts in 2019 U.S. dollars.

⁷The HRS (Health and Retirement Study) is sponsored by the National Institute on Aging (grant number NIA U01AG009740) and is conducted by the University of Michigan.

These charitable giving variables capture total donations. I do not have information on QCDs themselves, which means that I cannot separately track QCDs and other donations. However, I emphasize that total charitable giving is the primary outcome of interest, as my goal is to study whether the incentives from QCD rules cause people to donate more to charity than they otherwise would have. Moreover, the total donations variables capture any potentially-unintended additional effects of QCDs on donations that are important to account for, such as, for instance, learning about a particular charity because of the rules and then donating to that charity out of normal income throughout the year. In these respects, the ability to use survey data to track total donations (including gifts that are not itemized and thus not recorded in tax data) is advantageous.

One disadvantage of the survey-based measures though is that they are naturally subject to recall error. If this measurement error in my outcome variables is classical, then my point estimates would suffer from reduced precision but would not be biased. Of course, if the measurement error is non-classical, then it is more difficult to predict the sign and magnitude of any bias. A key concern from the perspective of my research design is whether any recall error might be differential at the QCD eligibility cutoff. While recall error may be more relevant in older populations like the one I study, it seems unlikely that it would change discontinuously at $70\frac{1}{2}$, and I show later that covariates that could be correlated with recall error, like education or marital status, evolve smoothly through the cutoff.

For a point of reference, Appendix Table A.1 compares estimates of aggregate donations from HRS data using household survey weights to aggregate charitable deduction estimates reported by the Internal Revenue Service (IRS Statistics of Income, 2021). The statistics are not perfectly comparable; for example, the HRS should capture annual donations for everyone, regardless of tax filing status, whereas the IRS estimates should capture deductions for people who itemize, and the IRS totals include carryover deductions from donations in previous years. Nonetheless, I note that the HRS estimates for aggregate donations appear to track IRS estimates of cash contributions reasonably well and, for the most part, better than total contributions, despite the fact that the survey asks about donations of property and possessions as well.

It is also important to note that the timing of the outcome variables means that the charitable giving behaviors reported are for the calendar year before the survey year. For example, wave 15 of the survey corresponds to 2020, so the charitable giving outcome variables for that wave reflect donations in 2019. I therefore define a new variable, tax year, which is equal to the year before the survey year.

I use several other variables to define my sample and implement my RD design. To define the analysis sample, I use variables on IRA ownership. Using the RAND Fat Files, I define an individual-level variable that takes the value of one if the individual owns an IRA.⁸ To construct the running variable in my RD design, which is age as of December in the tax year defined relative to age $70\frac{1}{2}$, I use individual-level variables for birth year and birth month. For additional outcomes analyzed when assessing the validity of the RD design, I use indicator variables for whether an individual receives Social Security retirement benefits, for whether an individual is retired, defined as having zero earned income, and for whether the individual withdrew money from an IRA. For subsample analyses, I use indicator variables for claiming the standard deduction and for itemizing deductions. Finally, for potential control variables, I use information on gender, race, marital status, and college education.

3.2 Analysis Sample

To construct my main analysis sample, I merge the RAND HRS Longitudinal File with the variables extracted from the RAND Fat Files (charitable giving outcomes, an individual-level indicator for owning an IRA, and indicators for claiming the standard deduction or itemizing deductions). I then make three sample restrictions. First, I keep observations from survey waves 5 through 16, which correspond to survey interview years 2000 through 2022 and tax years 1999 through 2021. Second, I keep individuals who own an IRA because QCD policy applies to these individuals. Third, and finally, I drop observations with missing values for charitable giving outcomes. Appendix Table A.2 documents the impact of these restrictions on the number of observations and individuals in my analysis. The sample contains 53,608 observations on 13,909 unique individuals who own IRAs.

Table 1 displays summary statistics for my analysis sample and for a comparison group of people in households without IRAs. For each group, the first two columns report means and standard deviations of variables, and the third column reports the number of person-survey-wave observations underlying these calculations. My analysis sample of IRA holders are more likely to be male, married, and white compared to those without IRAs. They are

⁸The survey asks respondents whether they or their spouse own an IRA and, if so, how many IRAs they own. The survey then asks about the ownership of the largest account, the second-largest account, and then finally the third account or all other accounts. I define a person to own an IRA if they own any of the household IRAs. However, in wave 5 of the data (tax year 1999), the survey only asks about ownership of the largest account, so that for that one pre-period year individual-level ownership is defined based solely on the largest account. The cleaned-and-processed Longitudinal File contains a variable that captures household-level IRA balances, and I use this variable to define a placebo sample of individuals within households where neither individual owns an IRA, defined as a household-level IRA balance of \$0.

also much more likely to have attended at least some college. On average, they are more likely to donate to charity, and they donate more money to charity conditional on donating.

4 Identification Strategy

4.1 Regression Discontinuity Design

I use a regression discontinuity (RD) design to estimate the causal effects of QCD eligibility. The idea is to derive identification from the policy-induced discontinuity at $70\frac{1}{2}$. My RD design tracks the evolution of outcomes as a function of age and estimates discontinuous jumps in those outcomes as people age into eligibility for QCDs.

The tax regulations allow people to make tax-free donations to charities once they reach age $70\frac{1}{2}$. These regulations mean that, for any given tax year, t , people face different incentives to give depending on when they were born. People born in June of year $t - 70$ turn $70\frac{1}{2}$ in December of year t and can thus make a QCD in that year. In contrast, people born in July of year $t - 70$ do not turn $70\frac{1}{2}$ until January of year $t + 1$, making them just-ineligible for QCDs in year t .

My approach is to compare the charitable giving behavior in a given year for people who turn $70\frac{1}{2}$ before the end of the year to otherwise-similar people who turn $70\frac{1}{2}$ at the start of the next year. I thus define the running variable in my RD design to be a person's age in December of the relevant tax year for the donations reported in the survey, and I define the cutoff to be $70\frac{1}{2}$.

To implement my RD design, I estimate separate regressions of the following form for each tax year:

$$y_i = \alpha + \beta \cdot 1[x_i \geq c] + \gamma \cdot x_i + \delta \cdot x_i \cdot 1[x_i \geq c] + \varepsilon_i, \quad (1)$$

where y_i is a charitable giving outcome variable for person i (such as an indicator for giving to charity), x_i is the running variable, monthly age in December of the tax year, defined relative to age $70\frac{1}{2}$, c is the cutoff, and ε_i is an error term. The parameter of interest is β , and I often refer to the estimate of this parameter as the “RD estimate.” It captures the discontinuity in the outcome variable at the cutoff and represents the reduced-form effect of aging into QCD eligibility.

The biennial structure of my data means that I can estimate the effects in odd-numbered years between 2007 (one year after QCDs were implemented) and 2019 (one year after the TCJA). In the baseline specifications, I use triangular weights, a common choice in practice

which assigns weights to observations that are highest at the cutoff and that decrease linearly with observations further from the cutoff (Cattaneo and Titiunik, 2022), and I cluster standard errors at the household level because two individuals from the same household can both be in the analysis sample if each owns an IRA. I also use the procedure from Calonico, Cattaneo and Titiunik (2014) to select a data-driven optimal bandwidth for each regression (that is, for each outcome-year combination). Later, I assess the robustness of my estimates to these choices.

4.2 Assessment of Validity and Interpretation

To interpret the RD estimates as causal, the identifying assumption is that the other factors that influence charitable giving do so smoothly as people reach $70\frac{1}{2}$ and age into QCD incentives. Importantly, this assumption applies to observable factors like education but also unobservables that certainly influence charitable giving like altruism. I carry out several validity checks to assess the relevance of threats to this assumption.

First, I assess the classical threat to the validity of RD designs, namely manipulation of the running variable. As the running variable in my design is age, which cannot be manipulated, typical problems associated with this threat are unlikely. However, my analysis sample consists of older IRA holders, so if the propensity to own an IRA or mortality changes at the cutoff, then my sample would be differentially selected on either side of the cutoff. This type of differential selection could lead to biased estimates. For example, suppose reaching age $70\frac{1}{2}$ led to a discontinuous increase in death and that the selected sample of people who do not experience an increase in mortality at $70\frac{1}{2}$ are more likely to give to charity than those who die for reasons unrelated to QCDs. In this case, the RD estimates for the change in giving at the cutoff would be biased upwards.

Therefore, following the logic of McCrary (2008) but using more-recent methods from Cattaneo, Jansson and Ma (2020), I test for discontinuities in the density of the running variable. Figure 2 plots a histogram of the running variable for 2019, the most recent tax year in my main analysis. The density appears to evolve smoothly through the cutoff. The p-value from the formal density test, where the null hypothesis is that there is no manipulation, or that there is continuity of the density functions for treated and control units at the cutoff, is 0.696. I therefore fail to reject the null hypothesis, which means there is no statistical evidence of a discontinuity in the density at the cutoff. Appendix Figure A.1 plots the histograms for the earlier years, and Appendix Table A.3 reports the t-statistics and p-values from density tests for each year. There is little to no evidence of problematic

discontinuities in the density of the running variable, although I note that one of the p-values (for the test in 2017), is 0.092. Moreover, to provide more evidence on the possibility of differential mortality, Appendix Figure A.2 plots a histogram for date of death that shows no evidence of a discontinuous change the probability of dying at $70\frac{1}{2}$. The p-value from the formal density test is 0.57.

Second, I conduct another standard check for RD designs, testing for discontinuities in control variables at the cutoff. It would be concerning if there were differences in covariates for people who turn $70\frac{1}{2}$ just before the end of the year compared to those who turn $70\frac{1}{2}$ just after the end of the year. To test for this possibility, I estimate my RD regression using control variables as outcomes. Appendix Table A.4 reports the results. Reassuringly, only one of the twenty-eight estimates is statistically different from zero at the 10-percent level. All other estimates are statistically indistinguishable from zero at conventional levels (including those in 2017) and are mostly small in magnitude.

Third, I investigate the relevance of threats to the interpretation of my estimates as being driven by the QCD-induced incentives to give to charity. If some other policy induces discontinuous changes in outcomes at the same cutoff, then any estimated discontinuities would be the result of QCD rules and the other policy. To my knowledge, there are two relevant policies to consider.

One is Social Security’s delayed retirement credit. In general, people who delay claiming their Social Security benefits past their Full Retirement Age receive an increase in their monthly benefit amount, up until age 70. The fact that this delayed retirement credit policy creates a cutoff close to $70\frac{1}{2}$ could be problematic for my design if it has its own impact on charitable giving. The concern is that the delayed retirement credits induce discontinuous changes in benefit receipt or retirement at 70, which could then impact donations. Importantly, my data allow me to investigate the relevance of this threat. Appendix Figures A.3 and A.4 present graphical evidence on the effects of reaching $70\frac{1}{2}$ during the main years of my analysis on the likelihood of receiving retirement benefits from Social Security and on the likelihood of being retired, respectively. Appendix Table A.5 presents the corresponding point estimates. The estimates are not consistent in sign and only one out of fourteen is statistically different from zero. Neither benefit claiming nor retirement appears to change discontinuously at $70\frac{1}{2}$. Notably, benefit receipt increases rapidly as people age through Social Security eligibility (which begins at 62), but few people delay their claiming as late as 70, which likely contributes to the smooth evolution of these outcomes.

Two is, as mentioned earlier, RMD rules. During my analysis time horizon, RMDs began

at age $70\frac{1}{2}$. These regulations strongly influence the drawdown of retirement assets (Poterba, Venti and Wise, 2013; Brown, Poterba and Richardson, 2017; Mortenson, Schramm and Whitten, 2019; Horneff, Maurer and Mitchell, 2023; Stuart and Bryant, 2024) and induce inter vivos transfers to children (Leganza, 2024). It is possible that requiring people to draw down assets in IRAs also induces them to donate money to charity, although I emphasize the rules do not require people to consume the assets; they could continue to save the funds in non-retirement accounts. If RMDs have their own independent impact on charitable giving, then my RD estimates that capture discontinuous changes in giving at $70\frac{1}{2}$ would reflect the combined effects of QCDs and RMDs.

Again, crucially, I can use my data to assess the relevance of this threat. Specifically, I use pre-period data that cover years for which RMDs were in place but QCDs were not (1999 through 2005) to account for any independent effects of mandated withdrawals. I first use my RD design to estimate discontinuities in outcomes during these pre-QCD tax years. Appendix Figure A.5 plots standard-style RD graphs for IRA withdrawals in these years and, as expected, shows that there is a large discontinuous increase in the likelihood of withdrawing money at $70\frac{1}{2}$. However, Appendix Figure A.6 plots RD graphs for all three of my main charitable giving outcomes in each of these pre-period years and shows no graphical evidence of an impact. Appendix Table A.6 presents the corresponding point estimates and also reveals a lack of evidence that RMDs have their own impact on donations. None of the point estimates are statistically different from zero, the sign of the estimates is not consistently positive or negative, and taking the magnitudes of these not-statistically-significant estimates at face value and averaging them would suggest only a modest 0.15 percentage point increase in the likelihood of donating, \$177 increase in donations, and \$173 increase in donations conditional on giving.⁹

4.3 Difference-in-Discontinuities

Nonetheless, I then more formally account for any pure effects of mandated withdrawals by estimating a difference-in-discontinuities regression similar to, e.g., Lalive (2008) and Grembi, Nannicini and Troiano (2016). The idea is to use the pre-period data (1999, 2001, 2003, and 2005), when RMDs were in place but QCDs were not, to difference out any pure effects of mandated withdrawals from the effects of the QCD-induced incentives to donate. For each post-period year separately, I implement a difference-in-discontinuities design by

⁹To put these magnitudes into perspective, recall that the main estimate for donations conditional on donating, which I document in detail in Section 5, is \$2,229.

pooling the four waves of pre-period data with the relevant wave of post-period data (either 2007, 2009, 2011, 2013, 2015, 2017, or 2019) and estimating regressions of the following form:

$$y_{it} = \alpha_0 + \beta_0 \cdot 1[x_i \geq c] + \gamma_0 \cdot x_i + \delta_0 \cdot x_i \cdot 1[x_i \geq c] \\ + \alpha_1 \cdot post_t + \beta_1 \cdot 1[x_i \geq c] \cdot post_t + \gamma_1 \cdot x_i \cdot post_t + \delta_1 \cdot x_i \cdot 1[x_i \geq c] \cdot post_t + \mu_{it}, \quad (2)$$

where y , x , and c are defined as previously, $post_t$ is a post-period indicator variable that takes the value of one if the tax year is the relevant post-period year, and μ_{it} is an error term. Here the parameter of interest is β_1 , and I often refer to the estimate of this parameter as the “diff-in-disc” estimate. It captures the difference in the discontinuity in the outcome variable at the cutoff in the post-period year in question with the discontinuity in the outcome at the cutoff over the pre-period. In the baseline difference-in-discontinuities specification, I continue to use triangular weights and cluster standard errors at the household level, and for each post-period-year-outcome combination, I use the same bandwidth as in the main analysis.

This difference-in-discontinuities approach relies on two key identification assumptions. One is related to the discontinuity component of the design and states that potential outcomes are continuous at the threshold; that is, in the absence of the policy changes at 70 $\frac{1}{2}$, outcomes would have evolved smoothly. The other is related to the differencing component of the design and states that the effect of the confounder (RMDs by themselves) is constant over time. One key way that identification would fail in this setup is if the effect of RMDs themselves on giving changes over time. If this were the case, then one could not reliably use pre-period discontinuities to difference out the effects of RMDs from the post-period estimates. Reassuringly, the fact that none of the pre-period RD estimates discussed in the previous section are statistically different from zero provides some indirect evidence supporting this assumption. To provide even more indirect evidence, I show that the density of the running variable was smooth through the cutoff in the pre-period as well (see Appendix Figure A.7 and Appendix Table A.8) and that estimating equation (2) on covariates produces diff-in-disc estimates of β_1 that are not statistically distinguishable from zero (see Appendix Table A.9), indicating that covariates around the cutoff are not changing over time.

4.4 Summary and Interpretation of the Empirical Approach

In sum, as it relates to the RD design, the density of the running variable is smooth, there is little to no evidence that covariates change at the cutoff, and while there are two potential

confounding policies to carefully consider, the evidence suggests that neither are problematic in practice. Moreover, I can use a difference-in-discontinuity design to difference out any effects of RMDs themselves on charitable giving; as I show later, and consistent with the lack of evidence of effects in the pre-period, the difference-in-discontinues estimates are very similar to the main estimates. Taken together, the evidence suggests that the main RD estimates should be interpreted as due to the extra incentives to donate to charity at $70\frac{1}{2}$, and not mandated withdrawals themselves. Therefore, I present the main year-by-year estimates in the next section and interpret those estimates as causal and as being driven by charitable giving incentives that arise because of QCD eligibility. However, I note that even though mandated withdrawals themselves do not appear to have an effect on giving, the effects of QCD eligibility could be different in a setting where one is also required to be drawing down retirement wealth versus a setting where mandated drawdown is not in place, an important point I return to in Section 5.5.

5 The Effects of Qualified Charitable Distributions on Donations

In this section, I estimate the effects of QCDs on donations. First, I document the effects across time, distinguishing between post-TCJA estimates in 2019 and pre-TCJA estimates earlier. Next, I assess the robustness of these estimates and carry out placebo tests. Finally, I provide suggestive evidence from 2021.

5.1 Main Results: Regression Discontinuity Estimates Across Years

Figure 3 illustrates the effects of QCDs on charitable giving in 2019. Each graph corresponds to a different outcome and provides a visual assessment of the causal effect of interest. Specifically, the graphs plot binned means of the outcomes against the running variable. The vertical lines denote the cutoff, age $70\frac{1}{2}$ in December. The regression lines and confidence intervals to the left and right of the cutoff come from estimating linear relationships between the outcome and the running variable using the underlying, unbinned data on observations within the optimal bandwidth and triangular weights.

Graph (a) depicts the effects of aging into QCD eligibility on the extensive-margin indicator for making any donations to charity. There is little to no graphical evidence that QCDs impacted decisions about whether to donate at all. The graph shows that between about 65% and 70% of people with IRAs on either side of the cutoff donated. In contrast, graph (b) depicts the effects of QCDs on donations, measured in dollars and including zeros,

and there is evidence of an effect. This outcome captures both extensive-margin decisions about whether to donate, as well as intensive margin decisions about how much to donate. The binned means to the left of the cutoff indicate that, before becoming eligible for QCDs, people with IRAs approaching 70 donated roughly \$2,500 to charity. The binned means to the right of the cutoff indicate that people eligible for QCDs donated more. The mean at the cutoff appears to be roughly \$4,000. Taken together, graphs (a) and (b) suggest an increase in giving that occurs on the intensive margin.

Graph (c) shows that, indeed, there was a large, clear, and discontinuous increase in donations for those who gave to charity. The graph plots donations in dollars for only those who donated. Before reaching $70\frac{1}{2}$, average donations for givers amounted to about \$4,000. After aging into QCD eligibility, average donations increased to about \$6,000 and then decreased as people reached more advanced ages. Overall, this graphical evidence indicates that QCDs induced people who donated to give more money to charity than they otherwise would have.

To quantify and assess the statistical significance of these discontinuities, I turn to the regression analysis. Table 2 reports the results. For each outcome, the table displays the RD estimate from estimating equation (1), the mean of the outcome for observations to the left of the cutoff, the optimal bandwidth (monthly ages) used to estimate the effects on the outcome, and both the number of clusters (households) and observations (individuals) used to estimate the regression. Consistent with the graphical evidence, the estimate in column (1) indicates a lack of a statistically significant effect of QCDs on the likelihood that a person donated to charity, whereas column (2) indicates a large and statistically significant effect on donations (including zeros) that amounts to \$1,331.

Column (3) presents the point estimate that corresponds to graph (c) of Figure 3. The estimate indicates that QCD eligibility increased donations for people who donated by \$2,229 on average. This estimate is large. Compared to the mean of \$4,138, it represents a 53.9% increase. People responded strongly to QCD incentives in 2019.

Next, I document the effects in earlier years, before the TCJA. During this time period, the standard deduction was lower and itemization rates were higher. More people itemizing means more people that can make tax-deductible gifts to charity, and thus more people that might find it less attractive to use QCDs. Consistent with this idea, I find little to no evidence that aging into QCDs impacted donations before the TCJA. Table 3 presents the estimates for all three outcomes in all years, and Appendix Figures A.8, A.9, and A.10 present the RD graphs. Each figure corresponds to a different outcome, and each graph within a

figure corresponds to a different year and indicates whether QCD rules were temporary or permanent at the time. None of the point estimates are statistically different from zero at the 5 percent level, and the patterns of the estimates do not consistently point to one sign.

However, there are two extensive-margin estimates worth highlighting that are statistically different from zero at the 10 percent level and larger in magnitude than the other estimates. The first is for 2007. This positive estimate could be consistent with stronger effects during the first year of the policy change, perhaps if awareness was higher with the initial introduction of the rules. The second is for 2017. While an effect in this year could in theory be linked to potential anticipatory behavior related to the TCJA, the negative sign of the estimate is difficult to reconcile. At face value, it would suggest that, above and beyond any change in the overall level of giving due to anticipatory effects, gaining access to an additional way to donate tax-free at $70\frac{1}{2}$ discouraged donations. Of course, this estimate is only one of many in the table, and it is only marginally statistically significant; indeed, in Section 5.3, I find that neither of these two extensive-margin estimates are statistically different from zero after adjusting p-values to account for multiple hypothesis testing.

To put the pieces together and help contextualize the estimates, Figure 4 displays RD estimates across years. Each graph corresponds to a different outcome. As the graphs show, there is little to no statistical evidence that reaching $70\frac{1}{2}$ impacts charitable giving outcomes in the pre-period, before QCDs are introduced, nor is there much evidence of an effect after QCDs are introduced but before the TCJA is enacted. After the TCJA though, there is a large increase in donations, driven by increases on the intensive margin.

5.2 Difference-in-Discontinuities Estimates Across Years

To formally account for the possibility that mandated withdrawals through RMD rules that also begin at $70\frac{1}{2}$ have their own impact on donations, I estimate difference-in-discontinuities regressions detailed in Section 4.2. Recall that the key estimate from each regression captures the difference in the discontinuity in the outcome at $70\frac{1}{2}$ in the relevant post-QCD-period year (when both exposure to QCDs and RMDs changes at the cutoff) from the discontinuity over the pre-QCD period, 1999 through 2005 (when only exposure to RMDs changes at the cutoff). Table 4 presents the difference-in-discontinuities estimates for 2019. Specifically, the table reports estimates of β_1 from equation (2). The number of clusters (households) and observations (person-waves) are greater in this analysis compared to the baseline RD analysis because here I pool each post-period year with the pre-period years. The point estimates are quite similar to the main RD estimates, consistent with the lack of evidence

that mandated withdrawals due to RMDs have their own impact on charitable giving. As in the main results, the difference-in-discontinuities estimates indicate no evidence of an increase in the extensive-margin decision to donate to charity but strong evidence of large, positive effects on (intensive-margin) donation decisions. The point estimate in column (3) indicates a statistically-significant \$2,367 average increase in donations for people who donated in 2019.

Appendix Table A.10 presents the difference-in-discontinuities results for earlier years, before the TCJA. Consistent with the main results for this time period, the estimates indicate little to no impact. The magnitudes are quite similar to the baseline magnitudes, none of the estimates are statistically different from zero at the 5 percent level, and there is no discernible pattern that might suggest either a positive or negative effect in these years.

To help put these estimates into perspective, Figure 5 plots the diff-in-disc estimates across years. In contrast to the analogous graph for the RD estimates, each diff-in-disc graph naturally starts with estimates in 2007 because these estimates capture the difference in the discontinuity in each post-period year compared to the discontinuity in the pre-period. As discussed earlier, I pool the four waves of pre-period data (covering odd-numbered tax years from 1999 to 2005) with the relevant wave of post-period data when estimating equation (2) separately for each post-period wave. These graphs highlight how the post-TCJA effects of QCD eligibility on donations stand in stark contrast to the pre-TCJA effects, both in magnitude and statistical significance.

5.3 Robustness and Specification Checks

Next, I conduct several robustness checks, focusing primarily on the main RD estimates. First, I investigate the sensitivity of the estimates to the choice of bandwidth. Figure 6 illustrates the results of this robustness check for the 2019 estimates. Each graph corresponds to a different outcome variable and plots RD estimates and 95% confidence intervals as I vary the bandwidth from 36 months to 180 months. The vertical lines denote the baseline estimates that use the data-driven optimal bandwidth. Graph (a) shows that the point estimate for the likelihood of donating to charity is quite stable and that none of the bandwidths result in an estimate for this outcome that is statistically different from zero. Graphs (b) and (c) show that the estimates for the donation outcomes fluctuate some when I use the smallest bandwidths, but are otherwise stable and consistently statistically significant. These results indicate that the takeaways from the baseline estimates are not sensitive to the choice of

bandwidth.¹⁰

Second, I probe the robustness of my estimates to various regression specification checks. Table 5 presents the results for 2019.¹¹ The columns of the table correspond to the different outcomes. The rows of the table correspond to the different robustness checks. Row A reproduces the baseline estimates. Row B includes control variables—indicators for being male, for being white, for being married, and for having attended at least some college—in the regressions. Row C drops the triangular weights and thus assigns equal weight to all observations within the optimal bandwidth. Row D uses survey weights instead of triangular weights for population-level representation. Row E clusters standard errors at the level of the running variable instead of at the household level. The results are quite robust to all of these standard specification checks and the takeaways do not change.

The next few rows address measurement of donations. In my baseline analysis, I winsorize donations at the 99.5th percentile to limit the influence of outliers. Row F does not winsorize. The estimates are positive and reasonably similar in magnitude to the baseline estimates, especially considering the standard errors, but they are much less precise. Row G winsorizes slightly more aggressively than the baseline approach, at the 99th percentile. The estimates are similar to the baseline estimates and are more precise. Limiting the influence of outliers appears to be important for my results, especially for precision. Finally, row H takes an alternative approach to winsorizing dollar amounts and instead uses log donations as the outcome. The point estimate is positive and statistically significant at the 1% level. It indicates a 38.1% increase in donations for givers, and the corresponding graph in Appendix Figure A.15 shows clear visual evidence of a discontinuous increase in log donations.

Third, I address multiple hypothesis testing. In my main analysis, I estimate the effects of QCDs on three outcome variables and across seven tax years (subsamples). I therefore use the approach developed by List, Shaikh and Xu (2019) and List, Shaikh and Vayalinkal (2023) to account for the fact that I am testing several hypotheses and to limit the likelihood of a false rejection explaining the main finding. Specifically, I use their procedure, which controls for the familywise error rate while exploiting dependence structures across test statistics, to

¹⁰Appendix Figure A.11 shows that the similar difference-in-discontinuities estimates for 2019 are also not sensitive to the choice of bandwidth. Moreover, Appendix Figures A.12, A.13, and A.14 present analogous robustness graphs for RD estimates for each of the three main outcomes variables, respectively, in the earlier, pre-TCJA years. The earlier-year estimates are also stable, and they are rarely statistically different from zero at the 5 percent level, which indicates that the lack of evidence for a response in earlier years is also not sensitive to the choice of bandwidth.

¹¹Appendix Tables A.11, A.12, A.13, A.14, A.15, and A.16 present analogous tables for each of the earlier tax years. The main estimates in those years, and the resulting takeaway that there is a lack of evidence of a response, are robust to specification checks.

adjust p-values for the family of RD estimates across years and outcomes. Table 6 presents the results. Each row indicates an outcome-year combination. Column (1) reports the RD coefficient of interest, column (2) reports the unadjusted p-value, and column (3) reports the adjusted p-value. Note that the coefficients are not identical to the main results reported in Tables 2 and 3 because the procedure for implementing the adjustment to p-values does not allow for weighted regressions. Instead, the coefficients correspond to those reported in Row C of the corresponding robustness tables, which implement the RD design without triangular weights, and which tend to be very similar to the main estimates.

There are two key takeaways. First and foremost, the main results survive this adjustment procedure. The main estimate is for donations conditional on giving in 2019 (the last row of the Table 6). This estimate is still statistically significant at the 1% level. Moreover, while the adjusted p-value for donations (including zeros) in 2019 is larger than the unadjusted p-value (0.076 compared to 0.005), that coefficient is still statistically distinguishable from zero at a conventional level. Second, none of the other estimates are statistically different from zero, including the estimates for donation indicators highlighted previously as marginally statistically significant and relatively large in size. Overall, this exercise establishes confidence in the main takeaway: QCDs led to a statistically significant increase in donations on the intensive margin in 2019.

5.4 Placebo Tests

Next, I conduct two placebo tests. First, I estimate RD regressions using placebo cutoffs around the true cutoff. Figure 7 plots the RD estimates and 95-percent confidence intervals for the three main outcome variables in 2019 as I vary the placebo cutoff used to conduct the analysis. Graph (a) is for the indicator for giving to charity, graph (b) is for donations (including zeros), and graph (c) is for donations conditional on giving. Reassuringly, none one of the estimates using placebo cutoffs are statistically different from zero. Moreover, the patterns in graphs (b) and (c) show how the estimates for the donations outcomes tend to shrink as I use placebo cutoffs further away from the true cutoff.¹²

Second, I estimate RD regressions using a placebo sample of people who do not own IRAs. These people cannot make a QCD, and therefore there should be no discontinuities in their donation outcomes at the age-70 $\frac{1}{2}$ cutoff. Table 7 presents the results for 2019, and

¹²For completeness, Appendix Figures A.16, A.17, and A.18 present analogous graphs for each of the three main outcomes variables, respectively, in the earlier, pre-TCJA years, even though the main results suggest a lack of effects in these years. The placebo estimates for these years are also, on the whole, not statistically different from zero.

Appendix Table A.17 presents the results for the earlier years. Reassuringly, I find little to no evidence of discontinuities in outcomes for this sample. Compared to the main results for 2019, the corresponding placebo estimates are meaningfully smaller in magnitude and none are statistically different from zero.

Overall, these two placebo tests lend additional confidence to the main results. The lack of evidence of discontinuities for placebo cutoffs and the lack of evidence of discontinuities for the placebo sample at the true cutoff supports the interpretation of the main results as being driven by IRA holders making use of QCD rules when they reach $70\frac{1}{2}$.

5.5 Suggestive Evidence from 2021

Before investigating mechanisms that help explain the main results, I provide suggestive evidence on the effects in 2021. However, this analysis faces three key limitations. First, RMD policy changed. Because of the SECURE Act, people in 2021 faced a new RMD age of 72 instead of $70\frac{1}{2}$. This change means there is another discontinuity in the vicinity of the QCD cutoff, which remained at $70\frac{1}{2}$, complicating the RD design setup. While the pre-period and difference-in-discontinuities analyses earlier suggest that RMDs did not have their own, independent effect on charitable giving, the effect of aging into both QCD eligibility and mandated drawdown at $70\frac{1}{2}$ (like in 2019) could be different than aging into QCD eligibility on its own (like in 2021), especially if people use QCDs as a way to reduce taxes associated with required withdrawals. In this case, we might expect to see increased donations at 72, when individuals age into mandatory drawdown and are still eligible to make a QCD.

Second, tax policy on charitable gifts changed. The CARES Act, signed into law in March of 2020, and the Consolidated Appropriations Act, signed into law in December of 2020, made it such that (i) taxpayers could deduct up to \$600 of charitable gifts (for married couples) without itemizing and (ii) itemizers could deduct gifts up to 100% of their adjusted gross income, instead of 60%. Importantly, both of these changes should be expected to weaken the effects of QCD eligibility. The above-the-line deduction makes it such that non-itemizers could benefit from modest charitable gifts without using QCDs. The dramatic increase in the cap on the tax-deductibility of gifts makes it such that itemizers could experience benefits similar to QCDs for even larger gifts. Indeed, popular outlets and institutions emphasized how the CARES Act meant that individuals could benefit even more than before from the strategy of withdrawing large sums from their IRA, contributing the cash to charity, and then offsetting taxes by taking a charitable deduction, now up to 100% of adjusted gross income, instead of making a QCD (Fidelity Charitable, 2021).

Third, the Covid pandemic in general might complicate the interpretation of results in this year. While any the effects of an economy-wide shock like Covid on charitable giving should be smooth through the age- $70\frac{1}{2}$ or age-72 cutoffs, it could be that households respond to additional charitable giving incentives during Covid in different ways than in other years, as major adverse events, negative employment shocks, or uncertainty about income could also affect the level of donations and the responsiveness to donation incentives

Taken together, these limitations suggest some caution is warranted when analyzing 2021 data. Still, I can use a similar empirical framework to provide graphical and regression evidence that assesses whether charitable giving changes at either $70\frac{1}{2}$ or 72 in 2021. Figure 8 presents the graphical evidence. Each graph corresponds to a different outcome and is constructed similarly to those in Figure 3, except here a key change is that I use two vertical dashed lines, one to denote $70\frac{1}{2}$ and one to denote 72.¹³ Graph (a) shows that the likelihood of donating to charity evolves smoothly, consistent with the results in 2019. Graphs (b) and (c) show some visual evidence of an increase in donations; however, the increase appears to happen more at 72 than $70\frac{1}{2}$. This pattern suggests that, at least in 2021, it was the combination of access to QCDs and required drawdown that led to increases in giving.¹⁴ For some reference on the magnitudes, Appendix Table A.7 presents point estimates from a donut RD specification that drops observations between $70\frac{1}{2}$ and 72 when estimating equation (1), which would suggest increases of \$1,185 and \$1,640 for donations and conditional donations, respectively. These estimates are still sizable, but smaller than the main estimates from 2019, perhaps as expected, given the changes to charitable giving policy.

On the one hand, this suggestive evidence emphasizes the relevance of QCDs allowing individuals facing required drawdown to offset associated taxes in a given year. On the other hand, the limitations of the 2021 analysis caution against drawing strong conclusions. For example, it could be that people who might otherwise have used QCDs at $70\frac{1}{2}$ (even without required drawdown) in 2021 to lower future IRA balances or to facilitate a large tax-advantaged gift might be more likely to delay these financial decisions because of Covid-related disruptions and uncertainty.

¹³Note too that because of the non-standard setup with two relevant cutoffs close to one another, I used the outcome-specific bandwidths from 2019 to chose an estimation widow over which to draw regression lines and I did not use a triangular kernel to put more weight on observations close to either one of the cutoffs.

¹⁴Appendix Figure A.19 shows sharp increases in withdrawals from IRAs at 72 instead of $70\frac{1}{2}$ in this year, consistent with the increase in the RMD age.

6 Mechanisms

I conclude my analysis with two additional sets of exercises that help to unpack the main estimates and shed some light on mechanisms. First, I investigate the size of the donations induced by QCDs that ultimately led to the \$2,229 increase in average donations in 2019. To explore donation amounts, I first plot the empirical cumulative distribution functions (CDFs) of donations for people above and below the cutoff. Figure 9 presents these CDFs. The black line is for people above the cutoff, who were eligible to make QCDs, whereas the gray line is for people below the cutoff, who were ineligible. The graph shows that those eligible for QCDs were not much, if at all, more likely to donate relatively smaller amounts of money compared to those who were ineligible. Instead, the fact that the CDFs separate more clearly in the \$5,000-to-\$15,000 range suggests that QCDs induced people to donate large sums.

I then estimate the effects of aging into QCDs on a series of indicator variables for donating various amounts to charity. Specifically, guided by the CDFs and the means of donation amounts in 2019, which indicate average donations of about \$3,000 and average donations conditional on donating of roughly \$5,000 (see Appendix Table A.18), I estimate the effects on the likelihood of donating (i) \$2,500 or more, (ii) \$5,000 or more, (iii) \$10,000 or more, (iv) \$15,000 or more, (v) \$20,000 or more, and (vi) \$25,000 or more. Figure 10 presents the RD graphs and Table 8 displays the corresponding regression results. Taken together, the graphs and regression estimates indicate some, perhaps weaker, evidence of increases in the probability of donating more than \$2,500 and \$5,000, strong and clear evidence of increases in the likelihood of donating more than \$10,000 and \$15,000, and little to no evidence of increases in the likelihood of donating very large amounts more than \$20,000 or \$25,000, although I emphasize that very large donations are rare in the sample. Importantly, Appendix Table A.19 shows that these takeaways are robust to adjusting p-values for multiple hypothesis testing to account for the six simultaneous tests.

Overall, the CDFs and these additional RD results highlight how the average increase in giving in 2019 due to QCDs reflects notable increases in the likelihood of donating large sums. For example, the estimate in column (3) of the table indicates that there was an 8.4 percentage point increase in the likelihood of donating more than \$10,000, which represents a near-doubling of the likelihood of donating this large sum when compared to the mean of 8.5%. In contrast, Appendix Table A.20 presents estimates for donation amount indicators in the earlier years. None of the estimates are statistically different from zero at the 5

percent level, indicating the null results on average donations for those years are unlikely to be masking significant increases in the probability of making large donations for some.

Second, I conduct subsample analyses to provide some insight into the bundle of incentives tied to QCD eligibility. One key feature of the tax rules is that they allow people who take the standard deduction to make tax-free gifts, and the pattern of the main estimates—large responses after the TCJA increased the standard deduction and decreased itemization rates but no evidence of responses earlier—provides some support to the idea that this feature is important in influencing the use of QCDs.

To further investigate the relevance of this channel, I estimate separate effects for people who claimed the standard deduction versus those who itemized.¹⁵ Tables 11 and 12 show the regression results. Sample sizes are quite small, and the graphical evidence is thus naturally noisier and less clear (see Appendix Figures A.20 and A.21), but the estimates point to stronger effects for people who claimed the standard deduction. At face value, the regression estimate for donations conditional on donating for this group is \$2,290 and statistically significant, whereas the corresponding estimate for people who itemized is \$876 and not statistically significant. These patterns thus provide some additional support to the idea that the standard deduction channel is relevant.

However, another key feature of the rules is that they allow people not relying on IRA withdrawals for financing consumption to reduce any tax-related burdens associated with RMDs. Recall that QCDs (i) count towards required withdrawals and (ii) reduce IRA balances and thus lower future RMDs. While these incentives are inherently bundled with the ability to make tax-free gifts, I can attempt to investigate the extent to which these incentives might also be important for households by looking at subsamples based on IRA balances. The idea is that people with larger IRA balances face greater RMDs and therefore may be more impacted by these additional incentives, whereas people with smaller IRA balances may be less impacted.

Tables 9 and 10 display the 2019 estimates for two groups: those with above-median IRA balances and those with below-median balances, respectively. Appendix Figures A.22 and A.23 display the corresponding RD graphs. The point estimates for donations among people with larger IRA balances are statistically significant and greater in magnitude than the main estimates. In contrast, while still positive, the point estimates for donations among people with smaller IRA balances are not statistically different from zero. They are also about

¹⁵Appendix Table A.21 shows no evidence of a change in the probability of claiming the standard deduction or itemizing deductions at 70 $\frac{1}{2}$.

four times smaller than the estimates for people with larger IRA balances. While I caution against drawing strong conclusions from these subsample estimates because (i) the sample sizes are small and (ii) IRA balances could of course be correlated with other important characteristics, the results nonetheless suggest that those with large IRA balances donated more when they aged into QCDs. They thus provide some supporting evidence to the idea that the RMD-related incentives to use QCDs to make tax-free donations at $70\frac{1}{2}$ entered the decision-making process for some people.

7 Conclusion

In this paper, I use data from the HRS and an RD design to provide novel causal evidence on charitable giving responses to tax rules for QCDs. The rules allow people to make tax-free transfers of funds from Individual Retirement Accounts to qualified charities. I find that QCD eligibility led to a \$2,229 increase in average donations for people who gave to charity in 2019, after the TCJA. I also show that QCDs induced more people to make large donations and that people with larger IRAs responded more than people with smaller IRAs. In contrast, I find no evidence that the tax rules impacted donations before the TCJA.

My findings have implications for tax policy. In a direct sense, my results highlight how QCDs can indeed increase total charitable donations, which is presumably the goal of these tax incentives. However, my findings across years highlight how the use of QCDs depends on the broader tax environment and underscore the importance of accounting for interactions between different sets of tax rules. When the standard deduction was low and more people itemized, QCDs were less effective at increasing donations, perhaps because many people could already make tax-advantaged gifts. In the current tax environment though, with greater standard deductions and thus fewer itemizers, the ability to transfer funds tax-free at $70\frac{1}{2}$ appears to be more valuable, and QCDs thus have more potential to increase donations. Any future changes to the standard deduction or to the amount of itemized deductions that households can take should be expected to influence the use of QCDs.

More generally, my results may be able to inform policy makers concerned with understanding whether making charitable gifts tax-deductible for more people would increase donations. At a high level, my results indicate that once people aged into the ability to donate in a tax-advantaged way that did not require itemization, donations increased. This takeaway thus lends some support to the idea that proposed policies to make charitable gifts tax-deductible for everyone and not just those who itemize—like the Universal Char-

itable Giving Act that was introduced in 2017 (H.R.3988) or the Charitable Act that was introduced in January 2025 (H.R.801)—would increase donations.

However, an important caveat of course is that my results are local to IRA holders around age $70\frac{1}{2}$, and the population of my study thus differs from the population targeted by these proposals. It is therefore crucial to reflect on the extent to which my findings might generalize across people, situations, stimuli, and time periods, as highlighted by List (2020). First, on people, those in my study are naturally older, wealthier, and more inclined to donate to charity. On the one hand, these features might make them more responsive to additional tax incentives compared to the full population. On the other hand, these features might also make them less responsive to new donation incentives if they already use other routes like itemized deductions to make tax-advantaged gifts. Second, on situations, making a QCD involves a trustee-to-trustee transfer of retirement wealth to a charity, which is a structurally-different and less-common process compared to donating cash and then taking a deduction on current income later, when filing taxes. This more involved process likely dampens responsiveness. Third, on stimuli, QCD and RMD regulations interact in a way that may amplify responses, since people bound by RMDs who would prefer not draw down funds from IRAs might look to QCDs to offset tax implications; this type of stimulus would likely be absent from a general policy like the Universal Charitable Giving Act. Finally, on time periods, as my results across years suggest, the responsiveness to QCD incentives appears to depend upon other tax rules, which continue to evolve. Overall, it is thus important for policy that empirical research continues to amass evidence on how responsive charitable giving is to different types of tax incentives in different settings and at different points in time.

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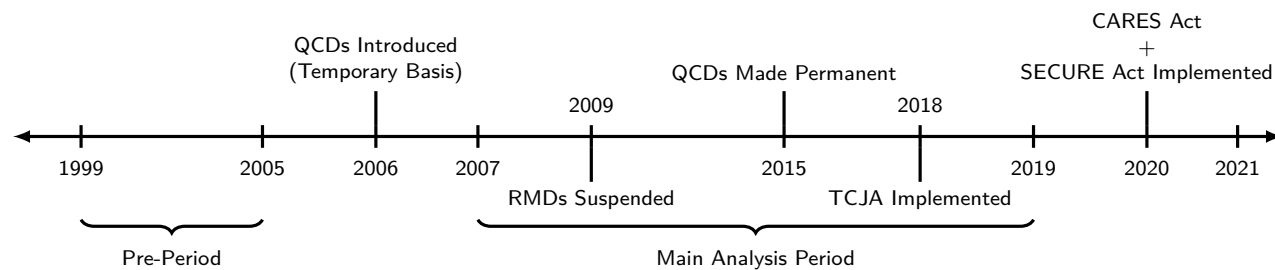
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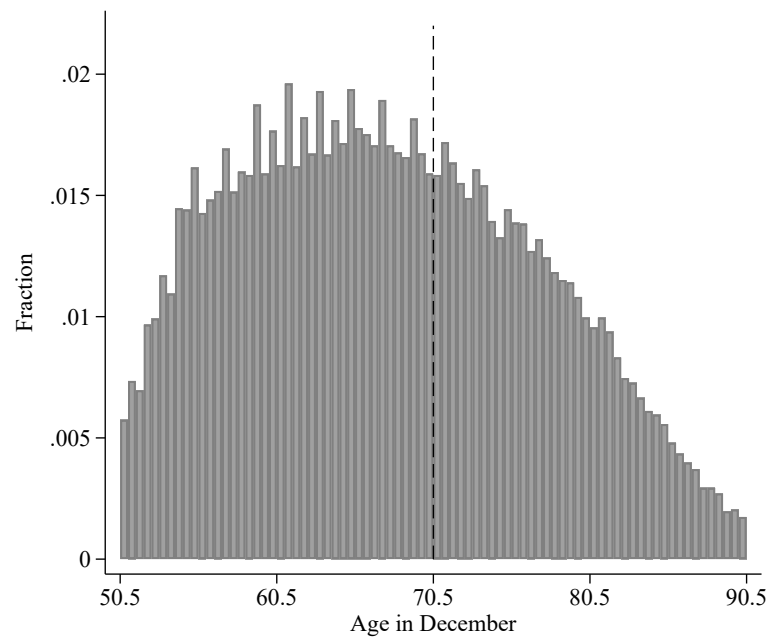
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Figure 1: Policy Timeline



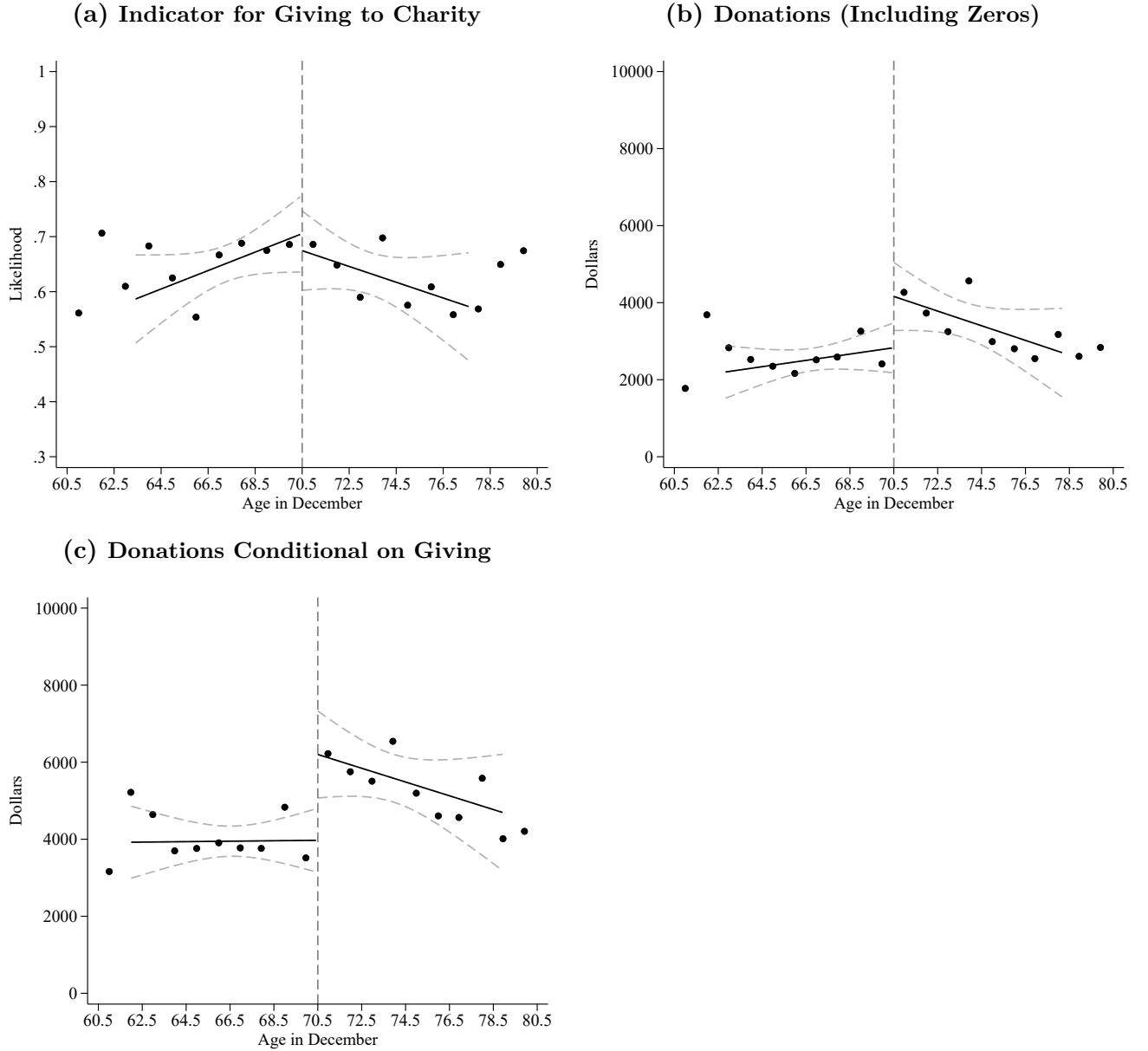
Notes: This figure plots a policy timeline that highlights key dates relevant for my analysis and previews the timing of the analysis data.

Figure 2: Histogram of the Running Variable



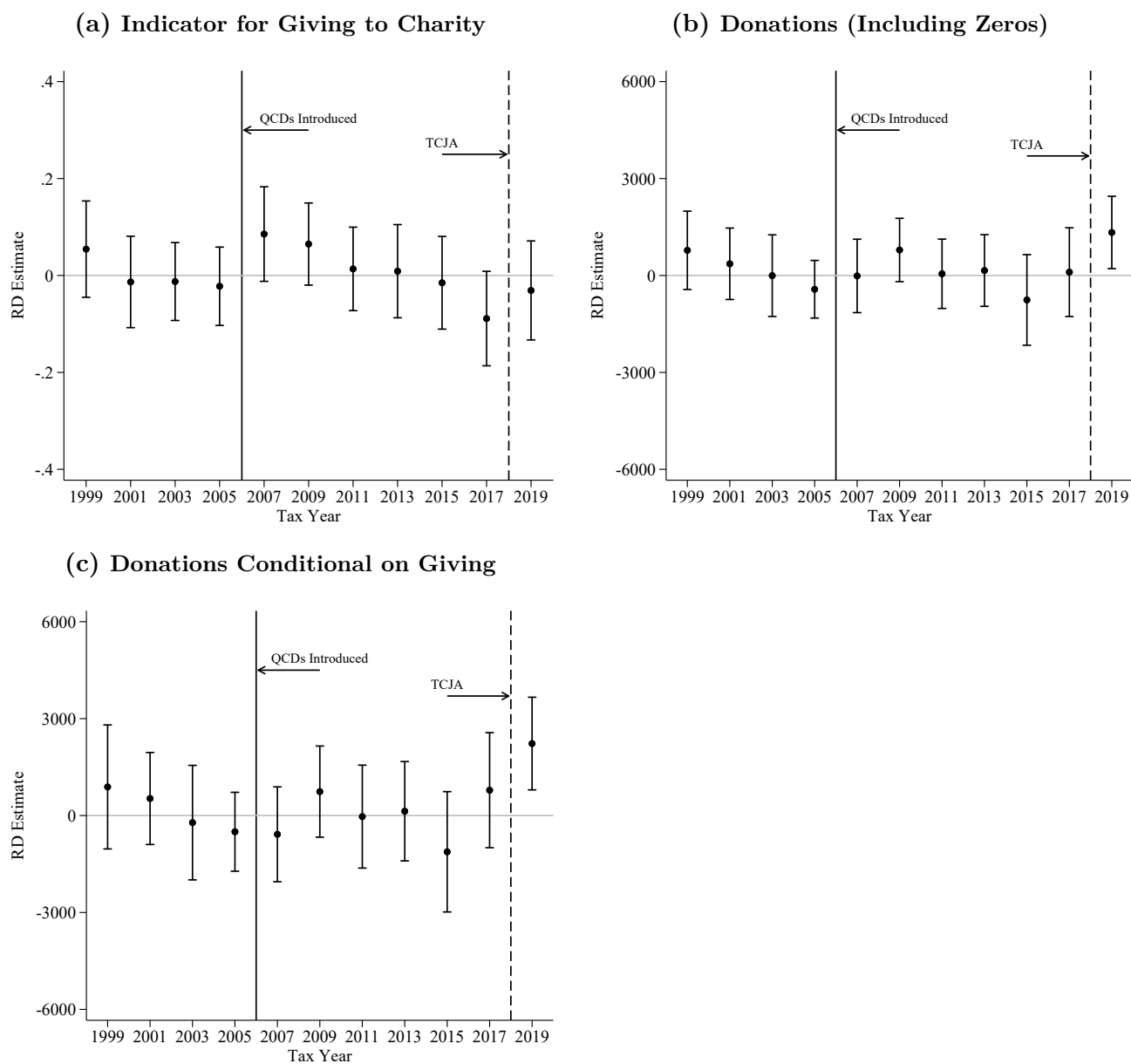
Notes: This figure plots a histogram of the running variable, age in December, for tax year 2019.

Figure 3: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act



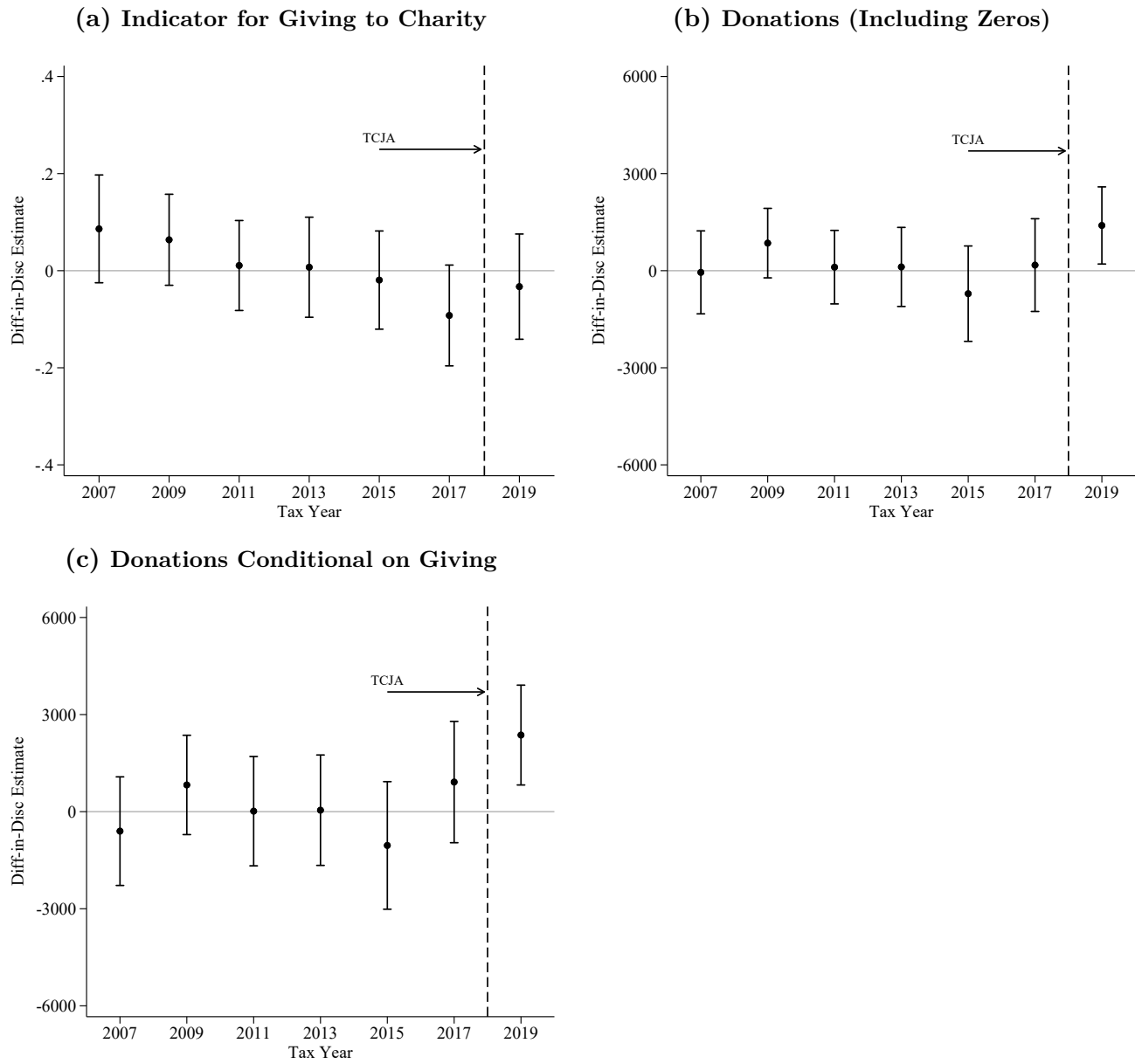
Notes: This figure illustrates the effects of aging into qualified charitable distributions on charitable giving in 2019. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving. Each graph is constructed as follows. The running variable along the horizontal axis is (monthly) age in December of the tax year to which the charitable giving outcomes correspond. The cutoff is $70\frac{1}{2}$ and is denoted by the dashed vertical line. The dots are average outcomes in 12-month bins. The superimposed regression lines and 95-percent confidence intervals are based on the underlying, unbinned data.

Figure 4: Effects of Qualified Charitable Distributions Over Time: Regression Discontinuity Estimates by Tax Year



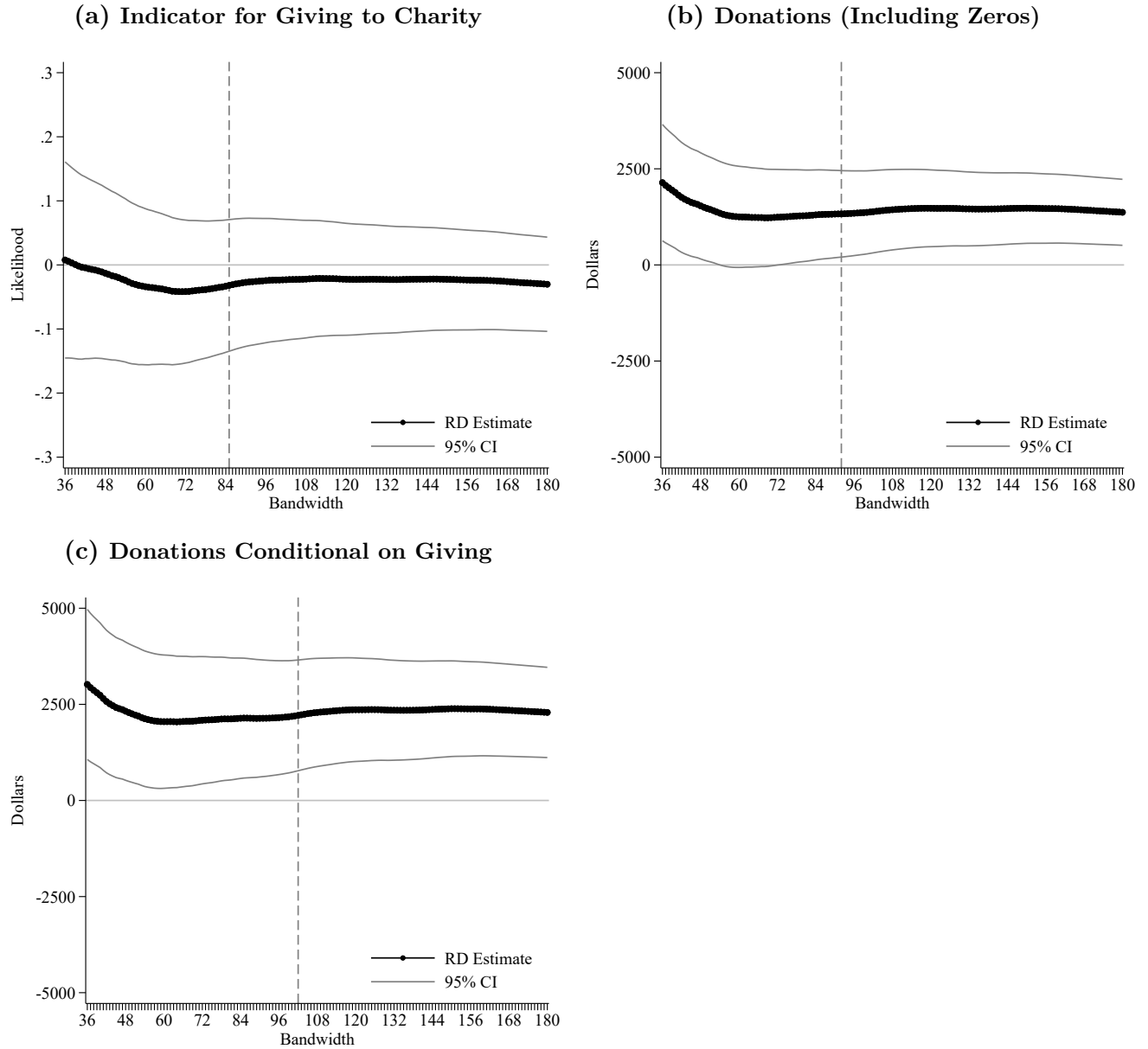
Notes: This figure plots regression discontinuity (RD) estimates over time. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving.

Figure 5: Difference-in-Discontinuities Estimates for the Effects of Qualified Charitable Distributions Over Time



Notes: This figure plots difference-in-discontinuity (diff-in-disc) estimates over time. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving.

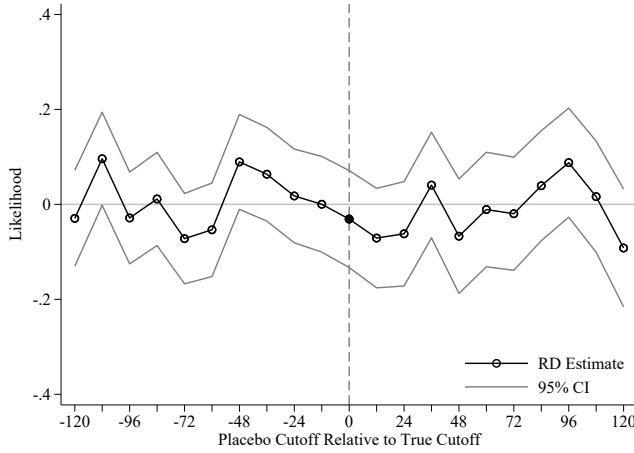
Figure 6: Robustness of Key Estimates to Bandwidth Selection



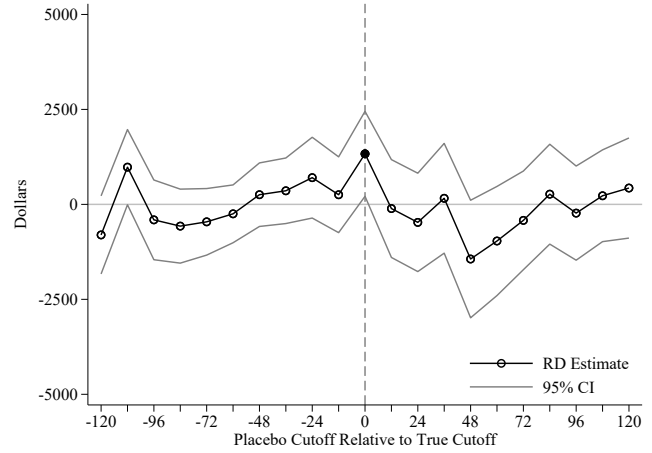
Notes: This figure illustrates the robustness of the main estimates for 2019 to varying the bandwidth. Each graph plots the regression discontinuity (RD) estimates and 95% confidence intervals that result from using bandwidths ranging from 36 months to 180 months. The vertical dashed lines denote the leading bandwidth.

Figure 7: Estimates Using Placebo Cutoffs Around the True Cutoff

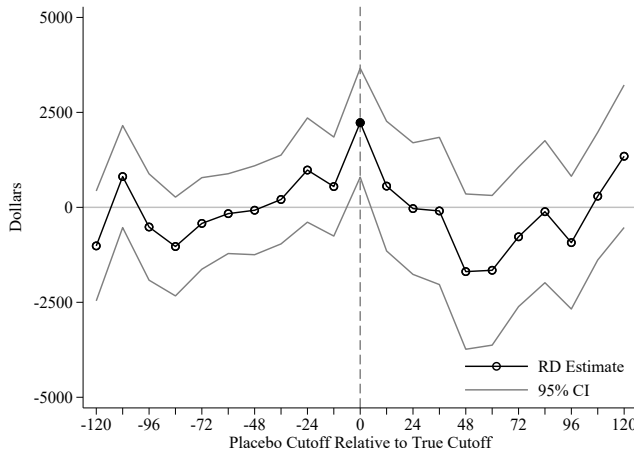
(a) Indicator for Giving to Charity



(b) Donations (Including Zeros)

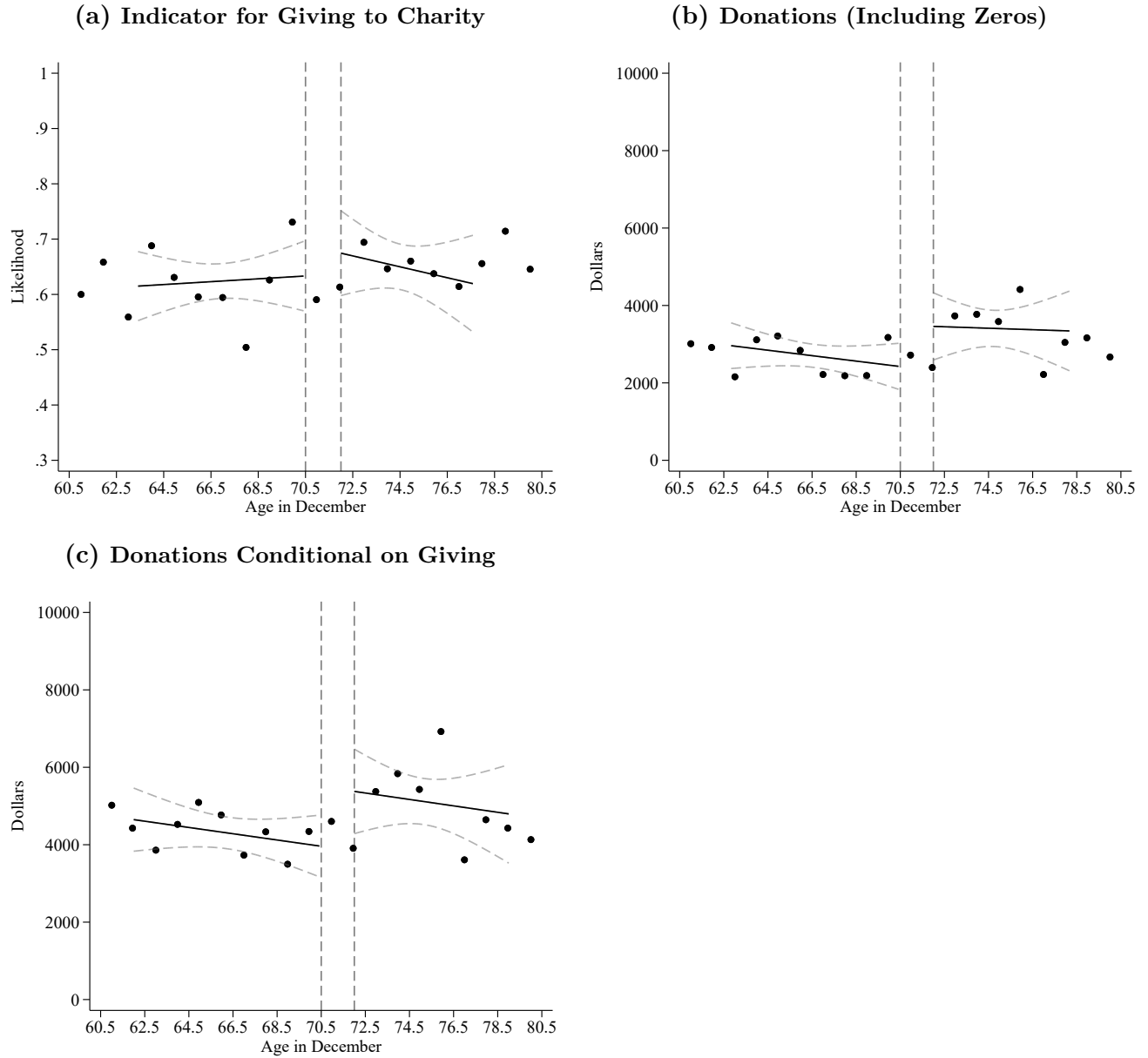


(c) Donations Conditional on Giving



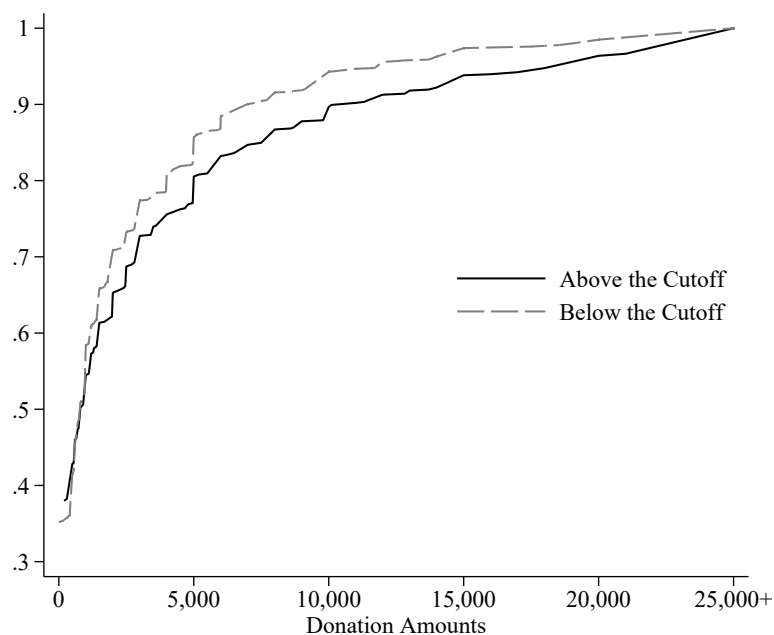
Notes: This figure illustrates how the regression discontinuity (RD) estimates for 2019 change when I use placebo cutoffs around the true cutoff. Each graph plots RD estimates and 95-percent confidence intervals from estimating my RD regression model using various placebo cutoffs ranging from 120 months before the true cutoff to 120 months after the true cutoff. The point estimate for the true cutoff is denoted by a solid circle, and the point estimates for the placebo cutoffs are denoted by hollow circles.

Figure 8: Suggestive Graphical Evidence on Charitable Giving in 2021



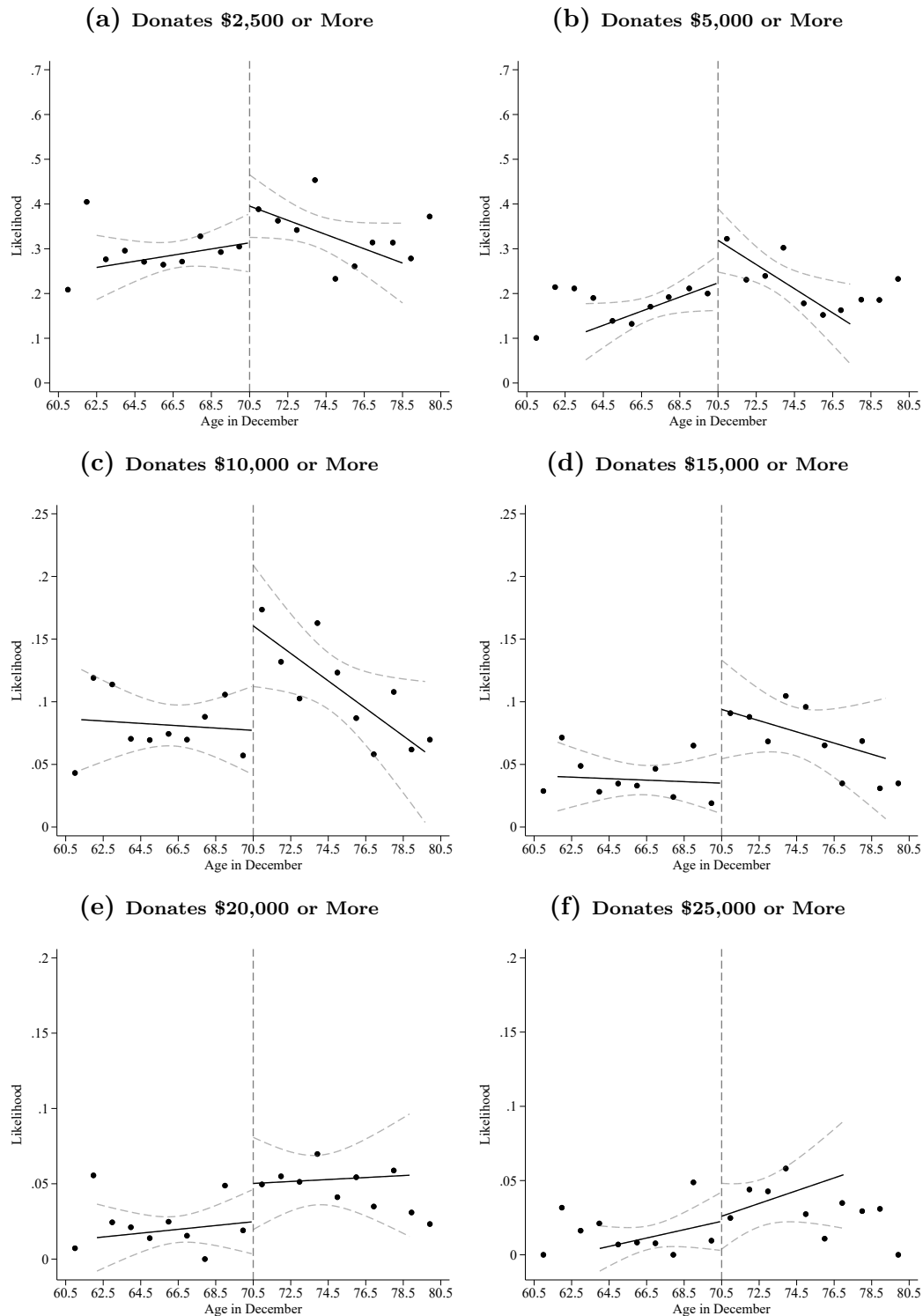
Notes: This figure illustrates how charitable giving evolves around ages $70\frac{1}{2}$ and 72 in 2021. Each graph is constructed similarly to those in Figure 3; however, there are two vertical lines to denote both cutoffs, and the regression lines and confidence intervals are drawn without using data for observations between ages $70\frac{1}{2}$ and 72.

Figure 9: Empirical Cumulative Distribution Functions of Donation Amounts in 2019, for People on Either Side of the Cutoff



Notes: This figure plots empirical cumulative distribution functions of donation amounts in 2019 for two groups: those above the age- $70\frac{1}{2}$ cutoff and those below the age- $70\frac{1}{2}$ cutoff. The solid black line is for those above the cutoff. The dashed gray line is for those below the cutoff. The underlying sample contains observations of people within 93 months of the cutoff on either side, which corresponds to the optimal bandwidth for the donation amounts variable (see Table 2).

Figure 10: Effects of Qualified Charitable Distributions on Indicators for Donating Various Amounts to Charity After the Tax Cuts and Jobs Act



Notes: This figure illustrates the effects of aging into qualified charitable distributions on indicator variables for donating various amounts to charity in 2019. Each graph corresponds to a different outcome. Graph (a) is for an indicator for donating \$2,500 or more. Graph (b) is for an indicator for donating \$5,000 or more. Graph (c) is for an indicator for donating \$10,000 or more. Graph (d) is for an indicator for donating \$15,000 or more. Graph (e) is for an indicator for donating \$20,000 or more. Graph (f) is for an indicator for donating \$25,000 or more. See the notes of Figure 3 for more details on how each graph is constructed.

Table 1: Summary Statistics

	People With IRAs			People Without IRAs		
	Mean (1)	SD (2)	Obs. (3)	Mean (4)	SD (5)	Obs. (6)
Age	67.41	10.11	53,608	67.32	11.84	134,801
Tax Year	2009.56	6.63	53,608	2009.91	6.72	134,801
Male	0.46	0.50	53,608	0.39	0.49	134,801
White	0.90	0.30	53,474	0.66	0.47	134,110
Married	0.70	0.46	53,581	0.54	0.50	134,621
College	0.65	0.48	53,492	0.36	0.48	134,482
Retired	0.59	0.49	53,608	0.68	0.47	134,801
Receives Social Security Benefits	0.58	0.49	53,608	0.56	0.50	134,801
Gives to Charity	0.64	0.48	53,608	0.30	0.46	134,801
Donations (Including Zeros)	2,907	5,152	53,608	880	2,640	134,801
Donations Conditional on Giving	4,516	5,828	34,505	2,933	4,150	40,435
IRA Balances	240,463	578,585	53,608	—	—	—
Individuals	13,909			31,611		

Notes: This table reports summary statistics for two groups. The underlying samples contain data from survey waves 5 through 16. The first three columns report means, standard deviations, and observations for people who own an IRA and make up my analysis sample. The next three columns report the same statistics for people in households where no member owns an IRA. The bottom row displays the number of unique individuals in each group.

Table 2: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2019	-0.031 (0.052)	1,331** (570)	2,229*** (731)
Mean	0.653	2,571	4,138
Bandwidth (Months)	86	93	103
Clusters (Households)	1,314	1,410	972
Observations (Individuals)	1,579	1,710	1,220

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Effects of Qualified Charitable Distributions on Charitable Giving Before the Tax Cuts and Jobs Act

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2007	0.085* (0.050)	-13 (580)	-580 (748)
Mean	0.661	3,224	4,764
Bandwidth (Months)	55	66	70
Clusters (Households)	1,310	1,501	1,028
Observations (Individuals)	1,529	1,805	1,273
RD Estimate for 2009	0.065 (0.043)	789 (500)	741 (719)
Mean	0.648	2,819	4,331
Bandwidth (Months)	80	89	80
Clusters (Households)	1,671	1,818	1,073
Observations (Individuals)	2,039	2,240	1,337
RD Estimate for 2011	0.014 (0.044)	52 (547)	-33 (812)
Mean	0.646	3,036	4,652
Bandwidth (Months)	91	83	76
Clusters (Households)	1,689	1,557	937
Observations (Individuals)	2,050	1,875	1,137
RD Estimate for 2013	0.009 (0.049)	154 (566)	134 (783)
Mean	0.653	3,447	5,291
Bandwidth (Months)	84	66	64
Clusters (Households)	1,555	1,228	787
Observations (Individuals)	1,897	1,459	948
RD Estimate for 2015	-0.015 (0.049)	-757 (715)	-1,122 (949)
Mean	0.671	3,433	5,019
Bandwidth (Months)	110	79	79
Clusters (Households)	1,803	1,292	867
Observations (Individuals)	2,237	1,562	1,067
RD Estimate for 2017	-0.089* (0.050)	104 (700)	785 (907)
Mean	0.674	3,009	4,483
Bandwidth (Months)	94	95	95
Clusters (Households)	1,447	1,456	956
Observations (Individuals)	1,744	1,759	1,170

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving before the Tax Cuts and Jobs Act, in odd-numbered years from 2007 to 2017. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Difference-in-Discontinuities Estimates for the Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
Diff-in-Disc Estimate for 2019	-0.033 (0.055)	1,398** (608)	2,367*** (787)
Mean	0.653	2,571	4,138
Bandwidth (Months)	86	93	103
Clusters (Households)	4,578	4,777	3,466
Observations (Person-waves)	10,468	11,250	7,977

Notes: This table reports difference-in-discontinuities estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019. The estimates come from estimating equation (2). The underlying sample contains person-wave observations covering pre-period tax years 1999, 2001, 2003, and 2005, as well as the post-period year 2019. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5: Robustness of Key Estimates to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	-0.031 (0.052)	1,331** (570)	2,229*** (731)
B. Include Controls	-0.048 (0.050)	1,157** (551)	2,117*** (711)
C. Drop Triangular Weights	-0.001 (0.049)	1,425*** (529)	2,718*** (701)
D. Use Survey Weights	0.005 (0.056)	1,080 (673)	2,381*** (883)
E. Cluster on Running Variable	-0.031 (0.046)	1,331*** (501)	2,229*** (606)
F. No Winsorizing	—	970 (987)	1,676 (1,357)
G. Winsorize More	—	1,236** (503)	2,120*** (594)
H. Log Donations	—	—	0.381** (0.149)

Notes: This table reports results from assessing the sensitivity of the 2019 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6: Robustness of Key Estimates to Adjusting p-Values for Multiple Hypothesis Testing

	Coefficient (1)	Unadjusted p-value (2)	Adjusted p-value (3)
Indicator for Giving in 2007	0.054	0.260	0.969
Indicator for Giving in 2009	0.054	0.175	0.916
Indicator for Giving in 2011	0.022	0.583	0.999
Indicator for Giving in 2013	-0.003	0.947	1.000
Indicator for Giving in 2015	-0.031	0.464	0.994
Indicator for Giving in 2017	-0.097	0.035	0.418
Indicator for Giving in 2019	-0.001	0.982	1.000
Donations (Including Zeros) in 2007	-102	0.847	1.000
Donations (Including Zeros) in 2009	751	0.091	0.720
Donations (Including Zeros) in 2011	174	0.730	1.000
Donations (Including Zeros) in 2013	69	0.903	1.000
Donations (Including Zeros) in 2015	-675	0.276	0.968
Donations (Including Zeros) in 2017	65	0.912	1.000
Donations (Including Zeros) in 2019	1,425	0.005	0.076
Donations Conditional on Giving in 2007	-640	0.332	0.972
Donations Conditional on Giving in 2009	703	0.288	0.964
Donations Conditional on Giving in 2011	-18	0.983	0.983
Donations Conditional on Giving in 2013	74	0.919	1.000
Donations Conditional on Giving in 2015	-1,045	0.208	0.943
Donations Conditional on Giving in 2017	793	0.324	0.972
Donations Conditional on Giving in 2019	2,718	0.000	0.000

Notes: This table reports results from assessing the sensitivity of the main regression discontinuity estimates to adjusting p-values for multiple hypothesis testing. I use the *mhtreg* Stata package Steinmayr (2020) to control for the familywise error rate, following the procedure developed by List, Shaikh and Xu (2019) and List, Shaikh and Vayalinkal (2023).

Table 7: Placebo Estimates for People Without Individual Retirement Accounts

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
Placebo RD Estimate for 2019	-0.004 (0.031)	130 (160)	469 (457)
Mean	0.272	708	2,588
Bandwidth (months)	104	111	110
Clusters	3,444	3,693	1,026
Observations	4,246	4,608	1,330

Notes: This table reports placebo regression discontinuity (RD) estimates for people in households with no Individual Retirement Accounts after the Tax Cuts and Jobs Act, in 2019. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8: Effects of Qualified Charitable Distributions on Indicators for Donating Various Amounts to Charity After the Tax Cuts and Jobs Act

	Indicator for Giving \$2,500 or More (1)	Indicator for Giving \$5,000 or More (2)	Indicator for Giving \$10,000 or More (3)	Indicator for Giving \$15,000 or More (4)	Indicator for Giving \$20,000 or More (5)	Indicator for Giving \$25,000 or More (6)
RD Estimate for 2019	0.082 (0.050)	0.095* (0.049)	0.083*** (0.031)	0.059** (0.024)	0.025 (0.019)	0.003 (0.017)
Mean	0.288	0.174	0.085	0.040	0.022	0.013
Bandwidth (Months)	97	84	111	106	101	79
Clusters (Clusters)	1,466	1,286	1,663	1,591	1,527	1,220
Observations (Households)	1,789	1,543	2,048	1,956	1,871	1,458

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on indicator variables for making large donations to charity after the Tax Cuts and Jobs Act, in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 9: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People with Above-Median Individual Retirement Account Balances

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2019	-0.005 (0.063)	2,094** (845)	2,949*** (1,108)
Mean	0.699	3,077	4,316
Bandwidth (Months)	95	93	86
Clusters (Households)	769	759	491
Observations (Individuals)	989	970	627

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019, for the subsample of people with above-median household Individual Retirement Account balances in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 10: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People with Below-Median Individual Retirement Account Balances

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2019	-0.054 (0.080)	381 (640)	741 (972)
Mean	0.581	2,015	3,166
Bandwidth (Months)	95	103	89
Clusters (Households)	677	741	352
Observations (Individuals)	762	836	402

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019, for the subsample of people with below-median household Individual Retirement Account balances in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 11: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People Who Claimed the Standard Deduction

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2019	0.002 (0.080)	1,427** (709)	2,290** (1,014)
Mean	0.570	1,443	2,623
Bandwidth (Months)	83	75	79
Clusters (Households)	594	540	308
Observations (Individuals)	717	648	388

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019, for the subsample of people who claimed the standard deduction in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

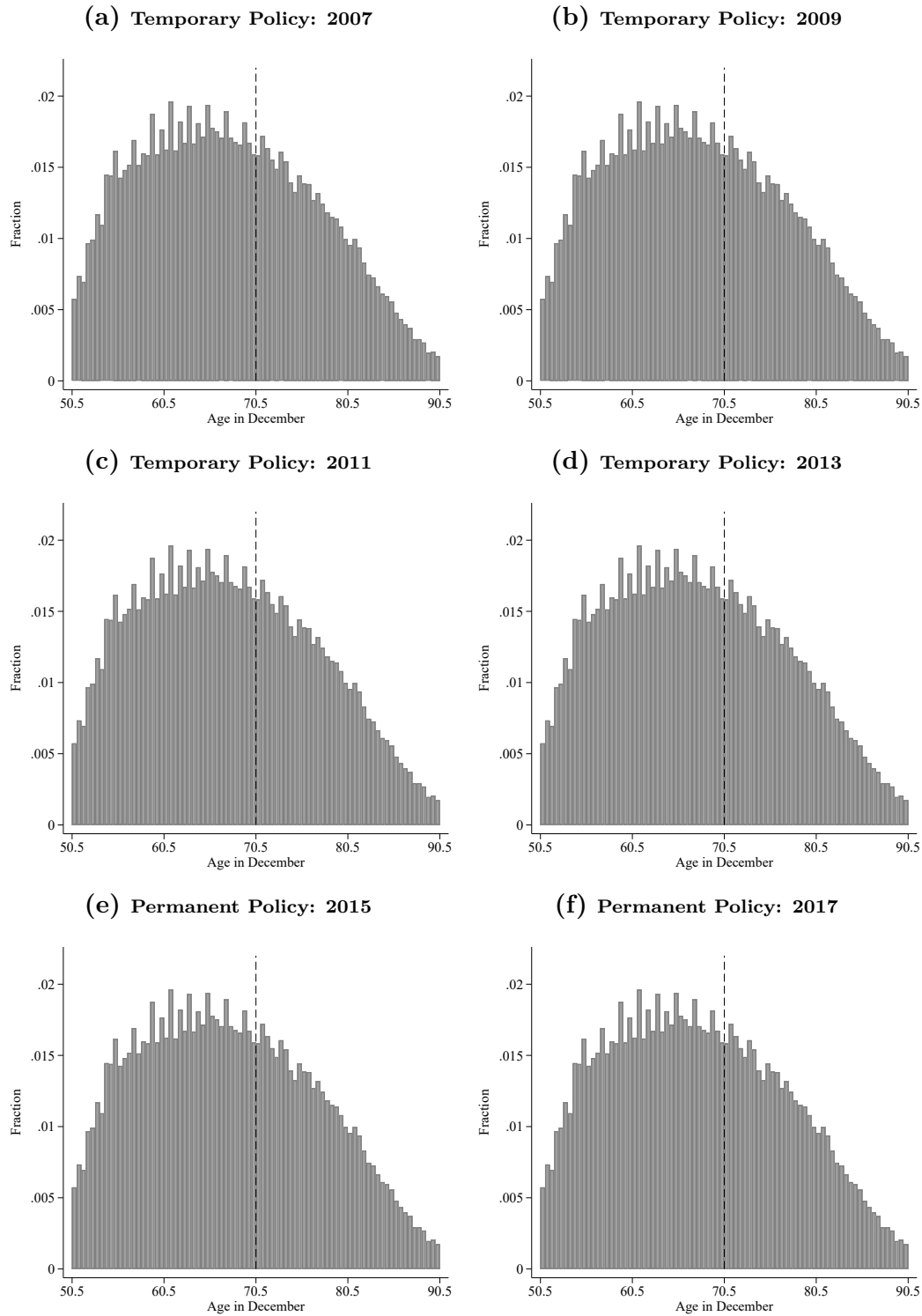
Table 12: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People Who Itemized Deductions

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2019	-0.094 (0.082)	969 (1,087)	876 (1,382)
Mean	0.756	3,912	5,186
Bandwidth (Months)	64	97	66
Clusters (Households)	377	560	293
Observations (Individuals)	439	678	346

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on charitable giving after the Tax Cuts and Jobs Act, in 2019, for the subsample of people who itemized deductions in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

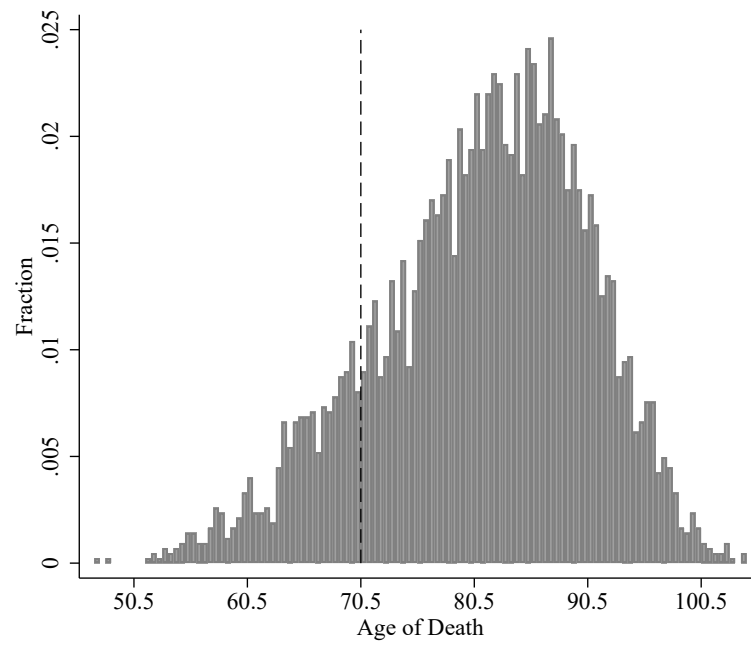
Appendix A Additional Figures and Tables

Figure A.1: Histograms of the Running Variable Before the Tax Cuts and Jobs Act



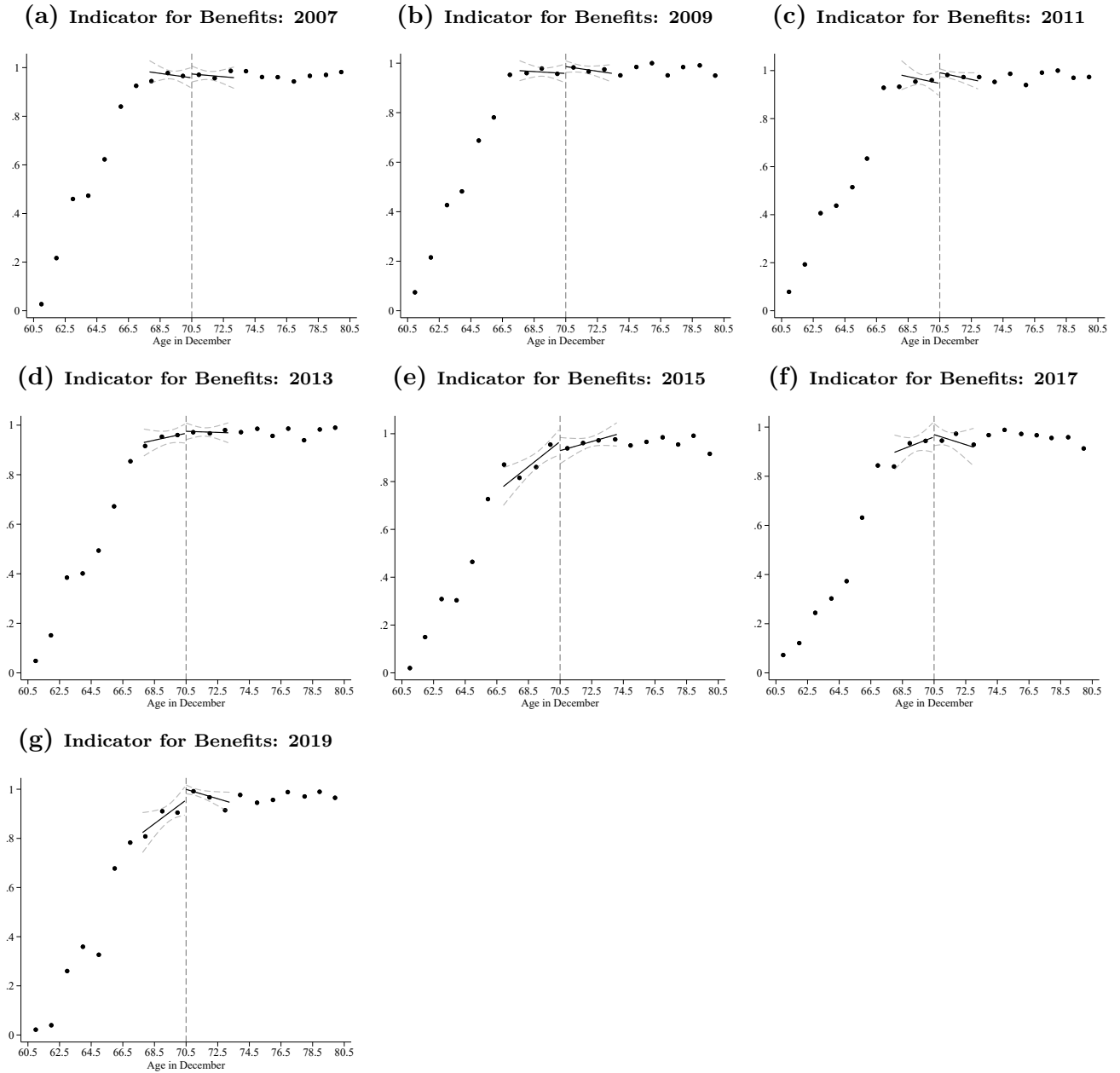
Notes: This figure plots histograms of the running variable, age in December, for tax years 2007 to 2017.

Figure A.2: Histogram of Age at Death



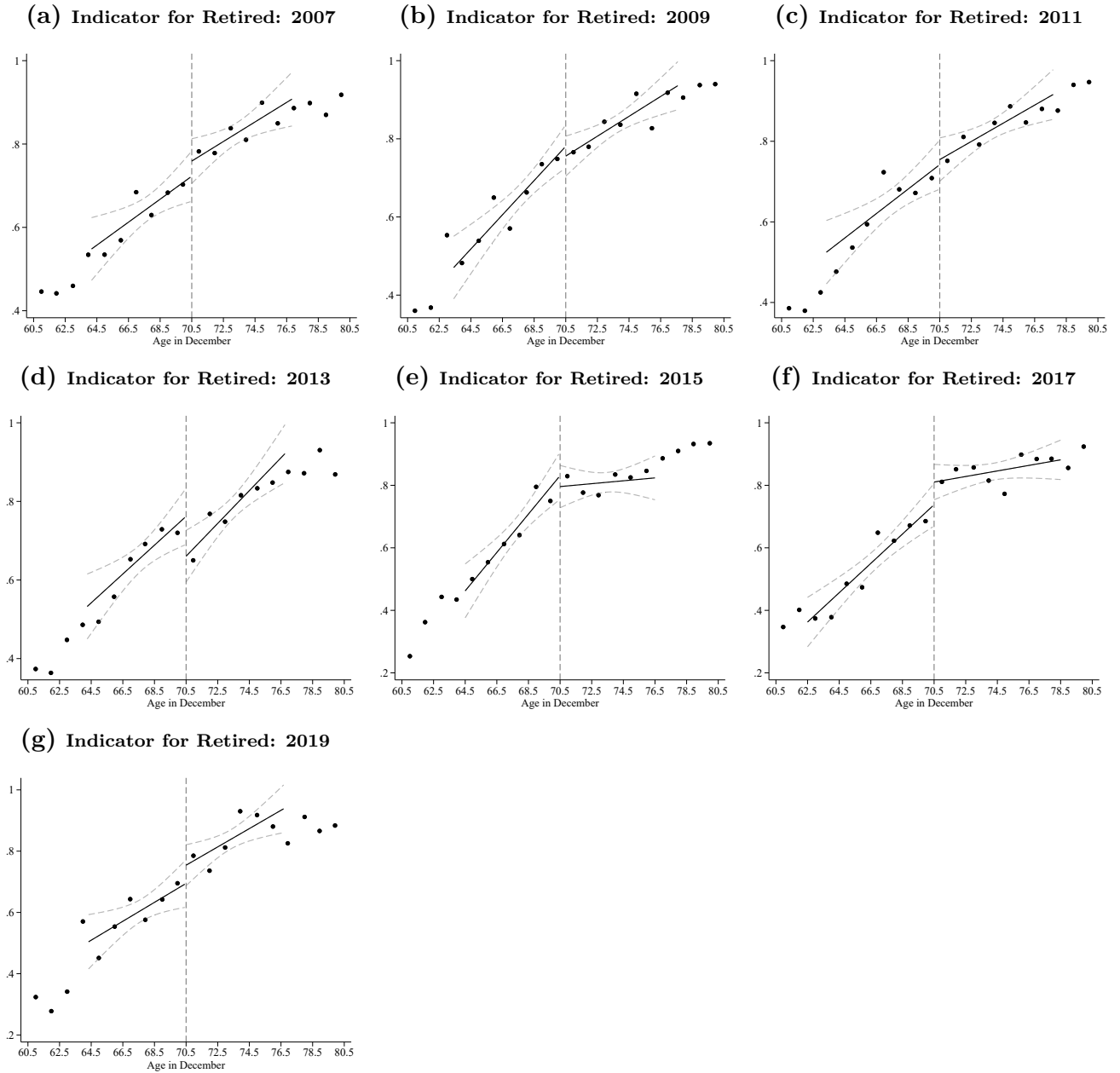
Notes: This figure plots a histogram of age at death for the subsample of people in my analysis data who die. The p-value from the formal density test that the distribution of age-at-death is smooth at $70\frac{1}{2}$ is 0.57.

Figure A.3: Effects of Reaching $70\frac{1}{2}$ on Receiving Social Security Benefits



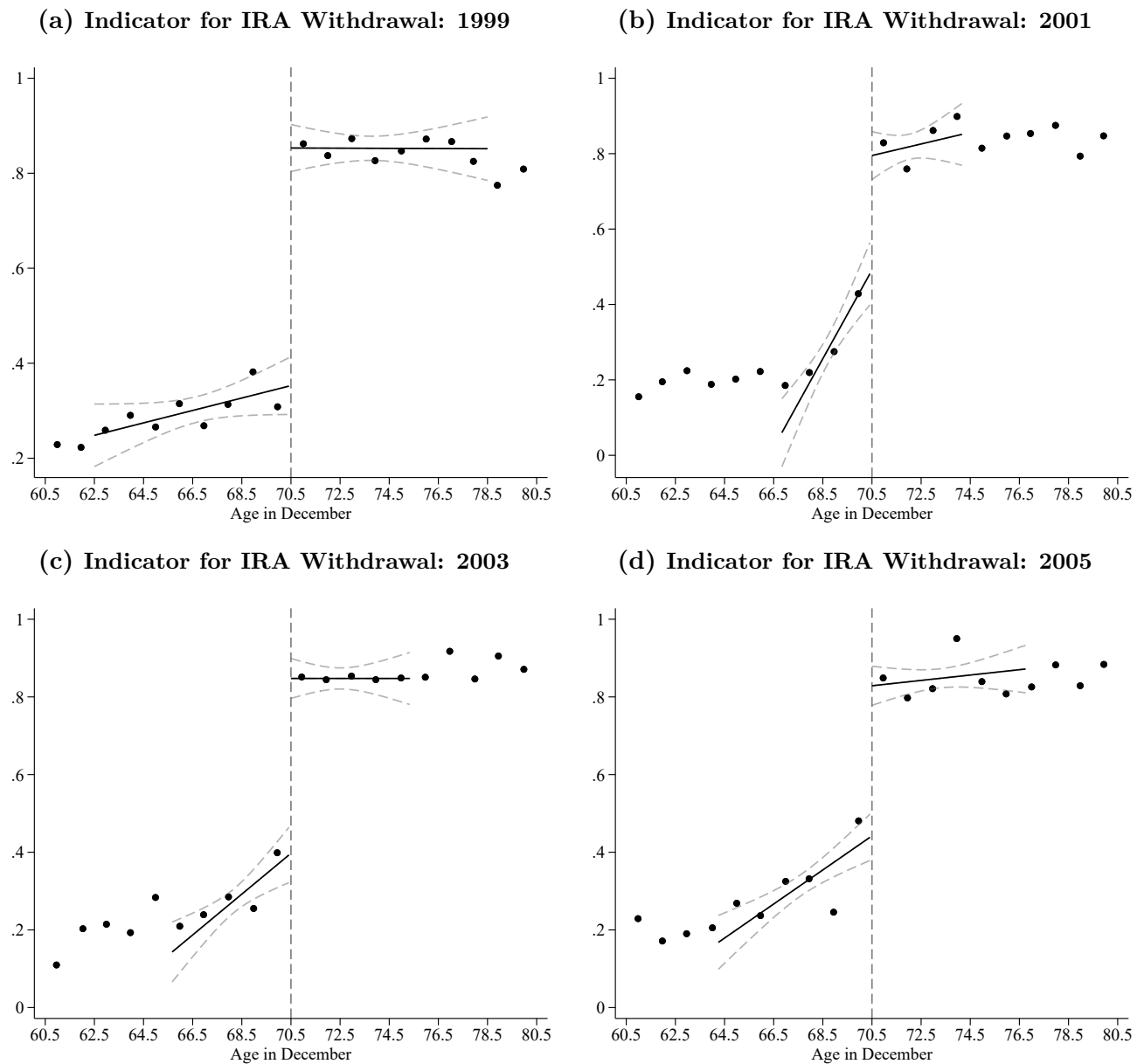
Notes: This figure illustrates the effects of reaching $70\frac{1}{2}$ on an indicator for receiving Social Security benefits. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.4: Effects of Reaching $70\frac{1}{2}$ on Retirement



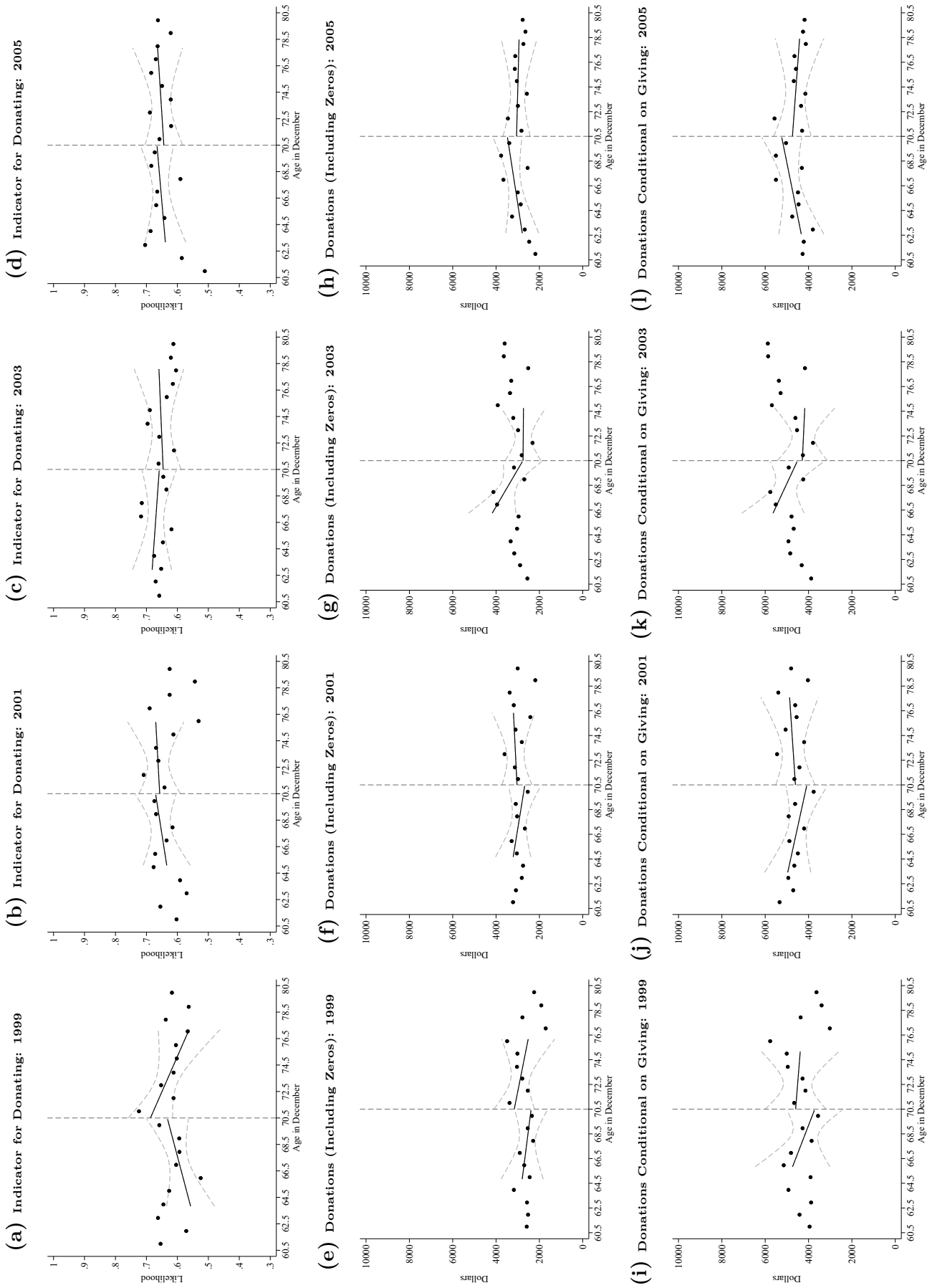
Notes: This figure illustrates the effects of reaching $70\frac{1}{2}$ on an indicator for being retired. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.5: Effects of Reaching $70\frac{1}{2}$ on Withdrawals from Individual Retirement Accounts in the Pre-Period, Before Qualified Charitable Distributions Existed



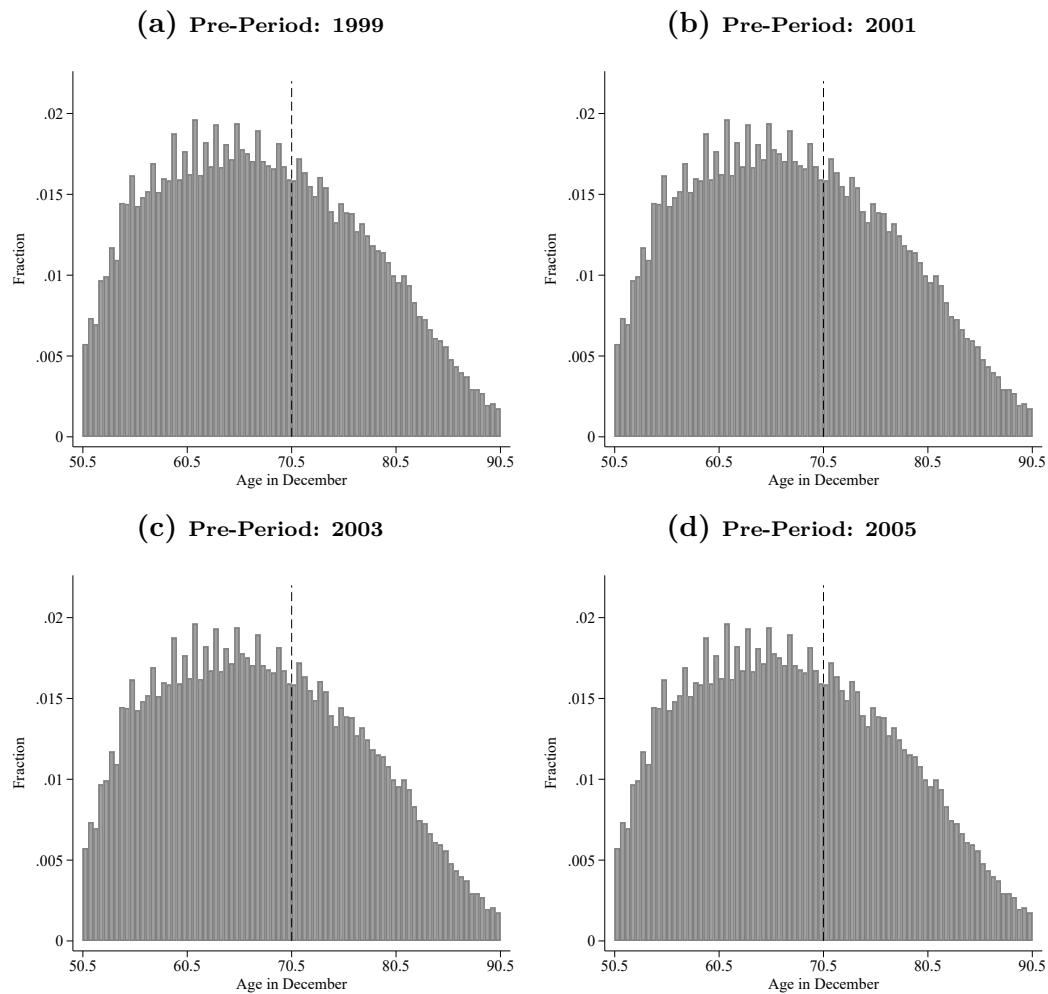
Notes: This figure illustrates the effects of reaching $70\frac{1}{2}$ on an indicator for withdrawing funds from an Individual Retirement Account in the pre-period, when required minimum distributions were in effect but before qualified charitable distributions existed. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.6: Effects of Reaching $70\frac{1}{2}$ on Charitable Giving in the Pre-Period, Before Qualified Charitable Distributions Existed



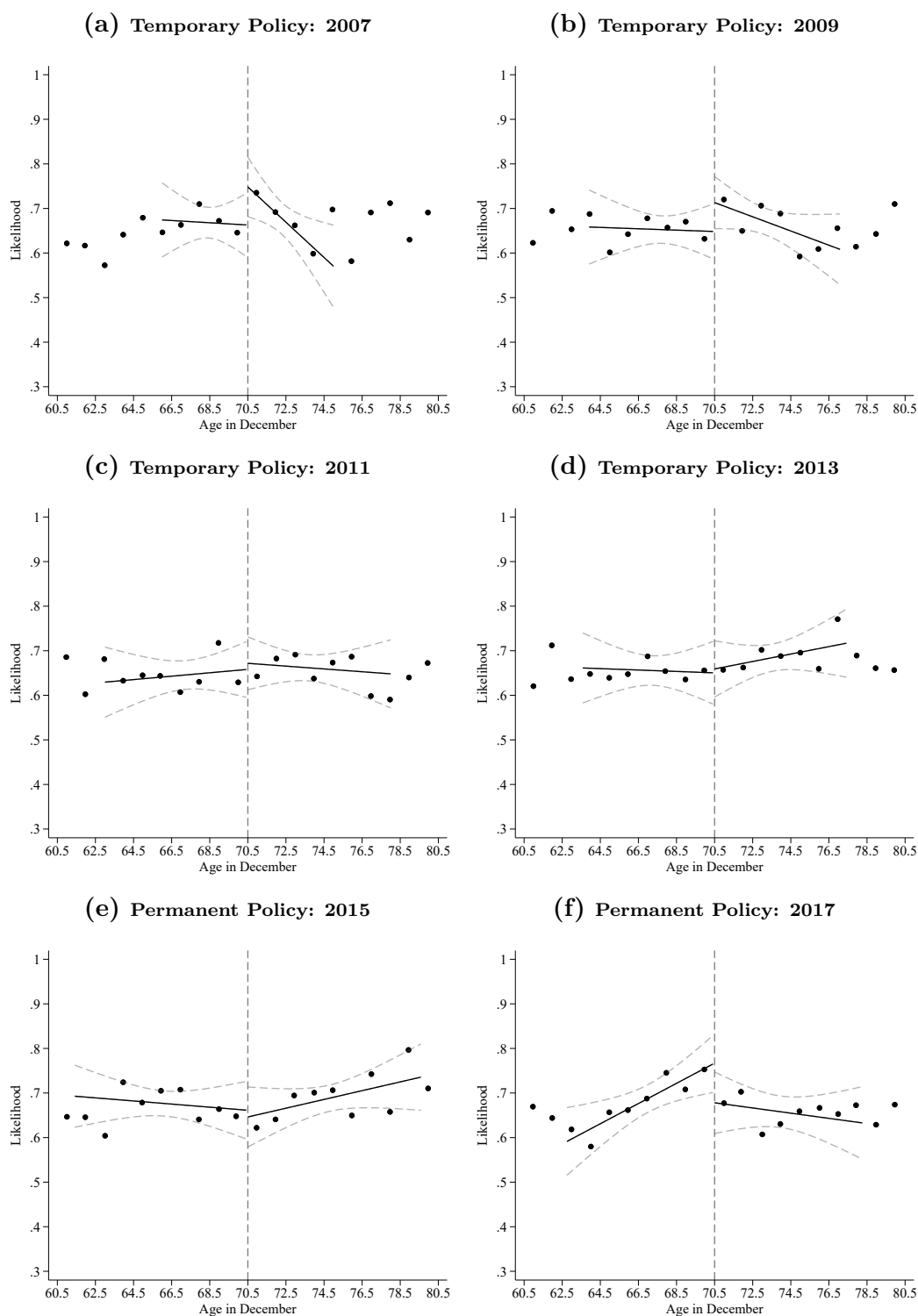
Notes: This figure illustrates the effects of reaching $70\frac{1}{2}$ on charitable giving outcomes during the pre-period, before qualified charitable distributions existed. Each graph corresponds to a different outcome-tax-year combination. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.7: Histograms of the Running Variable in the Pre-Period, Before Qualified Charitable Distributions Existed



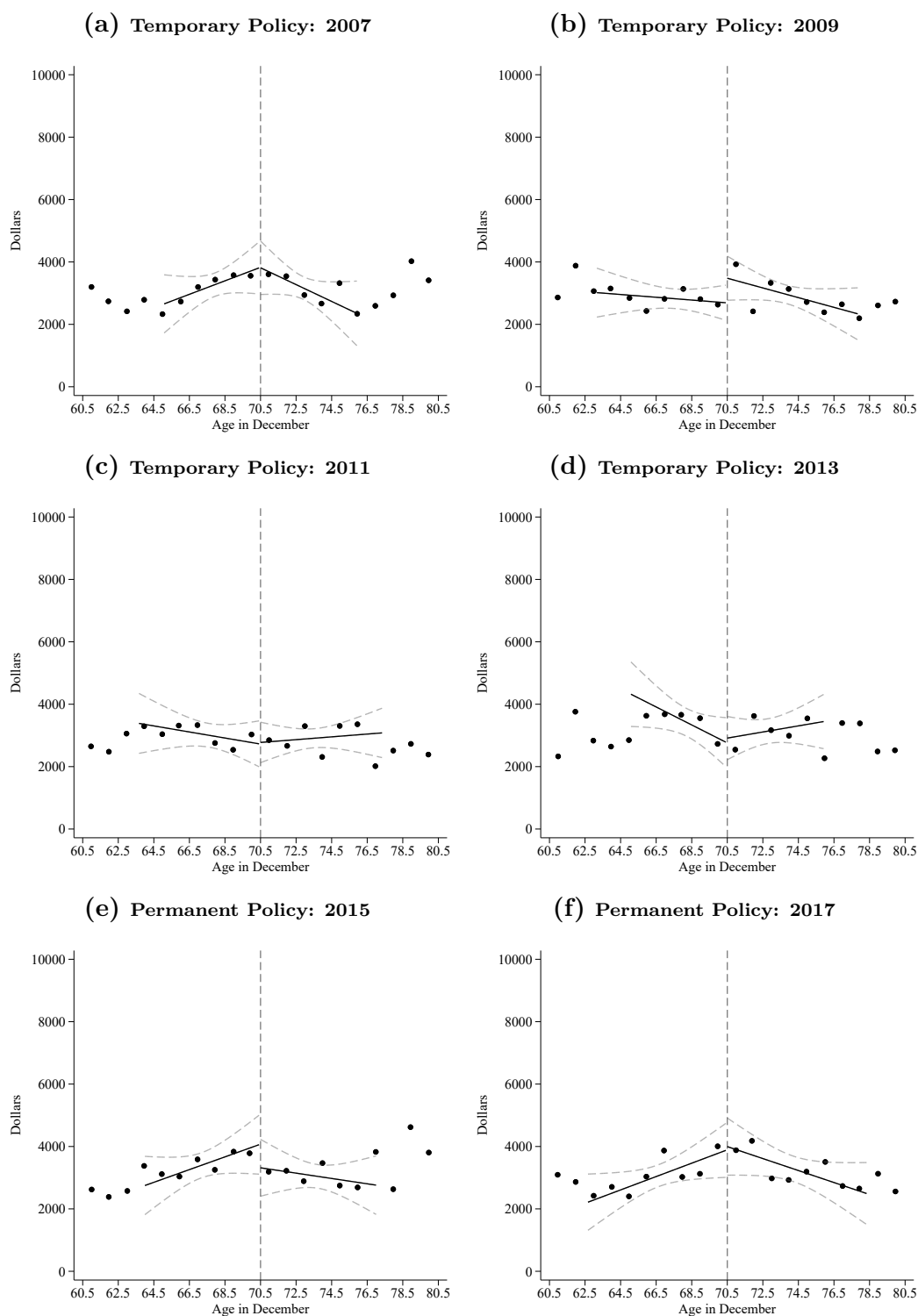
Notes: This figure plots histograms of the running variable, age in December, for tax years 1999 to 2005.

Figure A.8: Effects of Qualified Charitable Distributions on an Indicator for Giving to Charity Before the Tax Cuts and Jobs Act



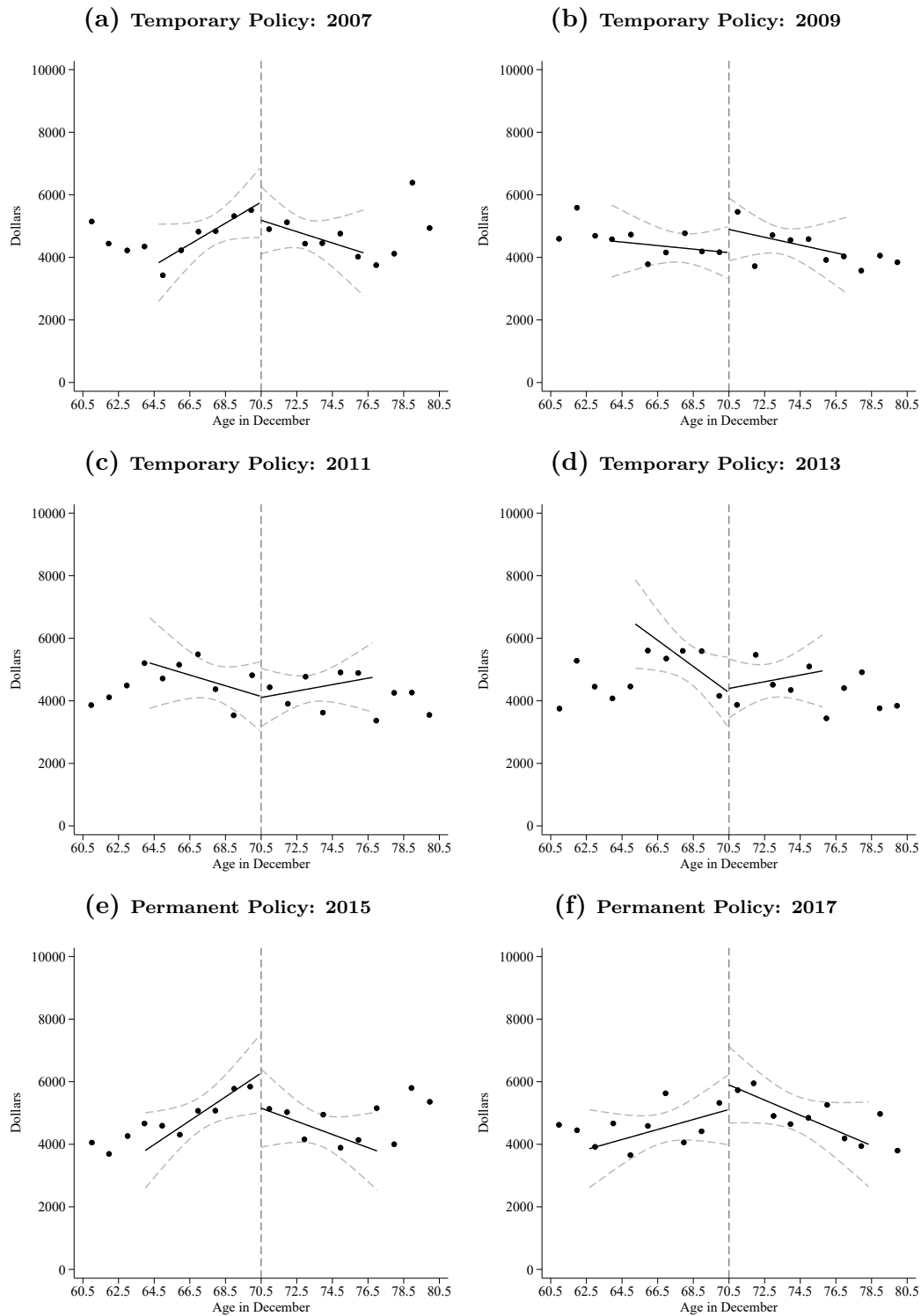
Notes: This figure illustrates the effects of aging into qualified charitable distributions an indicator for giving to charity before the Tax Cuts and Jobs Act. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.9: Effects of Qualified Charitable Distributions on Donations (Including Zeros) Before the Tax Cuts and Jobs Act



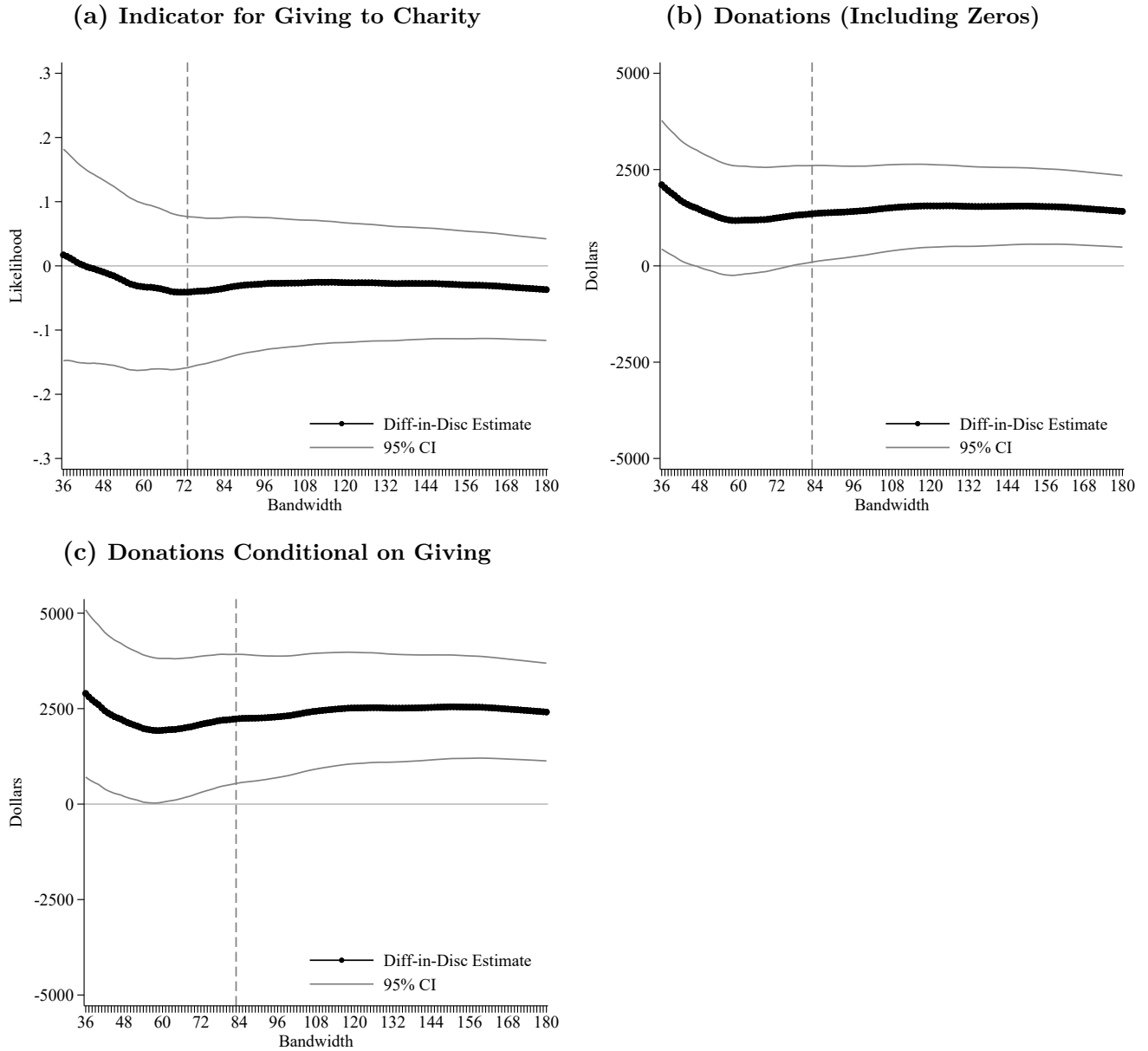
Notes: This figure illustrates the effects of aging into qualified charitable distributions on donations (including zeros) before the Tax Cuts and Jobs Act. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.10: Effects of Qualified Charitable Distributions on Donations Conditional on Giving Before the Tax Cuts and Jobs Act



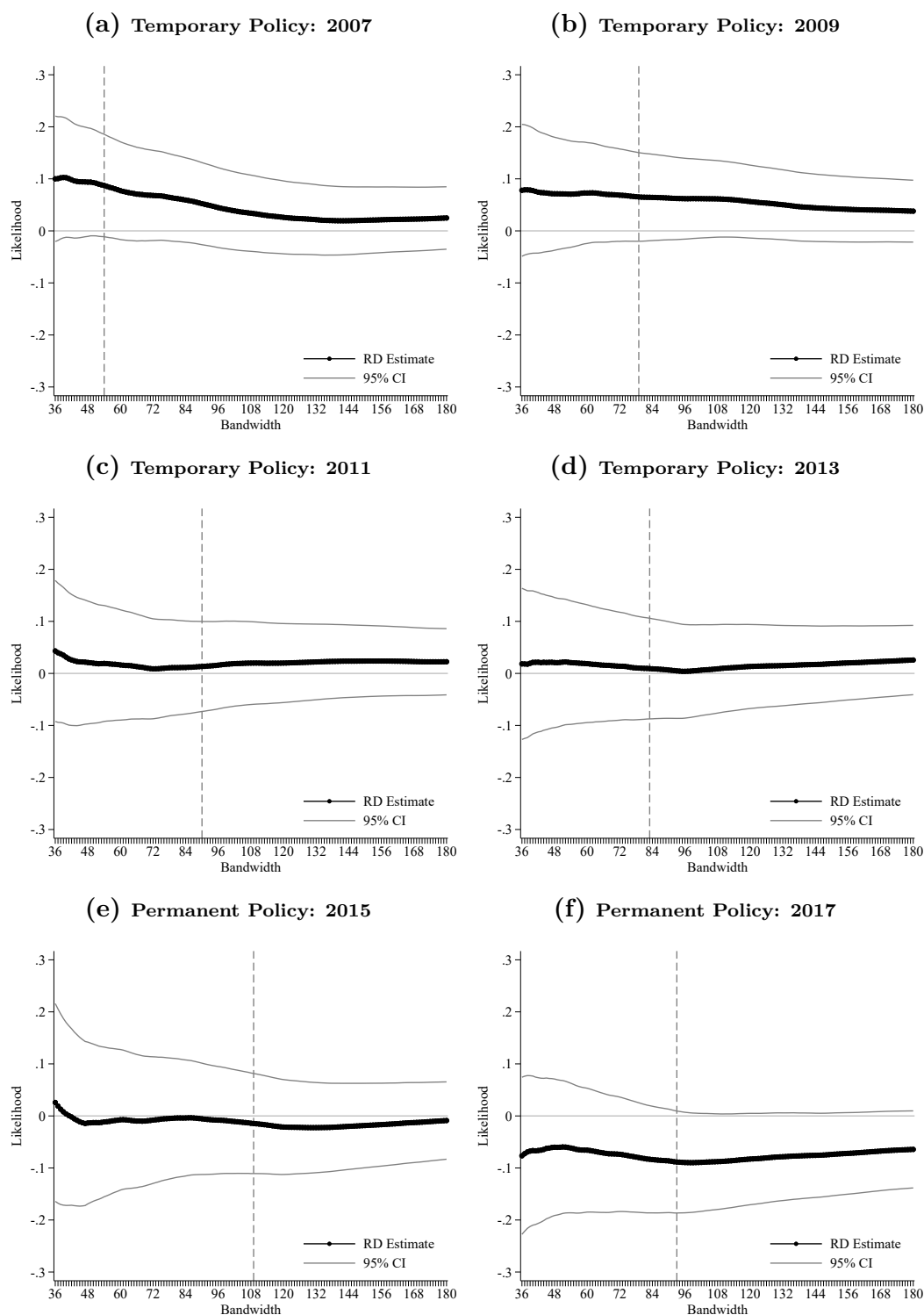
Notes: This figure illustrates the effects of aging into qualified charitable distributions on donations conditional on giving before the Tax Cuts and Jobs Act. Each graph corresponds to a different tax year. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.11: Robustness of Key Difference-in-Discontinuities Estimates to Bandwidth Selection



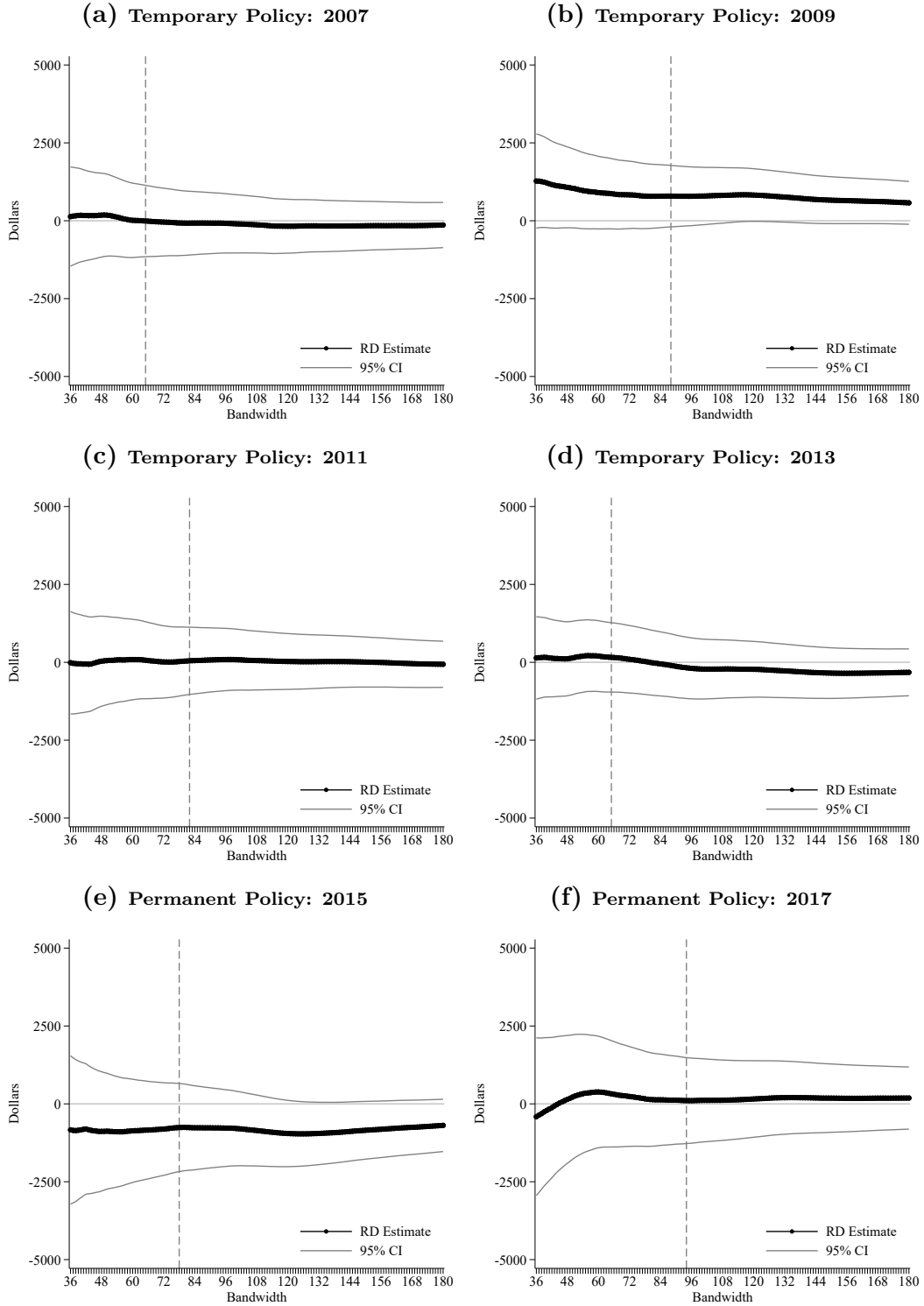
Notes: This figure illustrates the robustness of the difference-in-discontinuities estimates for 2019 to varying the bandwidth. Each graph plots the difference-in-discontinuities estimates and 95% confidence intervals that result from using bandwidths ranging from 36 months to 180 months. The vertical dashed lines denote the leading bandwidth.

Figure A.12: Robustness of Estimates for the Indicator for Giving to Charity in Earlier Years to Bandwidth Selection



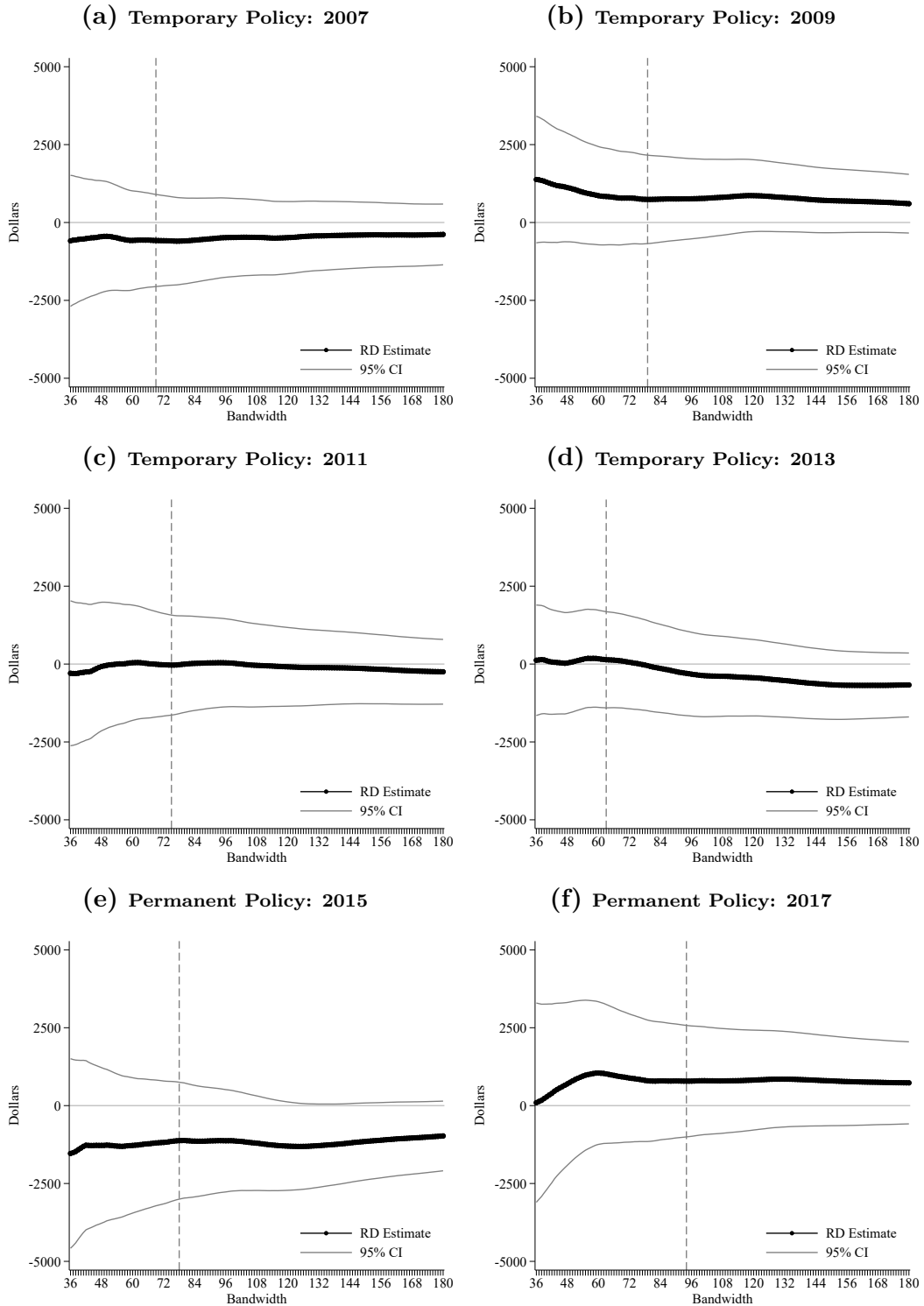
Notes: This figure illustrates the robustness of the estimates for the indicator for giving to charity in earlier years to varying the bandwidth. Each graph plots the regression discontinuity (RD) estimates and 95% confidence intervals that result from using bandwidths ranging from 36 months to 180 months. The vertical dashed lines denote the leading bandwidth.

Figure A.13: Robustness of Estimates for Donations (Including Zeros) in Earlier Years to Bandwidth Selection



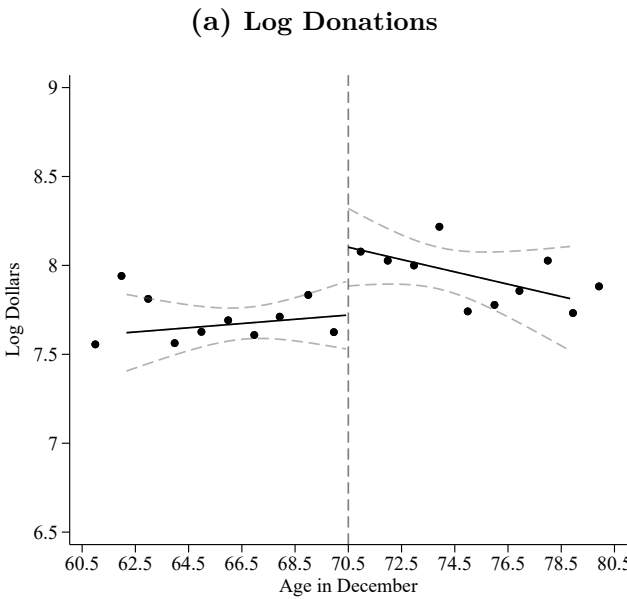
Notes: This figure illustrates the robustness of the estimates for donations (including zeros) in earlier years to varying the bandwidth. Each graph plots the regression discontinuity (RD) estimates and 95% confidence intervals that result from using bandwidths ranging from 36 months to 180 months. The vertical dashed lines denote the leading bandwidth.

Figure A.14: Robustness of Estimates for Donations Conditional on Giving in Earlier Years to Bandwidth Selection



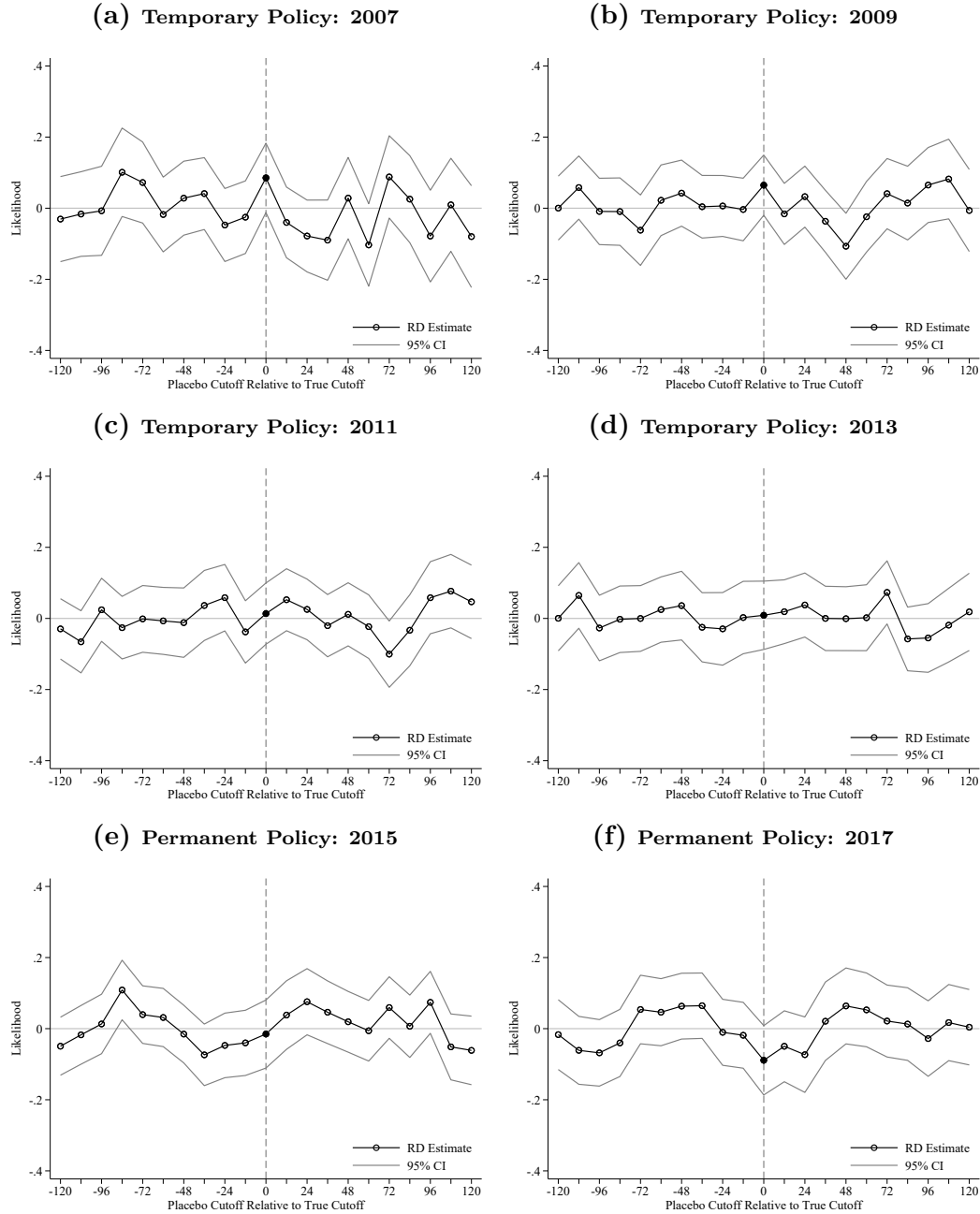
Notes: This figure illustrates the robustness of the estimates for the indicator for donations conditional on giving in earlier years to varying the bandwidth. Each graph plots the regression discontinuity (RD) estimates and 95% confidence intervals that result from using bandwidths ranging from 36 months to 180 months. The vertical dashed lines denote the leading bandwidth.

Figure A.15: Effects of Qualified Charitable Distributions on Log Donations After the Tax Cuts and Jobs Act



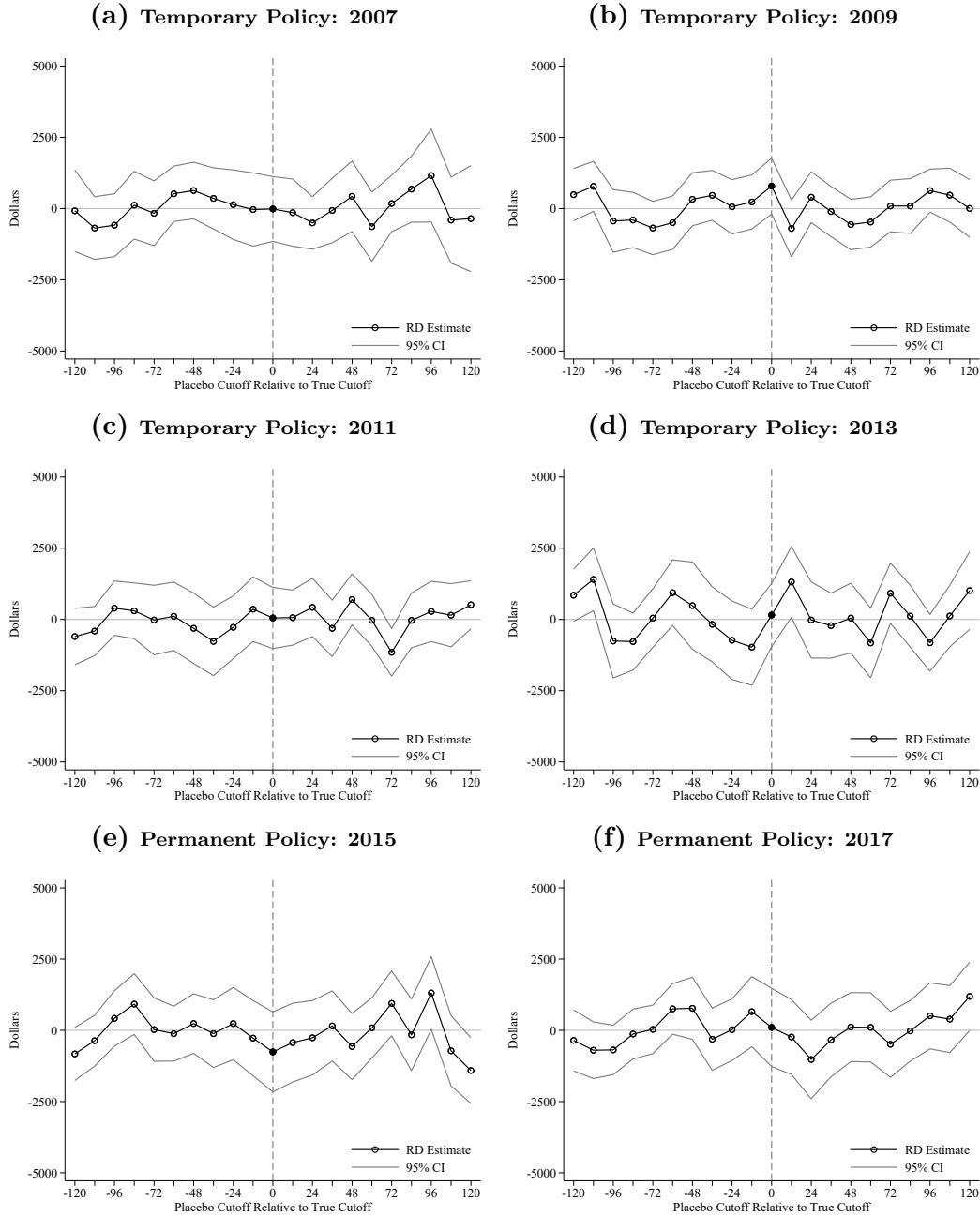
Notes: This figure illustrates the effects of aging into qualified charitable distributions on log donations in 2019. See the notes of Figure 3 for more details on how each graph is constructed.

Figure A.16: Estimates Using Placebo Cutoffs Around the True Cutoff for the Indicator for Giving to Charity in Earlier Years



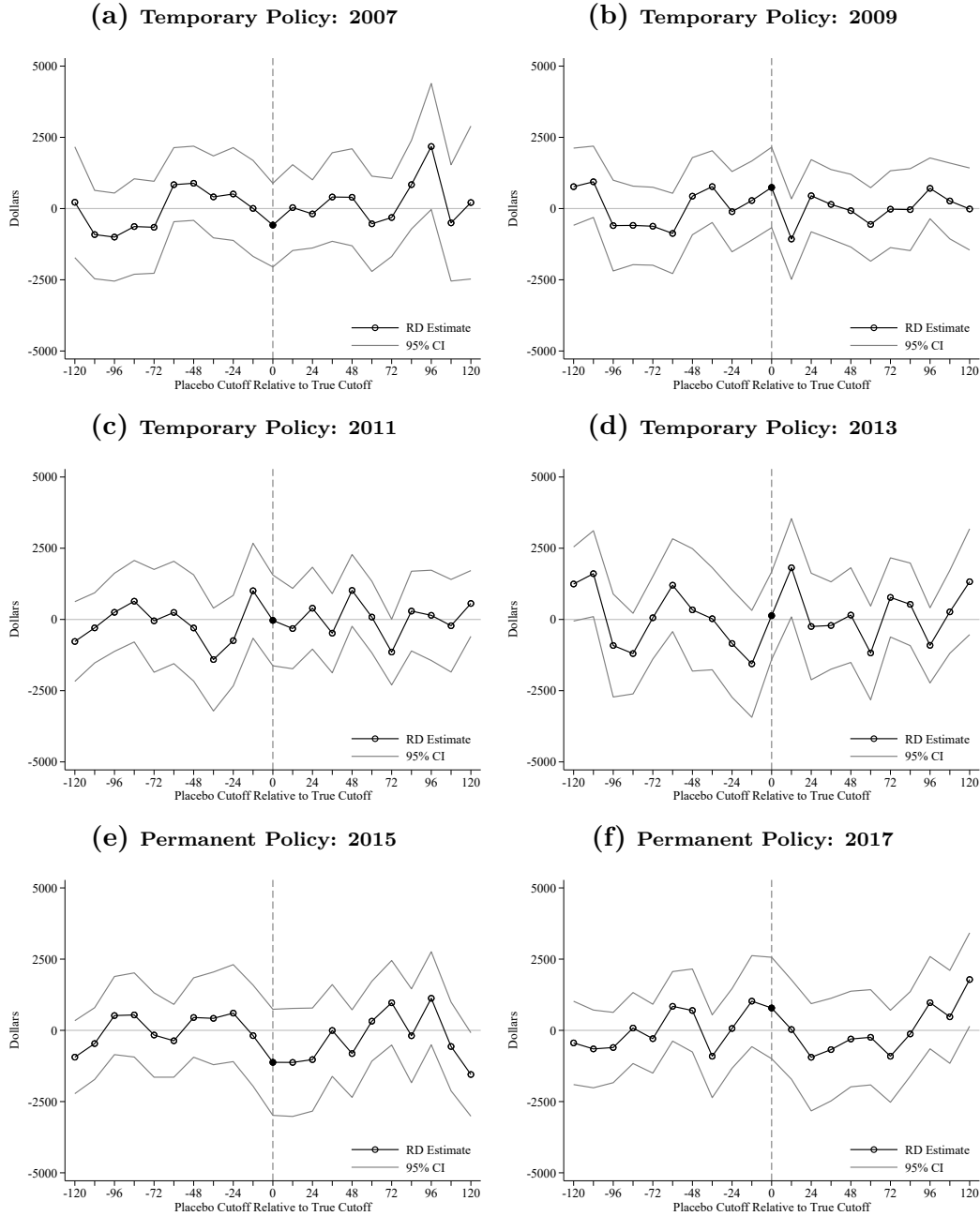
Notes: This figure illustrates how the regression discontinuity (RD) estimates for the indicator for giving to charity in earlier years change when I use placebo cutoffs around the true cutoff. Each graph plots RD estimates and 95-percent confidence intervals from estimating my RD regression model using various placebo cutoffs ranging from 120 months before the true cutoff to 120 months after the true cutoff. The point estimate for the true cutoff is denoted by a solid circle, and the point estimates for the placebo cutoffs are denoted by hollow circles.

Figure A.17: Estimates Using Placebo Cutoffs Around the True Cutoff for Donations (Including Zeros) in Earlier Years



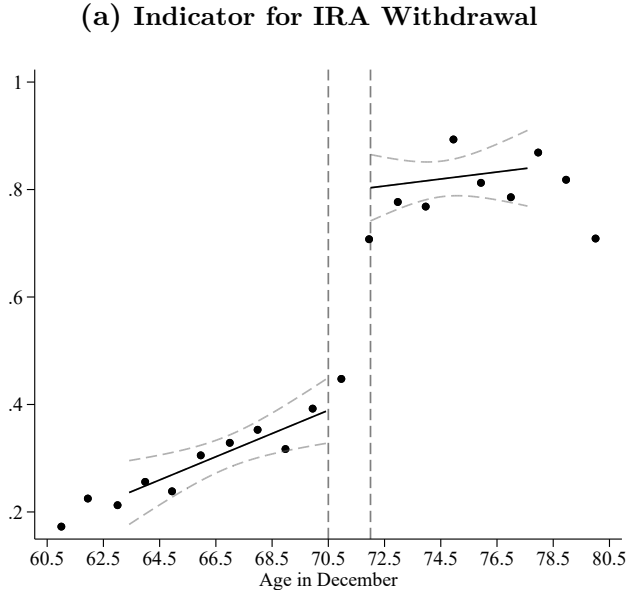
Notes: This figure illustrates how the regression discontinuity (RD) estimates for donations (including zeros) in earlier years change when I use placebo cutoffs around the true cutoff. Each graph plots RD estimates and 95-percent confidence intervals from estimating my RD regression model using various placebo cutoffs ranging from 120 months before the true cutoff to 120 months after the true cutoff. The point estimate for the true cutoff is denoted by a solid circle, and the point estimates for the placebo cutoffs are denoted by hollow circles.

Figure A.18: Estimates Using Placebo Cutoffs Around the True Cutoff for Donations Conditional on Giving in Earlier Years



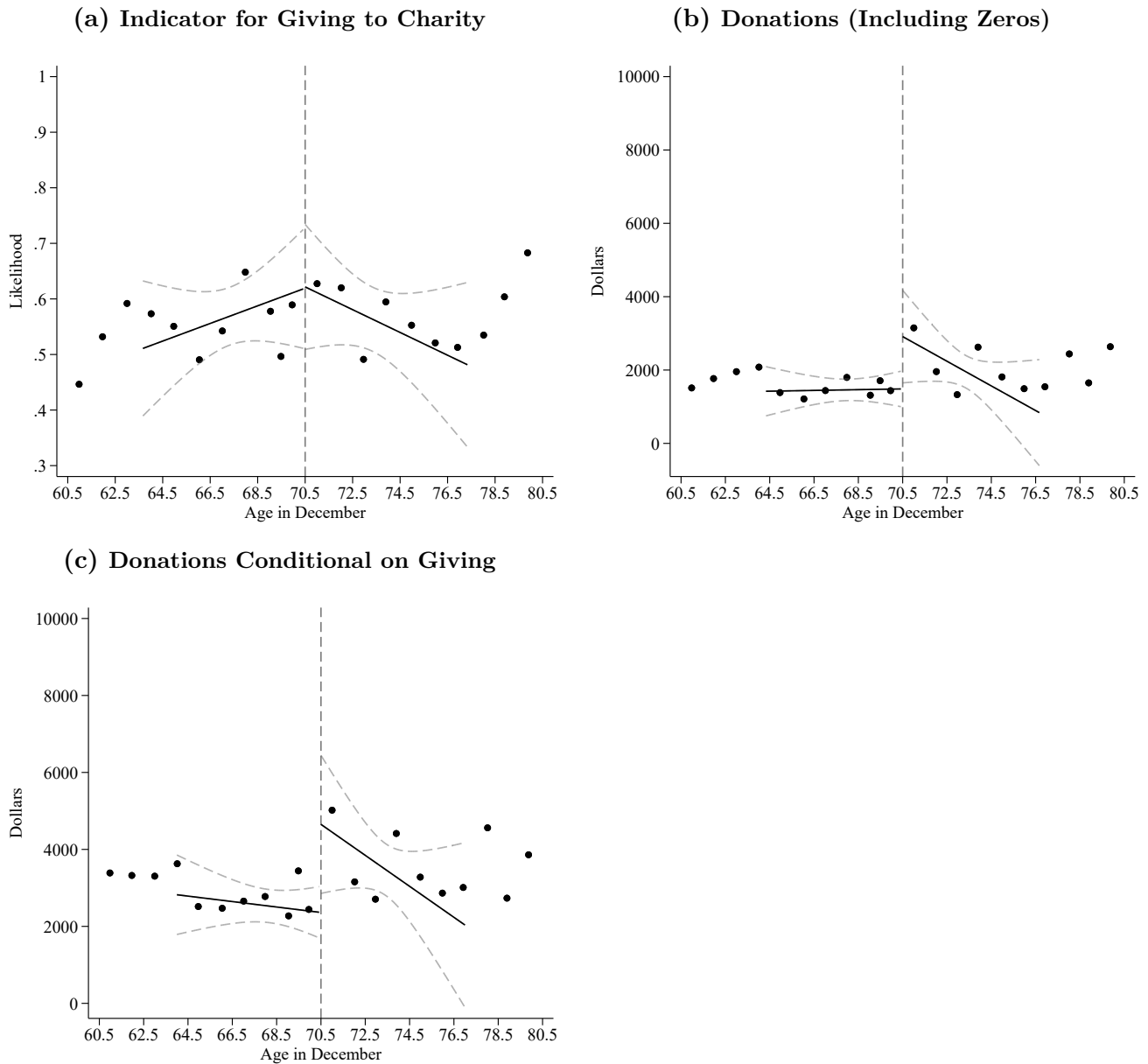
Notes: This figure illustrates how the regression discontinuity (RD) estimates for donations conditional on giving in earlier years change when I use placebo cutoffs around the true cutoff. Each graph plots RD estimates and 95-percent confidence intervals from estimating my RD regression model using various placebo cutoffs ranging from 120 months before the true cutoff to 120 months after the true cutoff. The point estimate for the true cutoff is denoted by a solid circle, and the point estimates for the placebo cutoffs are denoted by hollow circles.

Figure A.19: Suggestive Graphical Evidence on IRA Withdrawals in 2021



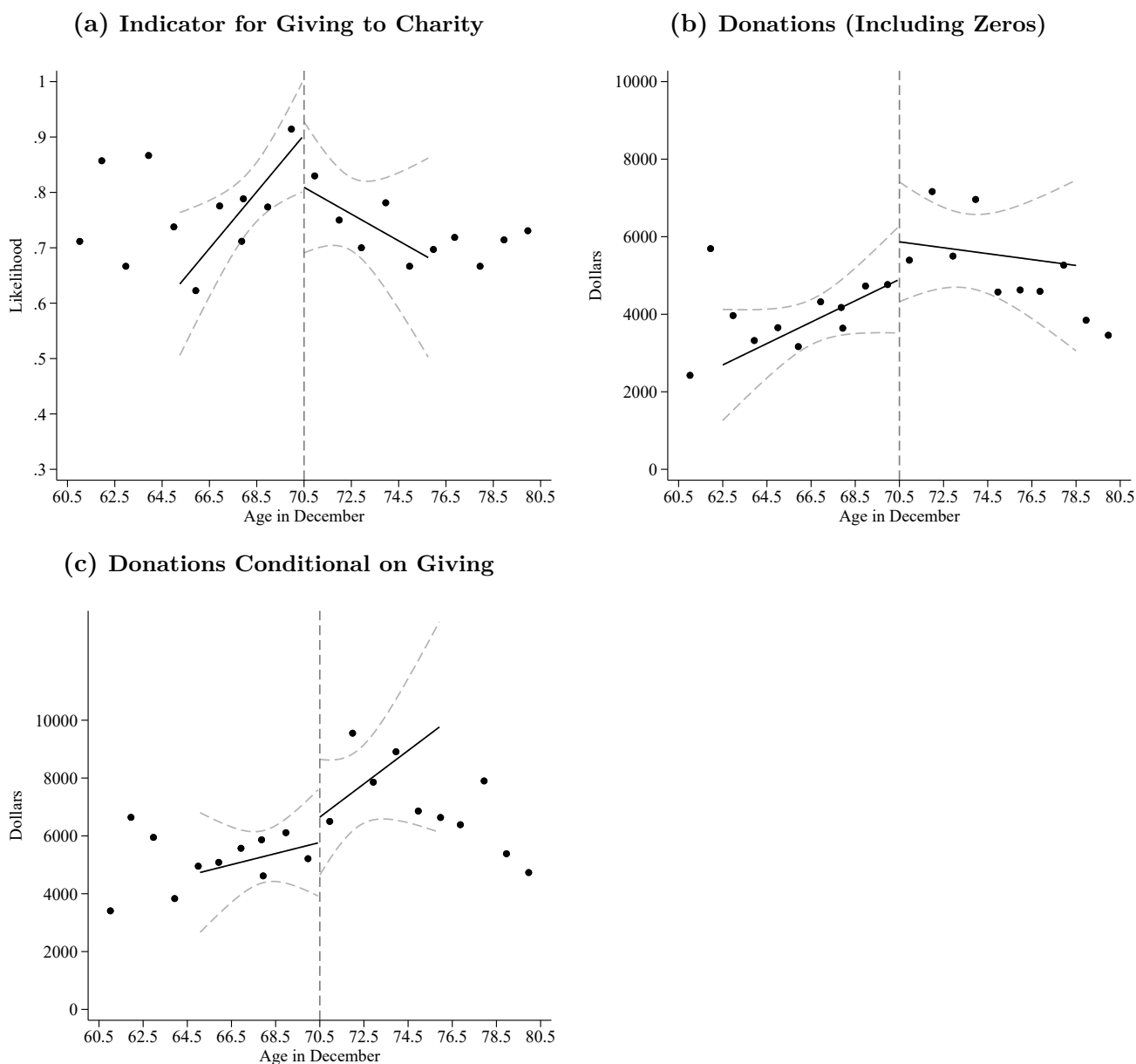
Notes: This figure provides suggestive graphical evidence on the effects of aging into qualified charitable distributions and required minimum distributions on IRA withdrawals in 2021. See the notes of Figure 8 for more details on how the graph is constructed.

Figure A.20: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People Who Claimed the Standard Deduction



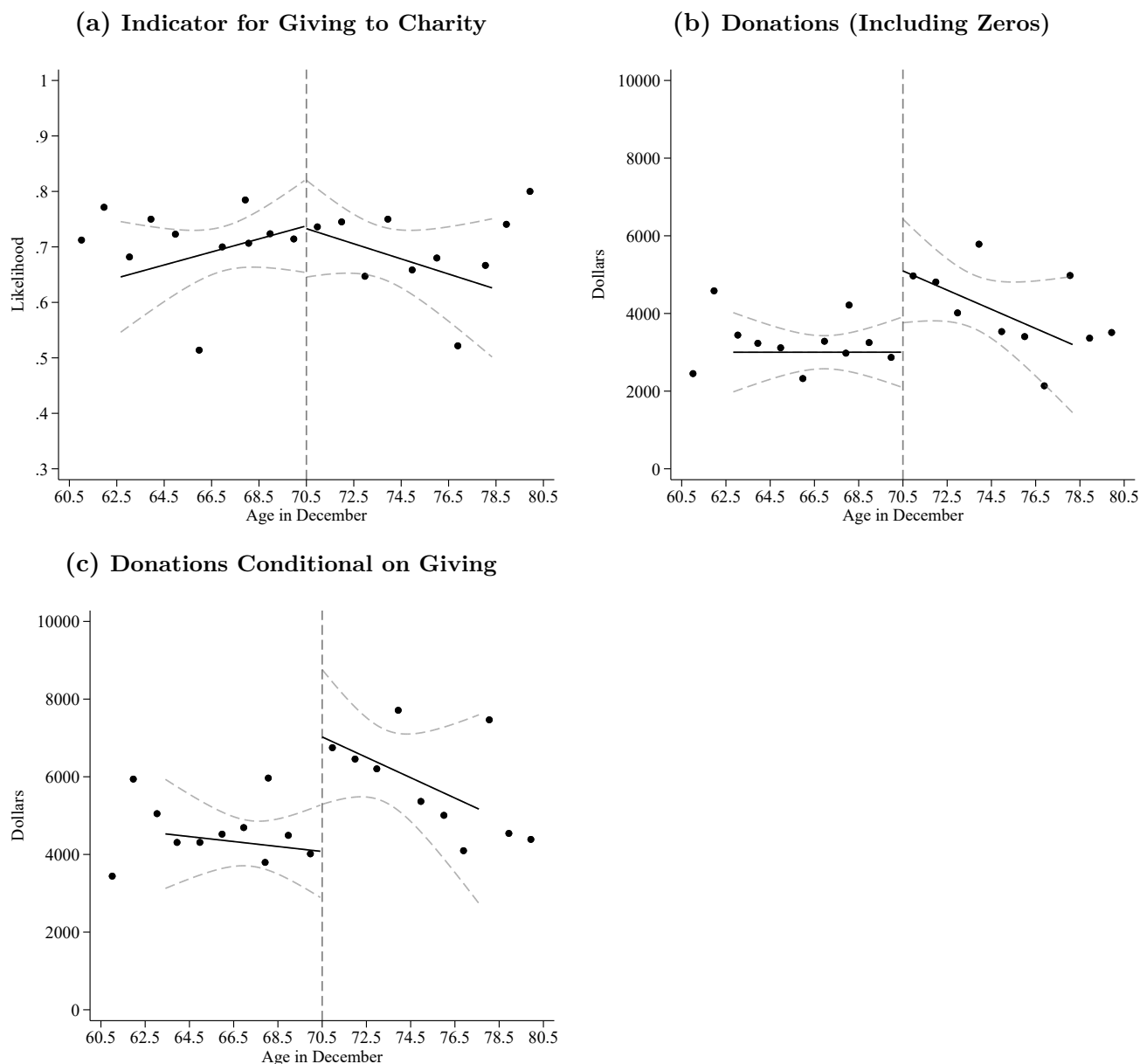
Notes: This figure illustrates the effects of aging into qualified charitable distributions on charitable giving in 2019 for the subsample of people who claimed the standard deduction. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving. See the figure notes of Figure 3 for more details on the construction of each graph.

Figure A.21: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People Who Itemized Deductions



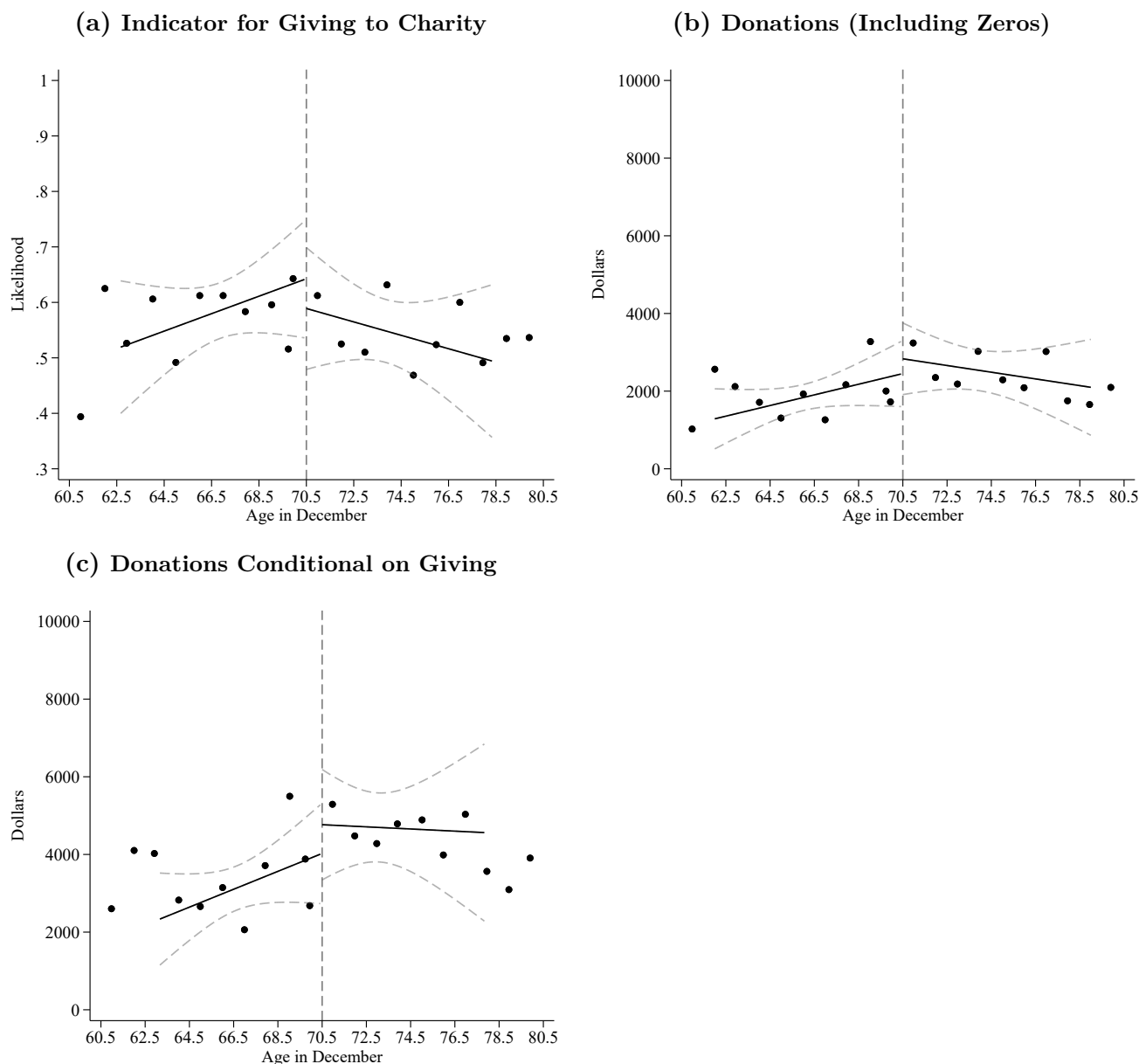
Notes: This figure illustrates the effects of aging into qualified charitable distributions on charitable giving in 2019 for the subsample of people who itemized deductions. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving. See the figure notes of Figure 3 for more details on the construction of each graph.

Figure A.22: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People with Above-Median Individual Retirement Account Balances



Notes: This figure illustrates the effects of aging into qualified charitable distributions on charitable giving in 2019 for the subsample of people who have above-median household Individual Retirement Account balances. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving. See the figure notes of Figure 3 for more details on the construction of each graph.

Figure A.23: Effects of Qualified Charitable Distributions on Charitable Giving After the Tax Cuts and Jobs Act for People with Below-Median Individual Retirement Account Balances



Notes: This figure illustrates the effects of aging into qualified charitable distributions on charitable giving in 2019 for the subsample of people who have below-median household Individual Retirement Account balances. Each graph corresponds to a different outcome. Graph (a) is for an indicator variable for giving to charity. Graph (b) is for donations in dollars, including zeros. Graph (c) is for donations in dollars, conditional on giving. See the figure notes of Figure 3 for more details on the construction of each graph.

Table A.1: Comparison of Aggregate Charitable Giving Estimates for Households 55 and Older

	HRS Data	IRS Publications	
	Total Donations (\$ Billions) (1)	Deductions for Cash Contributions (\$ Billions) (2)	Deductions for Total Contributions (\$ Billions) (3)
2007	77.18	73.80	102.25
2009	69.47	69.36	82.57
2011	82.21	79.05	98.20
2013	83.57	88.25	113.60
2015	98.06	98.71	133.38
2017	103.84	111.44	156.44
2019	112.20	82.08	121.23
2021	110.77	107.90	174.85

Notes: This table reports aggregate charitable giving estimates from both the Health and Retirement Study (HRS) data and Internal Revenue Service (IRS) publications. I compute the HRS-based estimates using observations of households in survey waves 9 through 15 with at least one spouse who is 55 or older. I calculate aggregate totals by summing donations in each year using household-level survey weights. The underlying HRS survey question used to produce these estimates asks respondents for the approximate value of their donations of money, property, or possessions that totaled \$500 or more. The IRS statistics are taken from Table 2.6 in Publication 1304 for each year (IRS Statistics of Income, 2021), which reports by-age estimates for taxpayers who itemize deductions. I report statistics for deductions from cash contributions and total (cash, non-cash, and carryover) deductions.

Table A.2: Sample Restrictions and Resulting Sample Sizes

	Number of Individuals (1)	Number of Observations (2)
Keep waves 5 through 16	39,358	224,530
Keep people with IRAs	14,312	57,716
Drop observations with missing donation outcomes	13,909	53,608

Notes: This table reports the resulting numbers of unique individuals and observations after implementing each of the three analysis sample restrictions. The final analysis sample consists of 49,963 observations on 13,180 individuals.

Table A.3: Density Test Results for Each Year After the Implementation of Qualified Charitable Distributions

	t-statistic (1)	p-value (2)
2007	1.192	0.233
2009	1.220	0.222
2011	-0.123	0.902
2013	0.368	0.713
2015	-1.339	0.181
2017	1.683	0.092
2019	-0.391	0.696
2021	-1.031	0.302

Notes: This table reports the resulting t-statistics and p-values from the formal density tests as proposed by Cattaneo, Jansson and Ma (2020) for tax years 2007 to 2021.

Table A.4: Regression Discontinuity Estimates Using Covariates as Outcome Variables

	Male (1)	White (2)	Married (3)	College (4)
RD Estimate for 2007	-0.009 (0.054)	0.019 (0.023)	-0.014 (0.044)	-0.011 (0.044)
Mean	0.518	0.919	0.747	0.598
Bandwidth (Months)	58	87	63	89
Clusters (Households)	1,355	1,833	1,444	1,861
Observations (Individuals)	1,600	2,276	1,726	2,316
RD Estimate for 2009	0.007 (0.045)	0.020 (0.025)	0.061 (0.041)	-0.036 (0.043)
Mean	0.444	0.917	0.743	0.631
Bandwidth (Months)	87	74	78	90
Clusters (Households)	1,784	1,587	1,641	1,837
Observations (Individuals)	2,194	1,935	1,997	2,267
RD Estimate for 2011	0.060 (0.048)	0.008 (0.030)	-0.044 (0.044)	0.022 (0.047)
Mean	0.433	0.914	0.757	0.684
Bandwidth (Months)	85	72	84	88
Clusters (Households)	1,589	1,384	1,575	1,643
Observations (Individuals)	1,917	1,656	1,900	1,987
RD Estimate for 2013	0.056 (0.052)	-0.001 (0.027)	0.040 (0.046)	0.092* (0.052)
Mean	0.431	0.886	0.748	0.709
Bandwidth (Months)	76	84	79	74
Clusters (Households)	1,420	1,554	1,469	1,375
Observations (Individuals)	1,712	1,896	1,781	1,656
RD Estimate for 2015	-0.013 (0.053)	-0.004 (0.030)	0.010 (0.046)	-0.049 (0.045)
Mean	0.459	0.868	0.738	0.766
Bandwidth (Months)	96	103	107	111
Clusters (Households)	1,594	1,694	1,755	1,821
Observations (Individuals)	1,947	2,091	2,169	2,264
RD Estimate for 2017	-0.069 (0.058)	0.005 (0.035)	0.036 (0.047)	-0.033 (0.041)
Mean	0.454	0.867	0.711	0.790
Bandwidth (Months)	75	89	95	115
Clusters (Households)	1,133	1,361	1,455	1,723
Observations (Individuals)	1,346	1,642	1,758	2,122
RD Estimate for 2019	-0.017 (0.059)	0.011 (0.036)	0.040 (0.051)	0.044 (0.045)
Mean	0.465	0.845	0.711	0.789
Bandwidth (Months)	71	84	87	87
Clusters (Households)	1,096	1,286	1,333	1,334
Observations (Individuals)	1,296	1,543	1,600	1,601

Notes: This table reports regression discontinuity (RD) estimates for the effects of reaching $70\frac{1}{2}$ on control variables in odd-numbered years from 2007 to 2017. Standard errors clustered at the household level are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.5: Effects of Reaching $70\frac{1}{2}$ on Social Security Benefits and Retirement

	Indicator for Social Security Benefits (1)	Indicator for Being Retired (2)
RD Estimate for 2007	0.016 (0.027)	0.036 (0.041)
Mean	0.968	0.635
Bandwidth (months)	33	77
Clusters	840	1,684
Observations	932	2,063
RD Estimate for 2009	0.028 (0.023)	-0.025 (0.039)
Mean	0.965	0.640
Bandwidth (months)	36	86
Clusters	905	1,764
Observations	1,028	2,167
RD Estimate for 2011	0.044 (0.029)	0.013 (0.042)
Mean	0.958	0.627
Bandwidth (months)	30	87
Clusters	635	1,624
Observations	709	1,966
RD Estimate for 2013	0.008 (0.027)	-0.102** (0.049)
Mean	0.945	0.625
Bandwidth (months)	33	76
Clusters	621	1,420
Observations	691	1,712
RD Estimate for 2015	-0.038 (0.041)	-0.035 (0.052)
Mean	0.875	0.631
Bandwidth (months)	44	73
Clusters	678	1,189
Observations	762	1,427
RD Estimate for 2017	0.009 (0.039)	0.072 (0.045)
Mean	0.916	0.536
Bandwidth (months)	31	97
Clusters	468	1,479
Observations	525	1,789
RD Estimate for 2019	0.043 (0.032)	0.059 (0.052)
Mean	0.879	0.588
Bandwidth (months)	34	75
Clusters	561	1,163
Observations	627	1,381

Notes: This table reports regression discontinuity (RD) estimates for the effects of reaching $70\frac{1}{2}$ on indicators for receiving Social Security benefits and for being retired. Each panel corresponds to a different year. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.6: Effects of Reaching $70\frac{1}{2}$ on Charitable Giving in the Pre-Period, Before Qualified Charitable Distributions Existed

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 1999	0.054 (0.051)	778 (618)	886 (978)
Mean	0.603	2,576	4,323
Bandwidth (Month)	81	69	57
Clusters (Households)	1,731	1,478	758
Observations (Individuals)	1,731	1,478	758
RD Estimate for 2001	-0.013 (0.048)	362 (564)	527 (726)
Mean	0.662	2,947	4,510
Bandwidth (Month)	66	71	86
Clusters (Households)	1,433	1,526	1,123
Observations (Individuals)	1,763	1,901	1,470
RD Estimate for 2003	-0.013 (0.041)	-5 (644)	-219 (903)
Mean	0.668	3,516	5,147
Bandwidth (Month)	92	52	52
Clusters (Households)	2,033	1,235	811
Observations (Individuals)	2,553	1,473	986
RD Estimate for 2005	-0.022 (0.041)	-427 (455)	-502 (623)
Mean	0.658	3,139	4,742
Bandwidth (Month)	89	95	96
Clusters (Households)	1,946	2,035	1,329
Observations (Individuals)	2,430	2,567	1,707

Notes: This table reports regression discontinuity (RD) estimates for the effects of reaching $70\frac{1}{2}$ on charitable giving in the pre-period, when required minimum distributions were in effect but before qualified charitable distributions existed. Each panel corresponds to a different pre-period year. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.7: Suggestive Evidence on Charitable Giving in 2021

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
RD Estimate for 2021	0.066 (0.068)	1,185* (667)	1,640** (822)
Mean	0.624	2,696	4,302
Bandwidth (Months)	86	93	103
Clusters (Households)	1,223	1,312	896
Observations (Individuals)	1,431	1,539	1,075

Notes: This table reports suggestive regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions and required minimum distributions in 2021. The RD estimates come from estimating equation (1) when excluding observations between $70\frac{1}{2}$ and 72. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.8: Density Test Results for the Pre-Period, Before Qualified Charitable Distributions Existed

	t-statistic (1)	p-value (2)
1999	0.492	0.623
2001	1.454	0.146
2003	1.109	0.267
2005	-1.603	0.109

Notes: This table reports the resulting t-statistics and p-values from the formal density tests as proposed by Cattaneo, Jansson and Ma (2020) for tax years 1999 to 2005.

Table A.9: Difference-in-Discontinuity Estimates Using Covariates as Outcome Variables

	Male (1)	White (2)	Married (3)	College (4)
Diff-in-Disc Estimate for 2007	0.009 (0.062)	0.025 (0.025)	-0.010 (0.050)	-0.036 (0.050)
Mean				
Bandwidth (Months)	58	87	63	89
Clusters (Households)	3,013	3,783	3,131	3,824
Observations (Individuals)	7,699	11,256	8,322	11,479
Diff-in-Disc Estimate for 2009	0.022 (0.049)	0.025 (0.026)	0.058 (0.045)	-0.061 (0.048)
Mean				
Bandwidth (Months)	87	74	78	90
Clusters (Households)	3,974	3,636	3,729	4,051
Observations (Individuals)	11,174	9,644	10,104	11,530
Diff-in-Disc Estimate for 2011	0.075 (0.050)	0.012 (0.031)	-0.049 (0.046)	-0.003 (0.049)
Mean				
Bandwidth (Months)	85	72	84	88
Clusters (Households)	4,057	3,674	4,033	4,139
Observations (Individuals)	10,697	9,180	10,603	11,056
Diff-in-Disc Estimate for 2013	0.073 (0.056)	0.004 (0.028)	0.036 (0.049)	0.066 (0.056)
Mean				
Bandwidth (Months)	76	84	79	74
Clusters (Households)	3,979	4,225	4,078	3,922
Observations (Individuals)	9,617	10,601	9,986	9,364
Diff-in-Disc Estimate for 2015	0.002 (0.056)	0.003 (0.031)	-0.007 (0.048)	-0.076 (0.048)
Mean				
Bandwidth (Months)	96	103	107	111
Clusters (Households)	4,649	4,848	4,956	5,063
Observations (Individuals)	11,780	12,579	12,989	13,434
Diff-in-Disc Estimate for 2017	-0.052 (0.061)	0.011 (0.036)	0.026 (0.050)	-0.060 (0.045)
Mean				
Bandwidth (Months)	75	89	95	115
Clusters (Households)	4,074	4,517	4,690	5,243
Observations (Individuals)	9,146	10,806	11,488	13,642
Diff-in-Disc Estimate for 2019	0.001 (0.063)	0.016 (0.037)	0.035 (0.053)	0.019 (0.048)
Mean				
Bandwidth (Months)	71	84	87	87
Clusters (Households)	4,001	4,512	4,613	4,613
Observations (Individuals)	8,727	10,248	10,578	10,580

Notes: This table reports difference-in-discontinuity (diff-in-disc) estimates for the effects of reaching $70\frac{1}{2}$ on control variables in odd-numbered years from 2007 to 2019. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.10: Difference-in-Discontinuities Estimates for the Effects of Qualified Charitable Distributions on Charitable Giving Before the Tax Cuts and Jobs Act

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
Diff-in-Disc Estimate for 2007	0.086 (0.057)	-52 (654)	-602 (856)
Mean	0.661	3,224	4,764
Bandwidth (Months)	55	66	70
Clusters (Households)	2,913	3,212	2,338
Observations (Individuals)	7,286	8,721	6,018
Diff-in-Disc Estimate for 2009	0.064 (0.048)	852 (548)	826 (783)
Mean	0.648	2,819	4,331
Bandwidth (Months)	80	89	80
Clusters (Households)	3,782	4,025	2,618
Observations (Individuals)	10,339	11,404	6,704
Diff-in-Disc Estimate for 2011	0.011 (0.047)	109 (579)	15 (862)
Mean	0.646	3,036	4,652
Bandwidth (Months)	91	83	76
Clusters (Households)	4,213	3,997	2,611
Observations (Individuals)	11,418	10,474	6,251
Diff-in-Disc Estimate for 2013	0.007 (0.053)	116 (623)	45 (870)
Mean	0.653	3,447	5,291
Bandwidth (Months)	84	66	64
Clusters (Households)	4,226	3,654	2,481
Observations (Individuals)	10,602	8,375	5,293
Diff-in-Disc Estimate for 2015	-0.019 (0.052)	-710 (752)	-1,044 (1,005)
Mean	0.671	3,433	5,019
Bandwidth (Months)	110	79	79
Clusters (Households)	5,038	4,138	2,851
Observations (Individuals)	13,329	9,769	6,382
Diff-in-Disc Estimate for 2017	-0.092* (0.053)	173 (731)	915 (956)
Mean	0.674	3,009	4,483
Bandwidth (Months)	94	95	95
Clusters (Households)	4,670	4,691	3,228
Observations (Individuals)	11,383	11,491	7,449

Notes: This table reports difference-in-discontinuities estimates for the effects of aging into qualified charitable distributions on charitable giving before the Tax Cuts and Jobs Act, in odd-numbered years between 2007 and 2017. The estimates come from estimating equation (2). The underlying sample in each panel contains person-wave observations covering pre-period tax years 1999, 2001, 2003, and 2005, as well as the corresponding post-period year for that panel. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.11: Robustness of Estimates for 2007 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	0.085* (0.050)	-13 (580)	-580 (748)
B. Include Controls	0.086* (0.048)	-16 (564)	-584 (742)
C. Drop Triangular Weights	0.054 (0.047)	-102 (518)	-640 (673)
D. Use Survey Weights	0.051 (0.050)	-201 (549)	-741 (715)
E. Cluster on Running Variable	0.085* (0.045)	-13 (499)	-580 (672)
F. No Winsorizing	—	-2,889 (2,089)	-4,804 (3,122)
G. Winsorize More	—	-75 (490)	-602 (613)
H. Log Donations	—	—	-0.089 (0.125)

Notes: This table reports results from assessing the sensitivity of the 2007 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.12: Robustness of Estimates for 2009 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	0.065 (0.043)	789 (500)	741 (719)
B. Include Controls	0.058 (0.042)	769 (490)	812 (714)
C. Drop Triangular Weights	0.054 (0.040)	751* (442)	703 (658)
D. Use Survey Weights	0.028 (0.045)	685 (494)	715 (754)
E. Cluster on Running Variable	0.065 (0.040)	789 (503)	741 (707)
F. No Winsorizing	—	2,190 (1,947)	2,810 (2,863)
G. Winsorize More	—	673 (417)	590 (596)
H. Log Donations	—	—	0.077 (0.140)

Notes: This table reports results from assessing the sensitivity of the 2009 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.13: Robustness of Estimates for 2011 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	0.014 (0.044)	52 (547)	-33 (812)
B. Include Controls	0.018 (0.043)	75 (538)	-85 (806)
C. Drop Triangular Weights	0.022 (0.041)	174 (488)	-18 (711)
D. Use Survey Weights	0.051 (0.047)	798 (541)	601 (771)
E. Cluster on Running Variable	0.014 (0.040)	52 (567)	-33 (873)
F. No Winsorizing	—	438 (1,421)	501 (2,135)
G. Winsorize More	—	57 (455)	27 (660)
H. Log Donations	—	—	0.076 (0.143)

Notes: This table reports results from assessing the sensitivity of the 2011 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.14: Robustness of Estimates for 2013 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	0.009 (0.049)	154 (566)	134 (783)
B. Include Controls	-0.003 (0.049)	-63 (557)	-193 (767)
C. Drop Triangular Weights	-0.003 (0.045)	69 (562)	74 (793)
D. Use Survey Weights	0.055 (0.052)	317 (620)	-14 (887)
E. Cluster on Running Variable	0.009 (0.041)	154 (502)	134 (701)
F. No Winsorizing	—	409 (807)	550 (1,032)
G. Winsorize More	—	65 (522)	1 (717)
H. Log Donations	—	—	-0.031 (0.158)

Notes: This table reports results from assessing the sensitivity of the 2013 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.15: Robustness of Estimates for 2015 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	-0.015 (0.049)	-757 (715)	-1,122 (949)
B. Include Controls	-0.008 (0.047)	-673 (698)	-1,195 (937)
C. Drop Triangular Weights	-0.031 (0.044)	-675 (624)	-1,045 (831)
D. Use Survey Weights	-0.060 (0.051)	-812 (767)	-1,009 (1,029)
E. Cluster on Running Variable	-0.015 (0.042)	-757 (704)	-1,122 (983)
F. No Winsorizing	—	-893 (848)	-1,327 (1,174)
G. Winsorize More	—	-725 (651)	-1,070 (845)
H. Log Donations	—	—	-0.245 (0.175)

Notes: This table reports results from assessing the sensitivity of the 2015 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.16: Robustness of Estimates for 2017 to Specification Checks

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
A. Baseline	-0.089* (0.050)	104 (700)	785 (907)
B. Include Controls	-0.094* (0.049)	116 (679)	837 (881)
C. Drop Triangular Weights	-0.097** (0.046)	65 (613)	793 (804)
D. Use Survey Weights	-0.102** (0.051)	-188 (749)	514 (960)
E. Cluster on Running Variable	-0.089* (0.045)	104 (719)	785 (896)
F. No Winsorizing	—	709 (1,290)	1,797 (1,981)
G. Winsorize More	—	166 (604)	828 (766)
H. Log Donations	—	—	0.235 (0.161)

Notes: This table reports results from assessing the sensitivity of the 2017 estimates to various specification checks. Each column corresponds to a main outcome variable. Each row indicates the robustness check. Row A reproduces the baseline estimates for comparison. Row B adds control variables for gender, race, marital status, and college education to the regressions. Row C drops the triangular weights. Row D uses respondent survey weights instead of triangular weights. Row E clusters standard errors at the level of the running variable instead of at the household level. Row F does not winsorize donations. Row G winsorizes donations at the 99th percentile instead of the 99.5th percentile. Row H uses log donations as the outcome variable instead of donations conditional on giving measured in dollars. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.17: Placebo Estimates for People Without Individual Retirement Accounts in Earlier Years

	Indicator for Giving to Charity (1)	Donations (Including Zeros) (2)	Donations Conditional on Giving (3)
Placebo RD Estimate for 2007	-0.024 (0.028)	-78 (142)	30 (398)
Mean	0.329	1,110	3,414
Bandwidth (Months)	90	109	96
Clusters (Households)	3,609	4,022	1,090
Observations (Individuals)	4,571	5,200	1,501
Placebo RD Estimate for 2009	-0.025 (0.028)	-207 (153)	-467 (385)
Mean	0.350	1,034	2,896
Bandwidth (Months)	95	103	114
Clusters (Households)	3,563	3,786	1,309
Observations (Individuals)	4,538	4,869	1,803
Placebo RD Estimate for 2011	-0.023 (0.028)	-250 (205)	-628 (551)
Mean	0.342	945	2,731
Bandwidth (Months)	111	103	87
Clusters (Households)	4,330	3,981	1,054
Observations (Individuals)	5,591	5,128	1,437
Placebo RD Estimate for 2013	0.027 (0.033)	413** (210)	989* (556)
Mean	0.323	921	2,733
Bandwidth (Months)	107	76	72
Clusters (Households)	4,040	2,793	840
Observations (Individuals)	5,088	3,438	1,089
Placebo RD Estimate for 2015	-0.007 (0.036)	221 (193)	710 (481)
Mean	0.312	818	2,685
Bandwidth (Months)	99	109	83
Clusters (Households)	3,666	4,037	929
Observations (Individuals)	4,545	5,056	1,200
Placebo RD Estimate for 2017	-0.012 (0.037)	150 (212)	455 (555)
Mean	0.291	764	2,620
Bandwidth (Months)	94	106	97
Clusters (Households)	3,066	3,501	941
Observations (Individuals)	3,762	4,354	1,229

Notes: This table reports placebo regression discontinuity (RD) estimates for people in households with no Individual Retirement Accounts before the Tax Cuts and Jobs Act, in odd-numbered years from 2007 to 2017. Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.18: Summary Statistics for 2019

	People With IRAs			People Without IRAs		
	Mean (1)	SD (2)	Obs. (3)	Mean (4)	SD (5)	Obs. (6)
Age	69.56	10.20	3,672	67.38	11.02	9,521
Tax Year	2019.00	0.00	3,672	2019.00	0.00	9,521
Male	0.45	0.50	3,672	0.39	0.49	9,521
White	0.85	0.36	3,671	0.57	0.50	9,466
Married	0.66	0.47	3,666	0.52	0.50	9,492
College	0.74	0.44	3,672	0.43	0.50	9,517
Retired	0.60	0.49	3,672	0.66	0.47	9,521
Receives Social Security Benefits	0.59	0.49	3,672	0.54	0.50	9,521
Gives to Charity	0.62	0.49	3,672	0.29	0.45	9,521
Donations (Including Zeros)	2,905	5,146	3,672	803	2,510	9,521
Donations Conditional on Giving	4,685	5,863	2,277	2,792	4,046	2,738
IRA Balances	334,691	532,413	3,672	—	—	—
Individuals			3,672			9,521

Notes: This table reports summary statistics for two groups. The underlying samples contain data from survey wave 15, which corresponds to survey year 2020 and contains information on charitable giving for tax year 2019. The first three columns report means, standard deviations, and observations for people who own an IRA and make up my analysis sample. The next three columns report the same statistics for people in households where no member owns an IRA. The bottom row displays the number of unique individuals in each group.

Table A.19: Robustness of Estimates for Indicators for Donating Various Amounts to Charity in 2019 to Adjusting p-Values for Multiple Hypothesis Testing

	Coefficient (1)	Unadjusted p-value (2)	Adjusted p-value (3)
Indicator for Giving \$2,500 or more	0.085	0.061	0.115
Indicator for Giving \$5,000 or more	0.110	0.011	0.037
Indicator for Giving \$10,000 or more	0.089	0.001	0.004
Indicator for Giving \$15,000 or more	0.069	0.001	0.005
Indicator for Giving \$20,000 or more	0.034	0.049	0.118
Indicator for Giving \$25,000 or more	0.016	0.372	0.372

Notes: This table reports results from assessing the sensitive of the results for indicators for donating various amounts to charity in 2019 to adjusting p-values for multiple hypothesis testing. I use the *mhtreg* Stata package developed by Steinmayr (2020) to control for the familywise error rate, following the procedure developed by List, Shaikh and Xu (2019) and List, Shaikh and Vayalinkal (2023).

Table A.20: Effects of Qualified Charitable Distributions on Indicators for Donating Various Amounts to Charity Before the Tax Cuts and Jobs Act

	Indicator for Giving \$2,500 or More (1)	Indicator for Giving \$5,000 or More (2)	Indicator for Giving \$10,000 or More (3)	Indicator for Giving \$15,000 or More (4)	Indicator for Giving \$20,000 or More (5)	Indicator for Giving \$25,000 or More (6)
RD Estimate for 2007	0.003 (0.041)	0.002 (0.035)	-0.018 (0.024)	-0.014 (0.021)	-0.010 (0.018)	0.019 (0.015)
Mean	0.291	0.161	0.066	0.040	0.026	0.013
Bandwidth (Months)	87	78	71	69	63	52
Clusters (Clusters)	1,833	1,700	1,593	1,559	1,444	1,244
Observations (Households)	2,276	2,086	1,932	1,883	1,726	1,443
RD Estimate for 2009	0.050 (0.040)	0.041 (0.034)	0.023 (0.021)	0.019 (0.017)	0.016 (0.015)	0.006 (0.012)
Mean	0.294	0.159	0.061	0.027	0.020	0.015
Bandwidth (Months)	92	83	87	91	96	104
Clusters (Clusters)	1,870	1,707	1,784	1,853	1,924	2,049
Observations (Households)	2,320	2,094	2,194	2,289	2,400	2,565
RD Estimate for 2011	0.036 (0.044)	0.029 (0.036)	-0.014 (0.026)	0.001 (0.020)	-0.001 (0.016)	0.003 (0.014)
Mean	0.273	0.171	0.083	0.044	0.027	0.015
Bandwidth (Months)	83	74	76	77	99	93
Clusters (Clusters)	1,557	1,415	1,443	1,463	1,816	1,715
Observations (Households)	1,875	1,692	1,734	1,755	2,225	2,087
RD Estimate for 2013	0.037 (0.053)	-0.019 (0.042)	0.018 (0.031)	-0.006 (0.023)	0.007 (0.016)	0.015 (0.010)
Mean	0.325	0.206	0.105	0.051	0.024	0.018
Bandwidth (Months)	67	68	67	84	73	69
Clusters (Clusters)	1,248	1,266	1,248	1,555	1,362	1,289
Observations (Households)	1,481	1,502	1,481	1,897	1,631	1,527
RD Estimate for 2015	-0.041 (0.060)	-0.092* (0.055)	-0.050 (0.039)	0.010 (0.027)	-0.018 (0.017)	0.001 (0.013)
Mean	0.350	0.218	0.099	0.043	0.027	0.016
Bandwidth (Months)	76	76	87	80	105	115
Clusters (Clusters)	1,236	1,236	1,443	1,306	1,724	1,885
Observations (Households)	1,492	1,492	1,756	1,580	2,129	2,347
RD Estimate for 2017	-0.000 (0.054)	0.024 (0.048)	-0.002 (0.038)	0.010 (0.030)	0.009 (0.026)	-0.004 (0.016)
Mean	0.319	0.198	0.097	0.040	0.028	0.020
Bandwidth (Months)	94	89	94	96	97	127
Clusters (Clusters)	1,447	1,361	1,447	1,469	1,479	1,887
Observations (Households)	1,744	1,642	1,744	1,778	1,789	2,340

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on indicator variables for making large donations to charity before the Tax Cuts and Jobs Act, in odd-numbered years from 2007 to 2017. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.21: Effects of Qualified Charitable Distributions on Indicators for Claiming the Standard Deduction and for Itemizing Deductions in 2019

	Takes the Standard Deduction (1)	Itemizes Deductions (2)
RD Estimate for 2019	-0.038 (0.068)	0.038 (0.068)
Mean	0.524	0.476
Bandwidth (Months)	71	71
Clusters (Households)	923	923
Observations (Individuals)	1,097	1,097

Notes: This table reports regression discontinuity (RD) estimates for the effects of aging into qualified charitable distributions on indicator variables for claiming the standard deduction and for itemizing deductions in 2019. The RD estimates come from estimating equation (1). Standard errors clustered at the household level are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$